FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Prescribing for Children

Drug handbook | Dr Nabatte

The paediatric drug handbook for the ward. Children are dosed by weight (mg/kg) up to a maximum adult dose, neonates handle drugs differently, and a decimal-point slip can kill. For each drug you need the mg/kg dose, route, frequency, maximum, what to monitor and what reverses it. Weight-based antibiotics, IV artesunate, seizure drugs, fluids and emergency drugs are examined every year.

**Must know before the ward round**

1. **Weigh every child** and write the weight on every chart. No scale? Estimate with **(age in years + 4) × 2** (1–10 years) and replace it with a real weight as soon as possible. **Never exceed the adult dose.**
2. **Gentamicin 7.5 mg/kg IV/IM once daily** (>1 week old); neonates in week 1: **5 mg/kg** (≥2.5 kg) or **3 mg/kg** (LBW) once daily. **Ampicillin 50 mg/kg IV 6-hourly**; **ceftriaxone 50–80 mg/kg daily, 100 mg/kg for meningitis** (max 4 g/day).
3. **Pneumonia (UCG 2023, WHO 2024)**: **amoxicillin DT 40 mg/kg twice daily × 5 days** (250 mg BD at 2–12 months, 500 mg BD at 1–3 years, 750 mg BD at 3–5 years).
4. **Severe malaria**: **IV artesunate 3 mg/kg if <20 kg** (2.4 mg/kg if ≥20 kg) at 0, 12 and 24 h, then daily until oral, then a full ACT. **Rectal artesunate 10 mg/kg** pre-referral only (≤6 years) and **always refer**.
5. **Convulsing child**: **diazepam 0.5 mg/kg rectally** (or 0.2–0.3 mg/kg IV), max 10 mg, **repeat once after 10 min**; then **phenobarbital 15–20 mg/kg** (neonates 20 mg/kg) or **phenytoin 15–20 mg/kg in saline over 20+ min**. Check glucose every time.
6. **Hypoglycaemia**: **10% dextrose 5 mL/kg IV** (neonates **2 mL/kg**). Never push 50% dextrose into a child: dilute 1 mL/kg of 50% with 4 mL/kg water.
7. **Shock (not SAM, not severe anaemia)**: **20 mL/kg Ringer's lactate or saline** (UCG; WHO 2016: 10–20 mL/kg over 30–60 min) and reassess. **SAM in shock: 15 mL/kg over 1 h** of a dextrose-containing fluid. **Severe anaemia: blood, not boluses.**
8. **Maintenance fluid (UCG)**: 100 mL/kg/day for the first 10 kg + 50 mL/kg for the next 10 kg + 25 mL/kg for each kg above 20. **Neonate day 1: 10% dextrose 60 mL/kg/day**, rising daily to 150 by day 4.
9. **Paracetamol 15 mg/kg 6-hourly** (max 60 mg/kg/day, 4 g); **ibuprofen 5–10 mg/kg 6–8-hourly** (>3 months, not if dehydrated). **No aspirin under 16** (Reye's) except Kawasaki/rheumatic fever. **No codeine or tramadol in children.**
10. **Anaphylaxis**: **adrenaline 1:1000 IM 0.01 mL/kg**: **0.15 mL (<6 y), 0.3 mL (6–12 y), 0.5 mL (>12 y)**; repeat every 5 min. **Cardiac arrest**: **0.1 mL/kg of 1:10,000 IV/IO** (10 mcg/kg).
11. **Antidotes**: paracetamol → **acetylcysteine 150/50/100 mg/kg**; organophosphate → **atropine 0.05 mg/kg doubling until the chest is dry**; opioid → **naloxone 0.01–0.1 mg/kg**; iron → **desferrioxamine 15 mg/kg/h**; kerosene → **no emesis, no lavage, no charcoal**.
12. **Never in children**: ceftriaxone with IV calcium in neonates, chloramphenicol in the first week of life, tetracyclines under 8 years, codeine under 12, aspirin under 16, metoclopramide as a routine antiemetic, IV quinine as a bolus.


1 In a nutshell
2 Definitions & classification
3 Epidemiology & Uganda context
4 Principles of prescribing for children
5 Clinical acumen
6 Clinical workup
7 Antimicrobials
8 Antimalarials

9 Anticonvulsants
10 Respiratory drugs
11 Fluids, electrolytes & nutrition
12 Analgesia & antipyretics
13 Emergency & resuscitation drugs
14 Neonatal drugs
15 TB & HIV drugs
16 Vaccines & supplements
17 Poisoning antidotes
18 Management of drug emergencies & toxicity
19 Emergency drug box for the paediatric acute area
20 Ward-round & exam pearls
21 Mnemonics & memory aids
22 High-yield summary

# In a nutshell

Children are not small adults. A 3 kg newborn, a 10 kg toddler and a 40 kg adolescent can all be on the same ward with the same diagnosis, and each needs a **different dose**, often a **different formulation**, sometimes a **different drug**. Most paediatric prescribing is a calculation: **dose (mg/kg) × weight (kg)**, capped at the **adult maximum**. Get the weight right, get the decimal point right, and you have avoided most of the harm.

This handbook is organised by drug group, the way you will meet them on the Mengo and Mulago paediatric wards: antimicrobials, antimalarials, anticonvulsants, respiratory drugs, fluids and nutrition, analgesia, emergency drugs, neonatal drugs, TB and HIV drugs, vaccines and supplements, poisoning antidotes, and finally the drug emergencies you must recognise. Key drugs have a **drug card** (class, mechanism, indications, mg/kg dose, route, frequency, maximum, side effects, contraindications, monitoring, toxicity and antidote).

**CLINICAL PEARL**

For any drug the examiner asks about, answer in this order: **what it is, why we give it, how many mg/kg, by which route, how often, the maximum, what we watch for, when we must not give it, and what reverses it.** Then do the sum out loud for the child in front of you: "He weighs 12 kg, so 50 mg/kg is 600 mg."

**UGANDA CONTEXT**

Doses here follow the **Uganda Clinical Guidelines (UCG) 2023**, the **WHO Pocket Book of Hospital Care for Children (2nd edition, 2013)** and newer WHO guidance: the **WHO 2024 guideline on pneumonia and diarrhoea in children**, the **WHO guidelines for malaria (10 September 2026 edition)**, the **WHO 2023 guideline on wasting and nutritional oedema**, the **WHO 2022 consolidated TB guidelines for children and adolescents**, and the **Uganda consolidated HIV guidelines (2022)**. Where Uganda practice differs from WHO or UK/US guidance, it is flagged. Ugandan wards follow the UCG; when an exam question is set at UCU, the UCG answer is the safe one.

**EXAM FAVOURITE**

The commonest fatal paediatric drug errors are **tenfold errors** (a misplaced decimal point: 0.5 mg written as 5 mg, or "5.0" read as "50"), **mL versus mg confusion** (paracetamol 120 mg/5 mL versus 250 mg/5 mL), and **using an adult ampoule without dilution** (50% dextrose, 1:1000 adrenaline IV, undiluted potassium chloride). Examiners ask how you prevent them.

# Definitions & classification

| Term | Meaning | Why it matters |
|---|---|---|
| **Neonate** | Birth to 28 days | Immature liver and kidneys: longer dosing intervals, avoid some drugs completely |

| Term | Meaning | Why it matters |
|---|---|---|
| **Infant** | 1–12 months | Rapid change in drug handling; doses per kg often **higher** than in adults |
| **Child** | 1–12 years | Weight-based doses, capped at the adult dose |
| **Adolescent** | 10–19 years (WHO) | Usually adult doses once ≥ about 40–50 kg; adherence and confidentiality issues |
| **Postnatal age (PNA)** | Days since birth | Sets neonatal dosing intervals (e.g. ampicillin 12-hourly in week 1, 8-hourly in weeks 2–4) |
| **Gestational age / postmenstrual age** | Weeks of pregnancy at birth / gestation + postnatal age | Preterm babies clear drugs more slowly than term babies of the same postnatal age |
| **mg/kg/dose** vs **mg/kg/day** | Dose per administration vs total daily dose | Confusing the two causes three- or four-fold errors |
| **Loading dose** | A bigger first dose to reach a target level quickly | Phenobarbital, phenytoin, quinine, caffeine, aminophylline |
| **Therapeutic index** | Gap between effective and toxic levels | Narrow: gentamicin, aminophylline, phenytoin, digoxin, chloramphenicol |
| **Off-label** | Used outside the licensed age, dose or indication | Common and often unavoidable in paediatrics; follow guideline doses |
| **Weight band** | A dose fixed for a weight range (e.g. AL 1 tablet for 5–14 kg) | Used for ACTs, TB FDCs, ARVs and cotrimoxazole to avoid calculations |

# Epidemiology & Uganda context

- **Medication errors** are about **three times commoner** in children than adults in hospital studies, and **tenfold errors** are almost unique to paediatrics. Most happen at prescribing (wrong dose, wrong weight), then at preparation (wrong dilution) and administration (wrong route, wrong time).
- In Uganda the **big killers** of under-fives are **malaria, pneumonia, diarrhoea, neonatal conditions (prematurity, birth asphyxia, sepsis), malnutrition and HIV**. The drugs for these few conditions make up most of what you will prescribe: **artesunate and ACTs, amoxicillin, ampicillin + gentamicin, ceftriaxone, ORS and zinc, Ringer's lactate, 10% dextrose, F-75/RUTF, cotrimoxazole and ARVs, phenobarbital and diazepam, blood**.
- **Availability by level**:
  - **HC II/III**: ORS, zinc, amoxicillin DT, AL, rectal artesunate (pre-referral), cotrimoxazole, paracetamol, albendazole, vitamin A, iron, diazepam (rectal), IM artesunate or artemether at HC III, first-dose IM gentamicin/ampicillin or ceftriaxone, RUTF at OTC sites.
  - **HC IV / general hospital**: IV fluids, IV antibiotics, IV artesunate, phenobarbital, blood transfusion, oxygen concentrators, nebulised salbutamol, F-75/F-100.
  - **Regional/national referral (Mulago, Mengo)**: IV phenytoin, caffeine, aminophylline, hydrocortisone, ceftazidime/meropenem, acetylcysteine, antivenom (patchy), drug levels (rarely), paediatric ICU drugs.
- **Formulations are a daily problem**: syrups run out, dispersible tablets have to be split, and caregivers often receive a bottle without a measuring syringe. A child "on amoxicillin" may be getting half the dose because a household teaspoon holds 3–7 mL, not 5.
- **Self-medication and herbal medicine** are common. Ask specifically about **local herbs** (some are hepatotoxic or nephrotoxic), **drugs bought at a drug shop**, and **left-over adult tablets**.

# Principles of prescribing for children

## 1. The weight is part of the prescription

- **Weigh every child** at admission, undressed (or subtract the clothes), on a calibrated scale, and **write the weight at the top of the drug chart**. Reweigh SAM, nephrotic and dehydrated children daily: their weight changes fast.
- **Estimating weight** when you cannot weigh (resuscitation):

| Age | APLS formula | Example |
|---|---|---|
| 0–12 months | (0.5 × age in months) + 4 | 6 months → 7 kg |
| 1–5 years | (2 × age in years) + 8 | 3 years → 14 kg |
| 6–12 years | (3 × age in years) + 7 | 8 years → 31 kg |
| Older formula (1–10 years) | (age + 4) × 2 | 3 years → 14 kg |

A **length-based tape** (e.g. Broselow) is more accurate than age and gives pre-calculated doses. Rules of thumb: birth weight **doubles by 5 months, triples by 1 year, quadruples by 2 years**.

- **Malnourished children** are lighter than their age suggests: always weigh. **Obese children**: for drugs that stay in water (gentamicin) or have a narrow margin, dose on **ideal body weight** or an adjusted weight, and never exceed adult doses.

## 2. Weight versus body surface area (BSA)

- **Most drugs: dose by weight (mg/kg).** Simple, good enough, and the basis of almost every UCG and WHO table.
- **BSA (mg/m²)** tracks metabolic rate and organ size better. It is used for **chemotherapy**, some ARVs historically, and **hydrocortisone replacement**. BSA (m²) = √(height in cm × weight in kg ÷ 3600) (Mosteller). A 10 kg, 72 cm child is about **0.45 m²**; an adult about **1.7–1.8 m²**.
- **Why mg/kg doses are often higher in small children**: per kg, a toddler has a bigger liver and higher metabolic rate than an adult, so drugs like **theophylline, phenytoin and paracetamol** are cleared faster. Dosing a 10 kg child at one-seventh of the adult dose (by weight) usually *under*-doses.
- **Never exceed the adult dose.** A 50 kg adolescent on "ceftriaxone 100 mg/kg" gets 4 g (the adult maximum), not 5 g.

## 3. How children handle drugs (pharmacokinetics)

| Stage | Neonate (especially preterm) | Infant / child | What it means at the bedside |
|---|---|---|---|
| **Absorption** | Gastric pH higher (less acid), gastric emptying slow and irregular; **thin skin absorbs topical drugs** | Near adult by 2 years | Oral penicillins absorbed well; **avoid** alcohol/iodine skin washes and steroid creams over big areas in neonates; IM absorption unreliable in shock |
| **Distribution** | **Total body water ~75–80%** (adult 60%) → bigger volume for water-soluble drugs; **low albumin** and **bilirubin competing for albumin** | Body water falls to adult values by ~1 year | Gentamicin needs a **higher mg/kg** dose for the same peak but a **longer interval**; **sulphonamides and ceftriaxone displace bilirubin → kernicterus** risk |
| **Metabolism** | **Immature liver enzymes**, especially **glucuronidation** | Liver clearance **higher than adults** per kg from about 1–2 years | **Chloramphenicol → grey baby syndrome**; caffeine half-life ~100 h in preterms (once-daily dosing); toddlers need relatively large paracetamol/phenytoin doses |

| Stage | Neonate (especially preterm) | Infant / child | What it means at the bedside |
|---|---|---|---|
| **Excretion** | **GFR low at birth** (about 20–30% of adult per m²), matures over the first year; lower in preterms | Adult by about 1–2 years | Longer intervals for **aminoglycosides, penicillins, vancomycin** in the first weeks; avoid nephrotoxins |
| **Blood–brain barrier** | More permeable | | Bilirubin, opioids and sedatives reach the brain more easily: **naloxone and caution with morphine**; apnoea after benzodiazepines/opioids |

CLINICAL PEARL

The two big ideas: **neonates clear drugs slowly** (so the interval gets longer), and **toddlers clear many drugs fast** (so the mg/kg dose is relatively big). Postnatal age matters as much as weight in the first month.

## 4. Formulations and administration

- **Choose the formulation** the family can give: **dispersible tablets** (amoxicillin DT, AL dispersible, TB FDCs, pDTG, ABC/3TC) are cheaper, stable without a fridge and easier to dose than syrups. Dissolve in a little clean water or breast milk.
- **Write the strength of liquids** ("paracetamol 120 mg/5 mL, give 7.5 mL") and **give an oral syringe**. Household spoons are inaccurate.
- **Excipients**: some syrups contain **sugar** (dental caries in chronic use), **alcohol** or **propylene glycol**; **benzyl alcohol** preservative in some injections caused the fatal **gasping syndrome** in preterm babies. Use preservative-free water for neonatal dilutions where possible.
- **IV drugs for small children** must be **diluted to a measurable volume** (e.g. 1 mL of 1:1000 adrenaline + 9 mL saline = 1:10,000) and **flushed** through: a small infant's line dead space can hold a whole dose.
- **IM injections**: anterolateral **thigh** in infants and toddlers (never the buttock: sciatic nerve), maximum about **1 mL per site** in infants. **IM artesunate and artemether go into the anterior thigh.**
- **Rectal route**: diazepam (using the IV solution with a syringe without needle, or a rectal tube), artesunate suppositories, paracetamol suppositories. Hold the buttocks together for a few minutes.
- **Taste and adherence**: bitter drugs (quinine, chloroquine, some ARVs) are often spat out. **Repeat the dose if the child vomits within 30 minutes** (for ACTs and many antibiotics); for rectal artesunate, re-insert if expelled within 30 minutes.

## 5. Writing a safe paediatric prescription

Every prescription must contain:

1. Child's **full name, age, date of birth, weight (kg) and date weighed**, inpatient number, ward; **allergies** ("NKDA" if none).
2. **Generic drug name** in capitals or clear print (no abbreviations like "AZT" on an inpatient chart where "azathioprine" exists; write zidovudine).
3. The **mg/kg dose and the calculated dose**: "Gentamicin 7.5 mg/kg = **90 mg** IV once daily".
4. **Route, frequency, start date, duration or review date**.
5. For liquids: **strength and volume** (mg **and** mL).
6. For infusions: drug, amount, diluent, final volume, rate in mL/h (and drops/min for gravity sets).
7. **Your name, signature, cadre and phone** so the nurse can call.

### Writing rules that prevent tenfold errors

- **No trailing zero**: write **1 mg**, never "1.0 mg" (read as 10 mg).

- **Always a leading zero**: write **0.5 mg**, never ".5 mg" (read as 5 mg).
- **Write "micrograms" and "units" in full**: "µg" is misread as "mg", "U" as "0".
- **Use whole units where possible**: 500 micrograms rather than 0.5 mg; 125 mg rather than 0.125 g.
- **Check the maximum**: is the calculated dose above the adult dose? If yes, give the adult dose.
- **Double check high-risk calculations** with a colleague: insulin, potassium, opioids, digoxin, IV antimalarials, aminophylline, anything given to a neonate.

### 6. The commonest errors (and how to stop them)

| Error | Example | Prevention |
|---|---|---|
| **Tenfold (decimal) error** | Morphine 0.5 mg written ".5 mg" read as 5 mg | Leading zero, no trailing zero, recalculate |
| **mg/kg/day given as mg/kg/dose** | Amoxicillin "80 mg/kg" given twice daily (160 mg/kg/day) | Write per dose and frequency |
| **Wrong weight** | Using last month's weight, or a guessed weight that was never corrected | Weigh, document, reweigh |
| **Wrong strength of liquid** | Paracetamol 250 mg/5 mL given in the mL calculated for 120 mg/5 mL (double dose) | Write mg and mL and the strength |
| **Adult ampoule not diluted** | 50% dextrose IV, 1:1000 adrenaline IV, undiluted KCl | Know the paediatric strength: 10% dextrose, 1:10,000 adrenaline, KCl only diluted in a bag |
| **Exceeding adult maximum** | 40 kg child given 100 mg/kg ceftriaxone = 4 g as a single non-meningitic dose | Cap at the adult dose |
| **Wrong interval in neonates** | Gentamicin 7.5 mg/kg daily in a 1.8 kg day-2 neonate | Check postnatal age and weight |
| **Drug–drug or drug–fluid incompatibility** | Ceftriaxone with Ringer's lactate (calcium) in a neonate; phenytoin in dextrose | Know the compatibility; flush lines |
| **Omitted doses** | First dose of antibiotic delayed 6 h waiting for the pharmacy | "Stat" doses given in the emergency area; check the chart on every round |

## Clinical acumen

### Focused history

Before prescribing anything to a child, ask (each answer changes the drug, the dose or the formulation):

- **Age and weight** (and **gestation and postnatal age** in neonates): sets the dose, interval and allowed drugs.
- **Allergies**: what exactly happened? A rash on amoxicillin during a viral illness is not penicillin anaphylaxis. **Sulphonamide allergy** matters for cotrimoxazole and SP; **previous Stevens–Johnson** with nevirapine, cotrimoxazole or anticonvulsants is a lifelong contraindication.
- **What has already been given**, when, and by whom: antimalarials at the HC III (quinine in the last 12 h changes the quinine loading dose), **rectal artesunate**, **diazepam doses** (two doses already → move to phenobarbital), antibiotics, paracetamol from home (cumulative overdose), **herbs**.
- **HIV status** (child and mother) and **ART regimen**: interactions (rifampicin with DTG, LPV/r and NVP), cotrimoxazole prophylaxis (do not add SP), abacavir hypersensitivity.
- **TB treatment**: rifampicin speeds up the metabolism of DTG (double DTG dose), LPV/r, phenytoin, steroids, and stains urine orange.
- **Nutrition**: SAM changes fluids (no standard boluses), antibiotics (always), iron (not in stabilisation), vitamin A (only if eye signs or measles) and drug handling (low albumin).

- **Kidney and liver problems**: oliguria (gentamicin, potassium), jaundice (ceftriaxone in neonates, paracetamol, TB drugs, NVP).
- **G6PD deficiency or sickle cell disease**: avoid oxidant drugs (primaquine, dapsone, nitrofurantoin, sulphonamides in known deficiency); SCD needs penicillin V, folic acid and hydroxyurea.
- **Can the child swallow and keep things down?** Vomiting everything = a danger sign and a reason for the parenteral route.
- **Who will give the medicine at home**, can they read, do they have a measuring syringe, can they afford it?

### Examination

- **Weight**, **length/height** (for BSA and Z-scores), **MUAC**, oedema.
- **ABCD and vital signs** against age norms: RR and $SpO_2$ before and after opioids or benzodiazepines; heart rate on salbutamol and aminophylline (stop aminophylline if pulse >180); capillary refill before fluids.
- **Hydration and perfusion**: decides oral versus IV fluids and the bolus.
- **Pallor**: severe anaemia changes fluids (blood, not crystalloid), and AZT, cotrimoxazole and chloramphenicol can cause marrow suppression.
- **Jaundice** in a neonate: avoid ceftriaxone and sulphonamides.
- **Level of consciousness and gag**: unconscious children get nothing by mouth; use IV/IM/rectal or NGT.
- **Skin and mucosae**: rash, target lesions, mucosal ulcers (SJS from cotrimoxazole, NVP, phenobarbital, carbamazepine).
- **Ears and hearing** before and during long aminoglycoside courses in older children.

### Red flags

RED FLAG

- **Slow or shallow breathing, $SpO_2$ falling** after diazepam, midazolam, phenobarbital or morphine: open the airway, oxygen, bag-mask ventilation; naloxone for opioids.
- **Stridor, wheeze, facial swelling, hypotension or collapse** within minutes of an injection (ceftriaxone, penicillin, blood, vaccines): **anaphylaxis**, adrenaline IM now.
- **Sweating, jitteriness, coma or a fit during IV quinine** (or any sick child not feeding): **hypoglycaemia**; check glucose.
- **Puffy face, crackles, rising RR and liver edge after fluids**: **fluid overload** (especially SAM, severe anaemia, severe malaria): stop the fluid, sit up, furosemide 1 mg/kg IV.
- **Bradycardia or arrhythmia during IV phenytoin, potassium or aminophylline**: stop the infusion.
- **Blistering rash with mouth/eye involvement** on cotrimoxazole, NVP or anticonvulsants: **Stevens–Johnson syndrome**; stop the drug for life.
- **Dark urine and new pallor 1–3 weeks after IV artesunate**: post-artesunate delayed haemolysis; check Hb.
- **Oculogyric crisis or neck twisting after metoclopramide** (or haloperidol): acute dystonia.

### Thinking like a clinician

- **"What does this child weigh, and is the dose per dose or per day?"** Say the sum out loud.
- **"Could the drug be the problem?"** A drowsy child after rectal diazepam, a hypoglycaemic child on quinine, a deaf child after weeks of gentamicin, a jaundiced child on TB drugs, a breathless SAM child after a bolus: think of the drug first.
- **"Is this the right drug for this age?"** Neonates: ampicillin + gentamicin, not chloramphenicol; no ceftriaxone with jaundice; phenobarbital, not diazepam, for neonatal fits. Under 8: no doxycycline. Under 12: no codeine. Under 16: no aspirin.

- **"What is the cheapest effective oral option?"** WHO 2024: chest-indrawing pneumonia without danger signs is treated with **oral amoxicillin at home**, not admission for injections.
- **"Is the child getting the same drug twice?"** Paracetamol syrup from home plus IV paracetamol; cotrimoxazole prophylaxis plus SP; two benzodiazepines from two clinicians.
- **Common student mistakes**: prescribing in mL without the strength; forgetting the maximum; using adult 50% dextrose; giving boluses to a child with SAM or severe anaemia; prescribing quinine "IV stat" as a push; giving gentamicin 8-hourly at the old 2.5 mg/kg dose; forgetting zinc with ORS; giving rectal artesunate and then not referring; stopping cotrimoxazole prophylaxis in an HIV-exposed infant before infection is excluded.

# Clinical workup

For drugs, the "workup" is the set of checks before you prescribe and the monitoring afterwards. Use it every time you write a paediatric chart.

### Step 0: First 5 minutes

In an emergency (convulsion, shock, anaphylaxis, hypoglycaemia, severe respiratory distress), **ETAT first, paperwork second**:

1. **Triage**: emergency signs (**ABCD**: obstructed breathing, severe respiratory distress, central cyanosis, shock, coma, convulsions, severe dehydration) → treat immediately.
2. **Estimate the weight** (formula or tape) if you cannot weigh, and say it out loud: "About 12 kg."
3. Give the **first-line drug from memory**: oxygen, **10% dextrose 5 mL/kg** if glucose low or unknown in a convulsing/unconscious child, **diazepam 0.5 mg/kg rectally**, **20 mL/kg Ringer's lactate** for shock (15 mL/kg over 1 h in SAM), **adrenaline 0.01 mL/kg of 1:1000 IM** for anaphylaxis, **salbutamol** for wheeze.
4. **Closed-loop communication**: "Diazepam 6 mg rectally, that is 1.2 mL of the 5 mg/mL ampoule. Please confirm." A second person checks the ampoule strength.
5. **Record the time and dose** of everything given before the child moves on.

### Step 1: Focused history

- [ ] **Age, weight (measured or estimated?), gestation and postnatal age** (neonates).
- [ ] **Allergies** and previous severe drug reactions (SJS, anaphylaxis, abacavir hypersensitivity).
- [ ] **Drugs already given** today (HC III referral note, rectal artesunate, diazepam, antibiotics, quinine in the last 12 h, home paracetamol, herbs).
- [ ] **HIV status and ART**, TB treatment, cotrimoxazole prophylaxis.
- [ ] **Nutrition** (SAM?), **sickle cell disease**, **kidney or liver disease**, **G6PD** if known.
- [ ] **Ability to swallow and vomiting**: decides the route.
- [ ] **Caregiver**: who gives the medicine, literacy, cost.

### Step 2: Focused examination

- **Weight, length, MUAC, oedema** (dose and nutrition).
- **ABCD and vital signs** against age norms; **$SpO_2$**.
- **Perfusion** (capillary refill, pulse volume, cold hands) and **hydration**.
- **Pallor and jaundice**.
- **Consciousness** (AVPU), **gag and swallow**.
- **Skin and mucosa** (baseline before cotrimoxazole, NVP, anticonvulsants).

### Step 3: Bedside tests

- **Blood glucose** (glucometer) in every convulsing, unconscious, severely ill, malnourished or quinine-treated child.
- **Malaria RDT** before antimalarials (treat severe malaria immediately if the test is delayed).
- **Hb (HemoCue)** before transfusion decisions, on AZT and in severe malaria.
- **Pulse oximetry** before and after oxygen, opioids and sedatives.
- **Urine output** (weigh nappies: 1 g ≈ 1 mL; normal ≥1 mL/kg/h) on gentamicin, potassium and fluids.
- **Urine dipstick** for glucose and ketones (DKA), blood (haemoglobinuria after artesunate or in G6PD crises).

### Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Serum creatinine / urea** | Before and during gentamicin, vancomycin, TDF; oliguria; severe malaria | Rise above age normal (e.g. >35 µmol/L in an infant, >60 in a school child) | Extend the gentamicin interval or switch to ceftriaxone; reduce renally cleared drugs; stop NSAIDs (regional referral; HC IV sometimes) |
| **Serum electrolytes (Na, K)** | IV fluids >24 h, SAM, diarrhoea, DKA, potassium replacement | Na <130 or >150; K <3 or >5.5 mmol/L | Change the fluid; correct K slowly; stop KCl if K high or oliguria |
| **Full blood count** | AZT, cotrimoxazole high dose, chloramphenicol, severe anaemia | Hb <8 g/dL on AZT; neutropenia | Switch AZT; stop chloramphenicol; transfuse by threshold |
| **LFTs (ALT, bilirubin)** | TB drugs, NVP, high-dose paracetamol, valproate, overdose | ALT >3× ULN with symptoms or >5× without | Stop hepatotoxic drugs, reintroduce one by one; acetylcysteine for paracetamol |
| **Paracetamol level** (4 h post-ingestion) | Paracetamol overdose (national referral; rarely available) | Above the treatment line | Acetylcysteine; if unavailable, treat on the history of dose (>150 mg/kg) |
| **Serum calcium, magnesium** | Neonatal fits, hypocalcaemic tetany, rickets, SAM | Ca <2.0 mmol/L (or ionised <1.0) | Calcium gluconate 10% slowly IV |
| **Drug levels** (phenytoin, gentamicin trough, digoxin, aminophylline) | Toxicity or poor response (rarely available outside national referral) | Gentamicin trough >1–2 mg/L | Lengthen interval; clinical monitoring where no levels |
| **Blood culture and CSF** | Before the first antibiotic dose if it does not delay treatment | Organism and sensitivity | Narrow or change the antibiotic (regional/national referral) |
| **Blood group and cross-match** | Severe anaemia (Hb <4 g/dL or <6 with distress) | | Transfuse 20 mL/kg whole blood or 10 mL/kg packed cells (TRACT: 30 mL/kg whole blood in some) |
| **HIV DNA PCR / rapid test** | Every admitted child of unknown status | Positive | Cotrimoxazole, ART, drug interactions |
| **Chest X-ray** | Before TB treatment if available; suspected aspiration (kerosene) | | Supports TB diagnosis; chemical pneumonitis |

### Step 5: Putting it together

**Model summary before prescribing:** "Baby Namata, 2 years, **12 kg** (WFH Z-score −1), HIV-unexposed, fully immunised, no allergies. Severe malaria with prostration and Hb 7.8 g/dL; glucose 4.6 mmol/L; tolerating oral fluids poorly. **Plan**: IV artesunate **3 mg/kg = 36 mg** (child <20 kg) at 0, 12 and 24 h then daily; switch to a full 3-day ACT when she can drink; paracetamol **15 mg/kg = 180 mg** 6-hourly; maintenance fluid **1,100 mL/day** (100 × 10 + 50 × 2) mostly orally; glucose 6-hourly; Hb check day 3 and at 2–3 weeks for delayed haemolysis; discharge on DP and post-discharge malaria chemoprevention."

**Worked example 1: a gentamicin dose** A 14-day-old baby weighing 2.8 kg with suspected sepsis. Weeks 2–4 of life → gentamicin **7.5 mg/kg once daily** (WHO) = **21 mg**. The vial is 40 mg/mL; 21 mg = 0.525 mL. Dilute 1 mL (40 mg) with 3 mL saline = 10 mg/mL; give **2.1 mL** IV over 3–5 minutes. Ampicillin 50 mg/kg = **140 mg** every **8 hours** (second week of life). Check that the baby passes urine.

**Worked example 2: is this paracetamol dose safe?** An 8 kg infant has been given "paracetamol syrup 10 mL 4-hourly" at home for 2 days. If the syrup is 120 mg/5 mL, each dose is **240 mg = 30 mg/kg**, six times a day = **180 mg/kg/day** (maximum 60). If it is the 250 mg/5 mL syrup, it is **375 mg/kg/day**. This is a **repeated supratherapeutic ingestion**: check LFTs, glucose and INR if possible, start **acetylcysteine** and seek senior advice.

**Worked example 3: converting a 50% dextrose ampoule** A 10 kg unconscious child, glucose 1.8 mmol/L. Dose: **10% dextrose 5 mL/kg = 50 mL**. If only 50% dextrose and water are available: **1 mL/kg of 50% (10 mL) + 4 mL/kg water (40 mL)** = 50 mL of 10% dextrose. Give over 5 minutes, recheck glucose in 30 minutes, then start a dextrose-containing maintenance fluid or feeds.

## Antimicrobials

Most antibiotics on a Ugandan paediatric ward are prescribed for **pneumonia, sepsis, meningitis, neonatal infection and SAM**. Learn the "ward six" (ampicillin, gentamicin, ceftriaxone, amoxicillin, cloxacillin, metronidazole) perfectly, then the rest.

### Antibiotics by indication (UCG 2023 / WHO)

| Indication | First line | Duration / notes |
|---|---|---|
| **Pneumonia, fast breathing or chest indrawing, no danger sign (2–59 months)** | **Amoxicillin DT 40 mg/kg orally 12-hourly** | **5 days** (WHO 2024: 3 or 5 days for fast breathing only; 5 days for chest indrawing), outpatient |
| **Severe pneumonia** (danger sign, $SpO_2$ <90%) | **Ampicillin 50 mg/kg IV 6-hourly** (or benzylpenicillin 50,000 IU/kg 6-hourly) **+ gentamicin 7.5 mg/kg once daily** | ≥5 days; switch to oral amoxicillin when improving. **Not better at 48 h → ceftriaxone 80 mg/kg daily**; staph suspected → **cloxacillin 50 mg/kg 6-hourly + gentamicin** |
| **Young infant (0–59 days) with possible serious bacterial infection** | **Ampicillin 50 mg/kg + gentamicin** (5 mg/kg week 1, 7.5 mg/kg after) | 7–10 days; meningitis 14–21 days. If referral is refused: **gentamicin IM daily + oral amoxicillin 50 mg/kg 12-hourly × 7 days** (WHO 2015 PSBI) |
| **Bacterial meningitis (>1 month)** | **Ceftriaxone 100 mg/kg IV once daily** (or 50 mg/kg 12-hourly), max 4 g/day | 10 days (7–14; longer for Gram-negatives, GBS) |
| **Sepsis (child >2 months)** | Ampicillin + gentamicin, or ceftriaxone 50–80 mg/kg daily | 7–10 days; add cloxacillin for skin/bone source |

| Indication | First line | Duration / notes |
|---|---|---|
| **Bloody diarrhoea (dysentery)** | **Ciprofloxacin 15 mg/kg orally 12-hourly × 3 days** (WHO 2024) | Second line: ceftriaxone 50–80 mg/kg daily × 3 days. **No antibiotics for acute watery diarrhoea** (except cholera) |
| **Cholera** | **Azithromycin 20 mg/kg single dose** (max 1 g) | Rehydration is the treatment; antibiotics shorten shedding |
| **SAM, uncomplicated** | **Amoxicillin 40 mg/kg 12-hourly × 5 days** (UCG) | WHO 2023: may be omitted in carefully monitored outpatients; Uganda still gives it |
| **SAM, complicated (inpatient)** | **Ampicillin 50 mg/kg 6-hourly × 2 days then oral amoxicillin × 5 days + gentamicin 7.5 mg/kg daily × 7 days** | Ceftriaxone if no improvement at 48 h or meningitis |
| **Osteomyelitis / septic arthritis** | **Cloxacillin 50 mg/kg IV 6-hourly** (+ gentamicin in young children) | IV until improving, then oral, total **3–6 weeks** |
| **Acute otitis media** | Amoxicillin 40 mg/kg 12-hourly | 5 days |
| **Streptococcal pharyngitis** | **Benzathine penicillin** single IM dose, or amoxicillin 50 mg/kg once daily × 10 days | Prevents rheumatic fever |
| **Pertussis** | **Azithromycin 10 mg/kg day 1, 5 mg/kg days 2–5** (or erythromycin 12.5 mg/kg 6-hourly × 10 days) | Treat contacts <1 year |
| **Typhoid** | Ciprofloxacin 15 mg/kg 12-hourly, or azithromycin 10–20 mg/kg daily, or ceftriaxone | 7–14 days depending on drug and severity |

## Drug card: Amoxicillin

| | |
|---|---|
| **Class** | Aminopenicillin (beta-lactam) |
| **Mechanism** | Binds penicillin-binding proteins → blocks cell-wall cross-linking → **bactericidal**; **time-dependent** killing (time above MIC matters, so give it regularly) |
| **Indications (paeds)** | Community pneumonia (outpatient and step-down), otitis media, sinusitis, pharyngitis, uncomplicated SAM, PSBI when referral is not possible, step-down after IV ampicillin |
| **Dose, route, frequency** | **Pneumonia: 40 mg/kg orally 12-hourly** (80 mg/kg/day) × 5 days. **UCG weight/age bands (250 mg DT)**: **2–12 months: 250 mg (1 tab) BD; 1–3 years: 500 mg (2 tabs) BD; 3–5 years: 750 mg (3 tabs) BD**. Young infant PSBI: **50 mg/kg 12-hourly × 7 days**. Older/milder infections (WHO 2013): 25 mg/kg 12-hourly |
| **Maximum** | **1 g per dose** (3–4 g/day) |
| **Side effects** | Diarrhoea, nappy candidiasis, rash (a non-allergic **maculopapular rash is common with EBV** infection), rarely anaphylaxis |
| **Contraindications / cautions** | True penicillin allergy (anaphylaxis, angioedema) |
| **Monitoring** | Clinical response at **day 3**: fever, RR, chest indrawing; return immediately if danger signs |
| **Toxicity & antidote** | Very safe; no antidote. Anaphylaxis → adrenaline |

CLINICAL PEARL

**Dispersible tablets** (250 mg) are the Ugandan standard: dissolve in a little water or breast milk in a cup. Show the caregiver, and ask her to show you back. The commonest cause of "treatment failure" in outpatient pneumonia is that the child got half the doses.

## Drug card: Ampicillin

| | |
|---|---|
| **Class** | Aminopenicillin (IV/IM) |
| **Mechanism** | Cell-wall synthesis inhibitor, bactericidal, time-dependent. Covers streptococci, *Listeria*, enterococci, many *H. influenzae* and some *E. coli*; **not** penicillinase-producing staph |
| **Indications (paeds)** | **Severe pneumonia, neonatal sepsis and meningitis** (with gentamicin), complicated SAM, sepsis of unknown source |
| **Dose, route, frequency** | **Child >1 month: 50 mg/kg IV/IM 6-hourly. Neonates (WHO): 50 mg/kg 12-hourly in week 1, 8-hourly in weeks 2–4** (meningitis: 100 mg/kg per dose at the same intervals). UCG neonatal sepsis lists **50 mg/kg 6-hourly** |
| **Maximum** | 2 g per dose (12 g/day in meningitis) |
| **Side effects** | Diarrhoea, rash, phlebitis; seizures with very high doses in renal failure |
| **Contraindications / cautions** | Penicillin anaphylaxis; reduce frequency in severe renal impairment |
| **Monitoring** | Temperature, feeding, RR, perfusion; culture results |
| **Toxicity & antidote** | None specific |

UGANDA CONTEXT

UCG 2023 gives ampicillin **6-hourly** for neonatal sepsis whatever the age; the WHO Pocket Book uses **12-hourly in the first week** and **8-hourly in weeks 2–4** because neonatal kidneys clear it slowly. Both are safe (ampicillin has a wide margin). Follow your unit chart at Mengo or Mulago, but know why the WHO interval is longer.

## Drug card: Gentamicin

| | |
|---|---|
| **Class** | Aminoglycoside |
| **Mechanism** | Binds the **30S ribosomal subunit** → misreading of mRNA; **concentration-dependent** killing with a **post-antibiotic effect** (hence once-daily dosing). Gram-negative cover; synergy with ampicillin against GBS, *Listeria*, enterococci |
| **Indications (paeds)** | Neonatal sepsis, severe pneumonia, complicated SAM, UTI/pyelonephritis, sepsis (always **with** a beta-lactam) |
| **Dose, route, frequency** | **Child >1 month (and neonates in weeks 2–4): 7.5 mg/kg IV/IM once daily. Neonates, week 1: 5 mg/kg once daily if ≥2.5 kg; 3 mg/kg once daily if low birth weight** (some units extend to 36–48-hourly in very preterm babies). Give IV slowly over 3–5 min or IM into the thigh |
| **Maximum** | Dose by mg/kg; in obese children use ideal/adjusted body weight. Course usually **≤7–10 days** |
| **Side effects** | **Nephrotoxicity** (usually reversible), **ototoxicity** (cochlear and vestibular, often permanent), neuromuscular blockade (apnoea with $MgSO_4$ or in botulism/myasthenia) |
| **Contraindications / cautions** | AKI/oliguria, myasthenia gravis; avoid with furosemide or other nephrotoxins; mitochondrial mutation (m.1555A>G) causes deafness after a single dose (rarely tested in Uganda) |

| | |
|---|---|
| **Monitoring** | **Urine output**, creatinine at baseline and every 2–3 days in longer courses, **trough level <1 mg/L** before the next dose where available (rare), hearing screen after long courses |
| **Toxicity & antidote** | No antidote; stop and switch to ceftriaxone. Neuromuscular block → ventilatory support; calcium may help |

DRUG DOSING

The old regimen of **2.5 mg/kg 8-hourly** is still seen on charts. Once-daily **7.5 mg/kg** gives a higher peak (better killing), **less kidney toxicity** (a drug-free trough lets the tubules clear it) and one injection instead of three. In a child with poor urine output or rising creatinine, **hold the dose** and switch to ceftriaxone rather than "reducing" it blindly.

## Drug card: Ceftriaxone

| | |
|---|---|
| **Class** | Third-generation cephalosporin |
| **Mechanism** | Beta-lactam, bactericidal; broad Gram-negative and streptococcal cover, crosses inflamed meninges well; **long half-life → once-daily dosing** |
| **Indications (paeds)** | **Meningitis**, sepsis, severe pneumonia failing first line, dysentery (second line), typhoid, gonococcal ophthalmia neonatorum, pre-referral single dose |
| **Dose, route, frequency** | **Sepsis/pneumonia: 50–80 mg/kg IV/IM once daily** (UCG pneumonia: 80 mg/kg). **Meningitis: 100 mg/kg once daily** (or 50 mg/kg 12-hourly). **Neonates: 50 mg/kg once daily (100 mg/kg for meningitis)**, only if not jaundiced. **Ophthalmia neonatorum: 50 mg/kg IM single dose (max 125 mg)**. IV over **30 minutes** in neonates (or 2–4 min bolus in older children); IM dissolved in 1% lignocaine (never give the lignocaine solution IV) |
| **Maximum** | **4 g/day** (meningitis); 2 g per dose otherwise |
| **Side effects** | Diarrhoea, rash, biliary sludge ("pseudolithiasis"), phlebitis, *C. difficile*, rarely immune haemolysis |
| **Contraindications / cautions** | **Neonates with hyperbilirubinaemia or preterm** (displaces bilirubin from albumin → kernicterus): use **cefotaxime** if available. **Never with IV calcium-containing fluids in neonates** (Ringer's lactate, calcium gluconate, TPN): fatal **ceftriaxone–calcium precipitates** in lungs and kidneys. In older children, flush the line or use separate lines. Cephalosporin anaphylaxis; caution with penicillin anaphylaxis |
| **Monitoring** | Fever curve, consciousness, neck stiffness, seizures (meningitis); jaundice in neonates |
| **Toxicity & antidote** | No antidote |

RED FLAG

**Ceftriaxone + Ringer's lactate in a neonate is a never event.** Use normal saline or dextrose to flush and dilute, and give calcium (if needed) through a different line after a 48 h gap in neonates.

## Drug card: Cloxacillin (and flucloxacillin)

| | |
|---|---|
| **Class** | Penicillinase-resistant (anti-staphylococcal) penicillin |
| **Mechanism** | Beta-lactam stable to staphylococcal penicillinase |
| **Indications (paeds)** | ***Staphylococcus aureus***: pneumonia with pneumatoceles/empyema or after measles, osteomyelitis, septic arthritis, pyomyositis, cellulitis, impetigo, neonatal skin/umbilical sepsis |

| | |
|---|---|
| **Dose, route, frequency** | **25–50 mg/kg IV/IM 6-hourly** (50 mg/kg for severe infection, bone, empyema). **Neonates: 25–50 mg/kg 12-hourly in week 1, 8-hourly in weeks 2–4.** Oral (flu)cloxacillin **12.5–25 mg/kg 6-hourly** |
| **Maximum** | 2 g per dose |
| **Side effects** | Phlebitis, diarrhoea, rash; cholestatic hepatitis (flucloxacillin, rare in children) |
| **Contraindications / cautions** | Penicillin allergy; does **not** cover MRSA (vancomycin or clindamycin) |
| **Monitoring** | Fever, local signs, CRP/ESR where available; X-ray for bone |
| **Toxicity & antidote** | None |

## Drug card: Metronidazole

| | |
|---|---|
| **Class** | Nitroimidazole |
| **Mechanism** | Reduced inside anaerobes and protozoa to radicals that break DNA |
| **Indications (paeds)** | Anaerobic cover (peritonitis, perforated appendix, brain abscess, aspiration), **necrotising enterocolitis** (with ampicillin + gentamicin), **neonatal tetanus**, **amoebic dysentery/liver abscess**, **giardiasis**, *C. difficile* |
| **Dose, route, frequency** | **7.5 mg/kg IV/oral 8-hourly** (anaerobes, tetanus; WHO). **Amoebiasis: 10 mg/kg 8-hourly × 5–10 days. Giardiasis: 5 mg/kg 8-hourly × 5–7 days.** Neonates: 7.5 mg/kg 12-hourly (first weeks) |
| **Maximum** | 500 mg per dose |
| **Side effects** | Nausea, metallic taste, dark urine; disulfiram reaction with alcohol (adolescents, alcohol-containing syrups); peripheral neuropathy on long courses |
| **Contraindications / cautions** | Severe liver disease (reduce dose) |
| **Monitoring** | Clinical response; LFTs if prolonged |
| **Toxicity & antidote** | None specific |

## Drug card: Co-trimoxazole (sulfamethoxazole + trimethoprim)

| | |
|---|---|
| **Class** | Sulfonamide + diaminopyrimidine (ratio **5:1**). Strengths: paediatric tablet **120 mg** (100/20), syrup **240 mg/5 mL** (200/40), adult **480 mg** (400/80) and **960 mg** (800/160) |
| **Mechanism** | Sequential block of **bacterial folate synthesis** (dihydropteroate synthase, then dihydrofolate reductase) |
| **Indications (paeds)** | **Prophylaxis (CPT)** for HIV-exposed infants from **6 weeks** and all children living with HIV (prevents PCP, toxoplasma, malaria, bacterial infections); **PCP treatment**; UTI (if sensitive); some shigella (resistance common) |
| **Dose, route, frequency** | **CPT once daily by weight (UCG): <5 kg: 120 mg; 5–14.9 kg: 240 mg; 15–29.9 kg: 480 mg; ≥30 kg: 960 mg. Treatment: 4 mg/kg trimethoprim (24 mg/kg combined) 12-hourly. PCP: 120 mg/kg/day combined (= 20 mg/kg/day trimethoprim) in 3–4 divided doses × 21 days** (UCG; WHO Pocket Book 8 mg/kg TMP 8-hourly), IV or oral, plus **prednisolone** if hypoxic |
| **Maximum** | 960 mg per dose (prophylaxis); treatment by mg/kg |
| **Side effects** | Rash, **Stevens–Johnson syndrome**, marrow suppression (neutropenia, anaemia), hepatitis, hyperkalaemia (high dose), kernicterus risk in jaundiced neonates |

| | |
|---|---|
| **Contraindications / cautions** | Sulfa allergy/SJS; **first 6 weeks of life** (bilirubin displacement); **severe anaemia or neutropenia**; G6PD deficiency (caution); **never with SP** (duplicate antifolate) |
| **Monitoring** | Rash at every visit (**stop at once if mucosal involvement**), FBC if pale |
| **Toxicity & antidote** | Marrow suppression: stop; **folinic acid** for severe antifolate toxicity. SJS: stop for life, supportive care, use **dapsone** as alternative prophylaxis |

**UGANDA CONTEXT**

HIV-exposed infants start CPT at **6 weeks** (when NVP prophylaxis dose changes) and continue **until HIV infection is excluded after breastfeeding ends**. Children living with HIV continue it **until at least 15 years** (UCG 2023 stopping rules: >15 years, on ART ≥1 year, suppressed VL, no stage 3/4 event, not pregnant).

## Drug card: Azithromycin

| | |
|---|---|
| **Class** | Macrolide (azalide) |
| **Mechanism** | Binds **50S** ribosomal subunit; long tissue half-life allows short courses |
| **Indications (paeds)** | **Pertussis**, cholera, atypical pneumonia (*Mycoplasma*), typhoid, trachoma, yaws, penicillin allergy |
| **Dose, route, frequency** | **Pertussis: 10 mg/kg once daily day 1 (max 500 mg), then 5 mg/kg (max 250 mg) days 2–5** (infants <6 months: 10 mg/kg daily × 5 days). **Cholera/trachoma: 20 mg/kg single dose (max 1 g). Typhoid: 10–20 mg/kg daily × 7 days.** Atypical pneumonia: 10 mg/kg daily × 3 days |
| **Maximum** | 500 mg/day (1 g single dose) |
| **Side effects** | Diarrhoea, vomiting, **QT prolongation**, **infantile hypertrophic pyloric stenosis** in the first weeks of life (smaller risk than erythromycin) |
| **Contraindications / cautions** | QT-prolonging drugs (quinine, some antimalarials), liver disease |
| **Monitoring** | Projectile vomiting in neonates (pyloric stenosis) |
| **Toxicity & antidote** | None |

## Drug card: Benzylpenicillin (crystalline penicillin, "X-pen")

| | |
|---|---|
| **Class** | Natural penicillin (IV/IM). **1 mega-unit (MU) = 600 mg** |
| **Mechanism** | Cell-wall synthesis inhibitor; excellent for streptococci, pneumococcus (mostly), meningococcus, *Treponema*, clostridia |
| **Indications (paeds)** | Severe pneumonia (with gentamicin), meningococcal disease, **symptomatic congenital syphilis**, tetanus, diphtheria |
| **Dose, route, frequency** | **50,000 IU/kg IV/IM 6-hourly** (pneumonia). **Meningitis: 100,000 IU/kg 6-hourly** (UCG: 100,000–150,000 IU/kg 4–6-hourly). **Neonates: 50,000 IU/kg 12-hourly in week 1, 8-hourly in weeks 2–4** (double for meningitis). **Congenital syphilis (symptomatic): 50,000 IU/kg IV 12-hourly for the first 7 days, then 8-hourly, total 10 days** (or procaine penicillin 50,000 IU/kg IM daily × 10 days) |

| | |
|---|---|
| **Maximum** | 2.4 g (4 MU) per dose |
| **Side effects** | Allergy, phlebitis, seizures (huge doses in renal failure), Jarisch–Herxheimer reaction in syphilis |
| **Contraindications / cautions** | Penicillin anaphylaxis |
| **Monitoring** | Clinical response |
| **Toxicity & antidote** | None |

## Drug card: Benzathine penicillin G

| | |
|---|---|
| **Class** | Long-acting depot penicillin (IM only) |
| **Mechanism** | Slow release from the muscle gives low penicillin levels for **2–4 weeks** |
| **Indications (paeds)** | **Asymptomatic congenital syphilis** (baby of an untreated/inadequately treated RPR-positive mother, normal examination); **streptococcal pharyngitis**; **secondary prophylaxis of rheumatic fever/RHD** |
| **Dose, route, frequency** | **Congenital syphilis (asymptomatic): 50,000 IU/kg deep IM single dose** (anterolateral thigh). **Pharyngitis: 600,000 IU if <30 kg, 1.2 MU if ≥30 kg, single dose. Rheumatic fever prophylaxis: 600,000 IU (<30 kg) or 1.2 MU (≥30 kg) IM every 4 weeks** (every 3 weeks in high-risk patients), for at least 10 years or until age 25 (UCG 2023; AHA: age 21), longer (to 40 or lifelong) with residual valve disease |
| **Maximum** | 2.4 MU per dose |
| **Side effects** | **Painful injection** (can be reconstituted with lignocaine), allergy |
| **Contraindications / cautions** | Penicillin anaphylaxis. **Never IV** (cardiorespiratory arrest, embolic gangrene). Never for neurosyphilis or symptomatic congenital syphilis (poor CSF levels) |
| **Monitoring** | Observe 30 min after injection; RHD clinic attendance |
| **Toxicity & antidote** | Anaphylaxis → adrenaline |

## Other antimicrobials you will meet

| Drug | Dose | Key point |
|---|---|---|
| **Chloramphenicol** | **25 mg/kg IV/oral 6-hourly** (max 1 g/dose) | Cheap, crosses into CSF; **contraindicated in preterms and neonates <7 days** (**grey baby syndrome**: vomiting, grey colour, hypotension, collapse from immature glucuronidation); idiosyncratic **aplastic anaemia**. Second-line in Uganda |

| Drug | Dose | Key point |
|---|---|---|
| **Cefotaxime** | Neonates **50 mg/kg 12-hourly (week 1), 8-hourly (weeks 2–4)**; children 50 mg/kg 6–8-hourly | Preferred over ceftriaxone in jaundiced neonates; less available |
| **Ciprofloxacin** | **15 mg/kg orally 12-hourly × 3 days** (dysentery) | Cartilage concern in animals is not a reason to withhold it for dysentery in children (WHO) |
| **Doxycycline** | 2 mg/kg 12-hourly (max 100 mg) | **Only >8 years** (tooth staining, bone); rickettsia, brucellosis, cholera in older children |
| **Erythromycin** | 12.5 mg/kg 6-hourly | Pertussis, chlamydia conjunctivitis/pneumonia in infants; pyloric stenosis risk in neonates |
| **Vancomycin** | 15 mg/kg 6-hourly (child) | MRSA; slow infusion over 60 min (red man syndrome); levels; national referral |
| **Fluconazole** | Oral thrush/oesophageal: 3–6 mg/kg daily; cryptococcal disease per HIV guidelines | Nystatin 100,000 IU 6-hourly for simple oral thrush |
| **Nystatin** | 100,000 IU (1 mL) 6-hourly after feeds × 7 days | Paint the mouth; treat the mother's nipples too |
| **Tetracycline eye ointment 1%** | Both eyes once at birth | Prophylaxis of ophthalmia neonatorum |

EXAM FAVOURITE

The examiner's favourite antibiotic questions in paediatrics: (1) the **two drugs and doses for a sick young infant** (ampicillin 50 mg/kg + gentamicin 5–7.5 mg/kg); (2) **why not ceftriaxone in a jaundiced neonate** (bilirubin displacement, calcium precipitation); (3) **why not chloramphenicol in the first week** (grey baby); (4) **why once-daily gentamicin** (concentration-dependent killing, post-antibiotic effect, less nephrotoxicity); (5) **oral amoxicillin dose for pneumonia** (40 mg/kg BD × 5 days).

# Antimalarials

Malaria is the leading cause of admission on most Ugandan paediatric wards. The drugs are simple; the art is in **recognising severe malaria**, giving **artesunate fast**, **treating hypoglycaemia and anaemia**, and **completing the oral course**.

## Uganda national malaria treatment policy (UCG 2023)

| Situation | First line | Alternatives |
|---|---|---|
| **Uncomplicated malaria** (all ages, including infants) | **Artemether–lumefantrine (AL)** | Artesunate–amodiaquine (AS-AQ); second line **dihydroartemisinin–piperaquine (DP)**; pyronaridine–artesunate; quinine tablets if none available |
| **Severe malaria** | **IV artesunate** | IM artesunate; **IV quinine** or **IM artemether** if artesunate unavailable |
| **Pre-referral** (no IV facilities) | **Rectal artesunate** (children **≤6 years** only) | IM artesunate, IM artemether or IM quinine |

| Situation | First line | Alternatives |
|---|---|---|
| **After severe malaria** | Complete with a **3-day ACT**; UCG 2023 discharges severe malaria on **DP**, then **monthly DP for 3 months** (post-discharge malaria chemoprevention, PDMC) | |

## Drug card: Artesunate (IV/IM)

| | |
|---|---|
| **Class** | Artemisinin derivative (water-soluble) |
| **Mechanism** | Endoperoxide bridge activated by haem iron → free radicals that kill **all blood stages, including young ring forms**, so it clears parasites fastest and prevents sequestration. Converted to **dihydroartemisinin (DHA)**; very short half-life (<1 h), so it must be followed by a partner drug |
| **Indications (paeds)** | **All severe malaria** (any age, any trimester), and uncomplicated malaria in a child who cannot take oral drugs |
| **Dose, route, frequency** | **Child <20 kg: 3 mg/kg; ≥20 kg: 2.4 mg/kg** IV (or IM into the anterior thigh) **at 0, 12 and 24 h, then once daily** until able to take oral medicine (minimum 24 h/3 doses), **then a full 3-day ACT** |
| **Preparation** | Vials of 30, 60 or 120 mg. Add the whole **1 mL of 5% sodium bicarbonate** (shake until clear, use within 1 h). **IV**: add **5 mL** of saline or 5% dextrose → **10 mg/mL**, give as a slow push over 1–2 (UCG: up to 5) minutes. **IM**: add **2 mL** saline → **20 mg/mL**. **Never water for injection** |
| **Maximum** | No fixed cap; dose by weight |
| **Side effects** | Very well tolerated; dizziness, nausea; **post-artesunate delayed haemolysis (PADH)** 1–3 weeks later (especially with high parasitaemia), rare allergy |
| **Contraindications / cautions** | Known artemisinin allergy (rare). Safe in neonates, infants and all trimesters |
| **Monitoring** | Consciousness, fits, **glucose** (malaria itself causes hypoglycaemia), Hb, urine output, parasitaemia at 48–72 h if available; **Hb at 2–3 weeks** (PADH) |
| **Toxicity & antidote** | None needed. PADH: transfuse if severe |

DRUG DOSING

**Worked example**: a 14 kg child with cerebral malaria. Dose **3 mg/kg × 14 = 42 mg**. Use one 60 mg vial: 1 mL bicarbonate + 5 mL saline = 6 mL at **10 mg/mL**; give **4.2 mL** IV. Repeat at 12 h and 24 h, then daily. If the child weighed 22 kg: **2.4 × 22 = 52.8 mg ≈ 53 mg** (5.3 mL).

EXAM FAVOURITE

Why **3 mg/kg** in children under 20 kg? Pharmacokinetic studies showed that small children have lower exposure to artesunate and DHA at 2.4 mg/kg, so WHO raised the dose in 2015 to give the same exposure as adults.

## Drug card: Rectal artesunate (RAS)

| | |
|---|---|
| **Class** | Artemisinin suppository (100 mg; 50 mg in some kits) |
| **Indications** | **Pre-referral treatment** of suspected severe malaria in children **≤6 years** when injectable treatment is not available (VHT, HC II, HC III without IV) |

| | |
|---|---|
| **Dose** | **10 mg/kg single dose. UCG bands: 5–<14 kg (4 months to <3 years): 1 × 100 mg; 14–19 kg (3 to <6 years): 2 × 100 mg** |
| **Technique** | Insert blunt end first, hold the buttocks together for **10 minutes**; **repeat the dose if expelled within 30 min** |
| **Key rule** | **RAS is not a treatment, it is a bridge.** The child must be **referred immediately** and receive **injectable artesunate and a full ACT**. In the CARAMAL study (2018–2020), children given RAS without completed referral and follow-on treatment did worse; WHO now stresses complete referral and restricts RAS to pre-referral use only |

## Drug card: Artemether–lumefantrine (AL, "Coartem")

| | |
|---|---|
| **Class** | ACT: artemether (fast, short-acting) + lumefantrine (slow, long-acting partner) |
| **Mechanism** | Artemether clears most parasites in 3 days; lumefantrine (half-life 3–6 days) kills the rest |
| **Indications (paeds)** | **First-line uncomplicated malaria** in Uganda (all ages, including infants); completion after parenteral treatment |
| **Dose, route, frequency** | Tablet 20/120 mg (dispersible for small children). **6 doses over 3 days: 0 h, 8 h, then twice daily on days 2 and 3. 5–14 kg: 1 tablet; 15–24 kg: 2 tablets; 25–34 kg: 3 tablets; ≥35 kg: 4 tablets** per dose |
| **Maximum** | 4 tablets per dose |
| **Side effects** | Nausea, vomiting, dizziness, sleep disturbance; QT effect minor |
| **Contraindications / cautions** | First trimester (adolescents: follow current pregnancy policy), severe malaria (needs parenteral treatment), QT-prolonging drugs |
| **Monitoring** | Give with **milk or fatty food** (lumefantrine absorption increases many-fold); **repeat the dose if vomited within 30 min**; return if danger signs |
| **Toxicity & antidote** | None |

UGANDA CONTEXT

For infants under 5 kg, the **WHO guidelines for malaria (September 2026 edition)** recommend an ACT at the same mg/kg target as for a 5 kg child and, where AL is used, the **new infant formulation of artemether–lumefantrine (1:12 ratio, approved in 2025 for babies <5 kg)**. UCG 2023 predates it and gives 1 standard tablet as the smallest band. Until the infant formulation is stocked, small or sick infants with malaria should be assessed for severe disease and treated in hospital under senior guidance.

**ACTs and ART (WHO)**: do not use **artesunate–amodiaquine** in children taking **efavirenz or zidovudine** (neutropenia, hepatotoxicity), and do not use **artesunate + SP** in children on **cotrimoxazole**.

## Other ACTs (UCG 2023 doses)

| ACT | Dose |
|---|---|
| **Artesunate–amodiaquine** (AS 50 mg + AQ 153 mg tablets, or co-formulated) | **Once daily × 3 days**: 0–11 months ½ + ½ tab; 1–6 years 1 + 1; 7–13 years 2 + 2; >13 years 4 + 4. (≈ AS 4 mg/kg + AQ 10 mg/kg daily) |
| **Dihydroartemisinin–piperaquine (DP)** 40/320 mg | **Once daily × 3 days**: 5–9.9 kg ½ tab; 10–20 kg 1; 20–40 kg 2; 40–60 kg 3 (WHO: DHA 4 mg/kg + PPQ 18 mg/kg; children <25 kg at least 2.5 + 20 mg/kg) |
| **Pyronaridine–artesunate** | Granules 60/20 mg: 5–<8 kg 1 sachet, 8–<15 kg 2, 15–<20 kg 3; tablets 180/60 mg: 20–<24 kg 1, 24–<45 kg 2 (once daily × 3 days) |

## Drug card: Quinine

| | |
|---|---|
| **Class** | Cinchona alkaloid (blood schizonticide) |
| **Mechanism** | Interferes with parasite haem detoxification; slower than artesunate and does not kill ring stages |
| **Indications (paeds)** | Severe malaria **when artesunate is not available**; oral quinine if no ACT available |
| **Dose, route, frequency** | **IV: 10 mg/kg quinine salt in 5–10 mL/kg of 5% dextrose over 4 h, every 8 h** (UCG 2023) until oral is possible, then complete 7 days with oral quinine (10 mg/kg 8-hourly) or a full ACT. **WHO Pocket Book**: **loading dose 20 mg/kg** in 10 mL/kg IV fluid over 2–4 h, then 10 mg/kg every 8 h; **omit the loading dose if quinine or mefloquine was given in the previous 12–24 h**. Rate never above **5 mg/kg/h**. IM: dilute and split between both anterior thighs |
| **Maximum** | Adult 600 mg salt per dose (maintenance) |
| **Side effects** | **Hypoglycaemia** (stimulates insulin release: the classic danger), **cinchonism** (tinnitus, deafness, nausea, blurred vision), **QT prolongation/arrhythmia**, hypotension if infused fast, sterile abscess IM |
| **Contraindications / cautions** | Never as an IV bolus; QT drugs; caution in G6PD deficiency |
| **Monitoring** | **Blood glucose every 4 h** (and if any change in consciousness or fits), pulse/BP, ECG if available |
| **Toxicity & antidote** | Hypoglycaemia → **10% dextrose 5 mL/kg**; arrhythmia → stop infusion, consider sodium bicarbonate; supportive |

RED FLAG

A child on IV quinine who becomes **sweaty, restless, fits or goes deeper into coma** is **hypoglycaemic until proven otherwise**. Check glucose immediately; run quinine in 5% (or 10%) dextrose; keep feeding by NGT if unconscious.

## Artemether (IM)

**3.2 mg/kg IM** (anterior thigh, never the buttock) on admission, then **1.6 mg/kg daily** until oral treatment is possible; oily injection, erratic absorption in shock. Use only if neither artesunate nor quinine is available.

## Malaria prevention drugs in children

| Drug | Use | Dose |
|---|---|---|
| **Seasonal malaria chemoprevention (SPAQ)** | Karamoja and other seasonal areas (campaigns) | SP + amodiaquine monthly during transmission season |
| **Perennial malaria chemoprevention / IPTi** | Some districts (pilot/programmes) | SP at EPI contacts |

| Drug | Use | Dose |
|---|---|---|
| **Post-discharge malaria chemoprevention (PDMC)** | WHO: children discharged after **severe anaemia** in moderate–high transmission areas; UCG 2023: all severe malaria discharges | **Monthly 3-day DP for 3 months** (UCG 2023) |
| **Malaria vaccine (R21/Matrix-M)** | Introduced into the Ugandan routine schedule in 2025 | WHO: **4-dose schedule from 5 months**, 4th dose 6–18 months after the 3rd (see Vaccines) |
| **Cotrimoxazole prophylaxis** | HIV-exposed or infected | Also protects against malaria: **never add SP** |

# Anticonvulsants

Convulsions in Ugandan children are most often due to **malaria (cerebral or febrile seizures), meningitis, hypoglycaemia, hypocalcaemia (neonates), hypoxic-ischaemic injury, epilepsy and electrolyte problems**. The drug is only part of the treatment: **ABC, glucose, oxygen, recovery position, treat the cause**.

## Convulsions: the step-by-step drug ladder (child >1 month)

1. **0 min**: ABC, recovery position, suction, oxygen; **check glucose** (treat if <2.5 mmol/L, or <3 mmol/L in SAM: 10% dextrose 5 mL/kg). Do not put anything in the mouth.
2. **Fitting ≥5 min (or still fitting on arrival)**: **diazepam 0.5 mg/kg rectally** (or **0.2–0.3 mg/kg IV** slowly over 1–2 min), **max 10 mg**. Alternative: **midazolam 0.3–0.5 mg/kg buccal/intranasal** (max 10 mg) or 0.15–0.2 mg/kg IV/IM.
3. **Still fitting 10 min after the first dose**: **second benzodiazepine dose** (same dose). **Never more than two doses** (respiratory depression).
4. **Still fitting 10 min later (status epilepticus)**: **phenobarbital 15–20 mg/kg IV over 15–20 min (or IM)**, or **phenytoin 15–20 mg/kg IV over 20–60 min in normal saline** (cardiac monitoring), or (where available) **levetiracetam 40 mg/kg IV over 5 min** (APLS/NICE 2022: as effective as phenytoin, fewer side effects).
5. **Still fitting**: second agent (e.g. phenytoin after phenobarbital), call anaesthesia/ICU (thiopentone or midazolam infusion with intubation at national referral).
6. **Throughout**: treat **fever** (paracetamol), **malaria** (artesunate), **meningitis** (ceftriaxone after LP if safe), **hypoglycaemia**; check sodium and calcium.

> **UGANDA CONTEXT**
> UCG 2023 (severe malaria complications) uses **diazepam 0.5 mg/kg rectally or 0.2 mg/kg IV (max 10 mg)**, then **phenobarbital 10–15 mg/kg loading** with **2.5 mg/kg** maintenance once or twice daily, or **phenytoin 15 mg/kg**. The WHO Pocket Book uses phenobarbital **15 mg/kg** in children and **20 mg/kg** in neonates. Buccal midazolam and IV levetiracetam are standard in UK/US algorithms but are not reliably stocked in Ugandan public hospitals.

## Drug card: Diazepam

| | |
|---|---|
| **Class** | Benzodiazepine (long-acting) |
| **Mechanism** | Enhances **GABA-A** receptor chloride current → neuronal hyperpolarisation; fast brain entry (lipophilic) but redistributes, so fits can recur |
| **Indications (paeds)** | **First-line for convulsions lasting ≥5 min** in children >1 month; febrile seizures; tetanus spasms (with other drugs) |

| | |
|---|---|
| **Dose, route, frequency** | **Rectal 0.5 mg/kg** (use the IV solution 5 mg/mL drawn into a syringe, needle removed, inserted 4–5 cm, or a rectal tube); **IV 0.2–0.3 mg/kg slowly over 2–3 min**. Repeat **once** after 10 min. WHO volume chart (10 mg/2 mL): 2–<4 months 0.5 mL; 4–<12 months 1.0 mL; 1–<3 years 1.25 mL; 3–<5 years 1.5 mL rectally |
| **Maximum** | **10 mg per dose**; **two doses** |
| **Side effects** | **Respiratory depression and apnoea** (worse with phenobarbital or after two doses), hypotension, sedation masking neurological signs, thrombophlebitis IV |
| **Contraindications / cautions** | **Neonates: avoid** (use phenobarbital); caution in respiratory failure; IM absorption is poor (do not give IM) |
| **Monitoring** | RR, $SpO_2$, airway, consciousness for at least 1 h; have a bag-valve-mask ready |
| **Toxicity & antidote** | **Flumazenil 10 micrograms/kg IV** (max 200 mcg, repeat to 1 mg) only for iatrogenic overdose with respiratory depression; **can provoke seizures**, so ventilate first in a fitting child |

## Drug card: Midazolam

| | |
|---|---|
| **Class** | Short-acting water-soluble benzodiazepine |
| **Indications** | Prolonged seizures (buccal/intranasal when no IV access), sedation for procedures, refractory status (infusion in ICU) |
| **Dose, route, frequency** | **Buccal 0.3–0.5 mg/kg** (max 10 mg): NICE/BNFc bands 3–11 months **2.5 mg**, 1–4 years **5 mg**, 5–9 years **7.5 mg**, ≥10 years **10 mg**. **IV/IM 0.15–0.2 mg/kg** (IM absorbed well, unlike diazepam). Infusion 1–5 micrograms/kg/min in ICU |
| **Maximum** | 10 mg single dose; two doses in total of any benzodiazepine |
| **Side effects / cautions** | Respiratory depression, hypotension |
| **Antidote** | Flumazenil (as above) |

## Drug card: Phenobarbital (phenobarbitone)

| | |
|---|---|
| **Class** | Barbiturate |
| **Mechanism** | Prolongs GABA-A chloride channel opening; also reduces glutamate effects. Long half-life (2–5 days; longer in neonates) |
| **Indications (paeds)** | **First-line for neonatal seizures**; second-line for status epilepticus in children; maintenance for neonatal and some childhood epilepsy (cheap, available) |
| **Dose, route, frequency** | **Neonates: 20 mg/kg IV over 15–20 min (or IM)**; if still fitting, further **10 mg/kg** doses to a total of **40 mg/kg**; maintenance **5 mg/kg once daily** (start 24 h after loading). **Children: 15–20 mg/kg IV/IM loading** (UCG: 10–15 mg/kg), then **2.5–5 mg/kg/day** in 1–2 doses. Dilute IV to 20 mg/mL or less; give no faster than 1 mg/kg/min |
| **Maximum** | Loading 1 g |
| **Side effects** | **Respiratory depression** (especially after benzodiazepines), sedation, hypotension; long-term: hyperactivity, behaviour and learning problems, rash/SJS; **enzyme inducer** (lowers levels of DTG, LPV/r, NVP, steroids, contraceptives) |
| **Contraindications / cautions** | Severe respiratory depression without airway support; porphyria; interactions with ART (UCG: avoid DTG with phenobarbital or use alternatives/dose adjustments) |
| **Monitoring** | RR, $SpO_2$, BP, sedation; level if available (therapeutic 15–40 mg/L) |

| | |
|---|---|
| **Toxicity & antidote** | No antidote: airway and ventilation; urine alkalinisation and multiple-dose activated charcoal enhance elimination in overdose |

## Drug card: Phenytoin

| | |
|---|---|
| **Class** | Hydantoin (sodium channel blocker) |
| **Mechanism** | Stabilises inactivated **voltage-gated sodium channels**; non-sedating (useful when you need to assess consciousness); **saturable (zero-order) kinetics** at higher levels → small dose increases cause big level rises |
| **Indications (paeds)** | Status epilepticus not responding to benzodiazepines; neonatal seizures not responding to phenobarbital |
| **Dose, route, frequency** | **Loading 15–20 mg/kg IV** (WHO 15–18 mg/kg) **diluted in 0.9% saline only** (to ≤10 mg/mL), infused over **20–60 min** at **≤1 mg/kg/min (max 50 mg/min)**, through a good IV line. Maintenance 2.5–5 mg/kg 12-hourly (neonates 2.5 mg/kg 12-hourly) |
| **Maximum** | Loading 1.5–2 g |
| **Side effects** | Infusion: **bradycardia, arrhythmias, hypotension** (from the propylene glycol vehicle and the drug); **"purple glove" syndrome** and tissue necrosis if it extravasates (pH 12). Chronic: gum hypertrophy, hirsutism, ataxia, nystagmus, rash/SJS, enzyme induction |
| **Contraindications / cautions** | **Never in dextrose** (precipitates), never IM; heart block or bradycardia; check whether the child is already on oral phenytoin before loading |
| **Monitoring** | **ECG/heart rate and BP during the infusion**, cannula site; level if available (10–20 mg/L) |
| **Toxicity & antidote** | No antidote; stop the infusion, supportive care. Nystagmus and ataxia = high level |

## Other anticonvulsants on the ward

| Drug | Typical paediatric dose | Notes |
|---|---|---|
| **Carbamazepine** | Start 5 mg/kg/day in 2 doses, increase to 10–20 mg/kg/day | Focal epilepsy; rash/SJS (HLA-B*15:02 in Asians), hyponatraemia, enzyme inducer; worsens absence and myoclonic seizures |
| **Sodium valproate** | 10–15 mg/kg/day in 2 doses → 20–40 mg/kg/day | Generalised epilepsies; **hepatotoxicity (especially <2 years, metabolic disease)**, pancreatitis, weight gain; **teratogenic: avoid in girls of childbearing potential** unless no alternative |
| **Levetiracetam** | 10 mg/kg 12-hourly → 30 mg/kg 12-hourly; IV 40 mg/kg for status | Few interactions (useful with ART); behaviour change |
| **Paraldehyde** | 0.4 mL/kg rectally (mixed with equal volume of oil) | Old second-line in some Ugandan wards; glass syringe needed |

CLINICAL PEARL

A febrile, convulsing Ugandan child has **cerebral malaria, meningitis or hypoglycaemia until proven otherwise**. The benzodiazepine stops the fit; the **dextrose, artesunate and ceftriaxone** treat the child. Simple febrile seizures (6 months–5 years, generalised, <15 minutes, single in 24 h, full recovery within an hour) need **no anticonvulsant prophylaxis**, only reassurance and fever control (antipyretics do not prevent recurrence).

# Respiratory drugs

The two big respiratory emergencies are **acute asthma/wheeze** and **croup**. Pneumonia is treated with antibiotics and **oxygen** (the most important respiratory "drug": give it when **$SpO_2$ <90%**, by nasal prongs at 0.5–1 L/min in infants and 1–2 L/min in older children; UCG targets $SpO_2$ >94% in sick young infants).

## Acute asthma: the drug ladder (GINA 2025, UCG 2023)

1. **Oxygen** to keep $SpO_2$ **94–98%**; sit the child up.
2. **Salbutamol** inhaler **100 micrograms/puff via a spacer** (with a mask under 4 years): **2–6 puffs (≤5 years) or 4–10 puffs (6–11 years) every 20 min for the first hour**; or **nebulised 2.5 mg (<5 years) / 5 mg (≥5 years)** in 3–4 mL saline, driven by oxygen, every 20 min × 3.
3. **Ipratropium bromide** for severe attacks: **250 micrograms nebulised (≤12 years; 500 micrograms ≥12)** every 20 min for the first hour, mixed with salbutamol.
4. **Systemic corticosteroid** within the first hour: **prednisolone 1–2 mg/kg orally once daily** (max **20 mg under 2 years, 30 mg at 2–5 years, 40 mg at 6–11 years**) for **3–5 days** (UCG: 1 mg/kg, 3 days), or **hydrocortisone 4 mg/kg IV 6-hourly** (max 100 mg) if vomiting or too breathless to swallow, or **dexamethasone 0.3–0.6 mg/kg** once (2 days).
5. **Not improving after 1 hour (severe or life-threatening)**: **$MgSO_4$ 40 mg/kg IV over 20 min (max 2 g)**; consider **aminophylline 5 mg/kg IV over at least 20 min** (omit if already on theophylline) then 5 mg/kg 6-hourly (WHO) or infusion 1 mg/kg/h; **IV salbutamol 15 micrograms/kg over 10 min** in HDU; call for help (intubation is dangerous in asthma).
6. **Discharge**: inhaler technique with spacer, controller treatment (low-dose ICS; GINA 2025 Track 1 for 6–11 years uses ICS-formoterol as maintenance and reliever at steps 3–4), written action plan, review in 1 week.

> **UGANDA CONTEXT**
> GINA 2025 discourages SABA-only treatment and promotes ICS-containing relievers (ICS-formoterol) for adolescents and adults, and ICS-formoterol maintenance-and-reliever therapy from 6 years. In Uganda, budesonide-formoterol and beclometasone inhalers are expensive and often unavailable in public facilities, so many children still go home on salbutamol only. At minimum, any child with more than occasional symptoms should leave with a **steroid inhaler and a spacer** (a cut plastic water bottle works when commercial spacers are unavailable).

## Drug card: Salbutamol

| | |
|---|---|
| **Class** | Short-acting $\beta_2$-agonist (SABA) |
| **Mechanism** | $\beta_2$ receptor → cAMP ↑ → bronchial smooth-muscle relaxation within minutes; also drives **potassium into cells** (used in hyperkalaemia) |
| **Indications (paeds)** | Acute asthma and wheeze, hyperkalaemia, (bronchiolitis: not recommended routinely) |
| **Dose, route, frequency** | **MDI 100 mcg/puff + spacer: 2–10 puffs** (by age/severity) every 20 min × 3, then 1–4-hourly. **Nebulised 2.5 mg (<5 y) or 5 mg (≥5 y)** every 20 min × 3, then 1–4-hourly; continuous in life-threatening attacks. **IV (HDU)**: 15 micrograms/kg bolus over 10 min (max 250 mcg), then 1–5 micrograms/kg/min. **Hyperkalaemia**: nebulised 2.5–5 mg. **Oral salbutamol is not recommended** (slow, weak, more side effects) |
| **Maximum** | Titrate to effect and side effects |
| **Side effects** | **Tachycardia**, tremor, restlessness, **hypokalaemia**, hyperglycaemia, lactic acidosis with high doses |

| | |
|---|---|
| **Contraindications / cautions** | Tachyarrhythmias; monitor potassium with repeated nebs |
| **Monitoring** | RR, work of breathing, $SpO_2$, heart rate, ability to talk/feed; potassium if frequent/IV |
| **Toxicity & antidote** | Reduce dose; correct potassium; beta-blockers are **not** used in asthmatics |

CLINICAL PEARL

**A spacer delivers as much drug as a nebuliser** for most acute attacks, faster and with less tachycardia, and it works without electricity. Use the nebuliser for children too breathless to use a spacer, or who need oxygen to drive it.

## Drug card: Ipratropium bromide

| | |
|---|---|
| **Class** | Short-acting antimuscarinic bronchodilator |
| **Mechanism** | Blocks M3 receptors on bronchial smooth muscle → bronchodilatation and less mucus; additive to salbutamol in the first hour of severe asthma |
| **Dose** | **Nebulised 250 micrograms (<12 years) / 500 micrograms (≥12 years)** every 20 min for 3 doses, then 4–6-hourly; MDI 20 mcg/puff, 4–8 puffs |
| **Side effects** | Dry mouth, blurred vision if mist enters the eyes (mydriasis, can precipitate angle-closure in adolescents) |
| **Cautions** | Little benefit after the first few hours or in bronchiolitis |

## Drug card: Prednisolone

| | |
|---|---|
| **Class** | Oral glucocorticoid |
| **Mechanism** | Reduces airway inflammation, oedema and mucus; restores $\beta_2$ responsiveness; onset **4–6 hours**, so give it **early** |
| **Indications (paeds)** | Acute asthma, **nephrotic syndrome**, TB meningitis/pericarditis (adjunct), severe PCP, croup (alternative), autoimmune/inflammatory disease |
| **Dose, route, frequency** | **Asthma: 1–2 mg/kg once daily × 3–5 days** (max 20/30/40 mg by age as above). **Nephrotic syndrome (KDIGO 2021): 60 mg/m²/day (or 2 mg/kg/day, max 60 mg) for 4 weeks, then 40 mg/m² (1.5 mg/kg) on alternate days for 4 weeks** (some protocols 6 + 6 weeks), then stop. **TB meningitis**: 2 mg/kg/day (max 60 mg) × 4 weeks then taper over 2 weeks. **PCP with hypoxia**: 1 mg/kg 12-hourly days 1–5, then taper over 21 days |
| **Maximum** | 60 mg/day |
| **Side effects** | Short course: hyperglycaemia, mood change, appetite ↑, gastritis. Long course: **adrenal suppression**, hypertension, growth suppression, infection risk, cushingoid features, cataracts |
| **Contraindications / cautions** | Untreated serious infection; **live vaccines** contraindicated on ≥2 mg/kg/day (or ≥20 mg/day) for ≥14 days; screen for TB and treat worms (strongyloides hyperinfection) before long courses |

| | |
|---|---|
| **Monitoring** | BP, glucose, weight, growth; taper if >2 weeks |
| **Toxicity & antidote** | Adrenal crisis on abrupt withdrawal after long courses → hydrocortisone |

## Drug card: Dexamethasone

| | |
|---|---|
| **Class** | Long-acting potent glucocorticoid (no mineralocorticoid effect) |
| **Indications (paeds)** | **Croup**, acute asthma (alternative to prednisolone), **bacterial meningitis in high-income settings** (not routinely recommended in low-income settings: WHO), cerebral oedema from tumours, airway oedema before extubation |
| **Dose, route, frequency** | **Croup: 0.15 mg/kg orally or IM single dose** (UCG; mild–moderate), up to **0.6 mg/kg** (WHO Pocket Book; severe), max 10–16 mg; or **prednisolone 1 mg/kg** or **nebulised budesonide 2 mg**. **Asthma: 0.3–0.6 mg/kg once daily × 1–2 days** |
| **Side effects** | As prednisolone; a single dose is very safe |
| **Cautions** | Never give steroids for **cerebral malaria** (they worsen outcome) |

## Croup: adrenaline nebuliser

**Severe croup** (stridor at rest, chest indrawing, agitation or lethargy): **nebulised adrenaline 1:1000, 0.4–0.5 mL/kg (max 5 mL)** undiluted or with saline (UCG: 0.4 mg/kg, max 5 mg), repeated after 30 min if needed, **plus dexamethasone**. Effect lasts about 2 hours: **observe for at least 2–4 hours** after the last dose (rebound). **Do not examine the throat** of a child with stridor and drooling (epiglottitis) and avoid distressing procedures.

## Drug card: Aminophylline

| | |
|---|---|
| **Class** | Methylxanthine (theophylline + ethylenediamine) |
| **Mechanism** | Phosphodiesterase inhibition and adenosine antagonism → bronchodilatation, respiratory centre stimulation (apnoea of prematurity), diaphragm contractility |
| **Indications (paeds)** | Severe asthma not responding to salbutamol + steroids (second line); **apnoea of prematurity** when caffeine is unavailable |
| **Dose, route, frequency** | **Asthma: loading 5–6 mg/kg IV (max 300 mg) over at least 20 min (preferably 1 h), then 5 mg/kg 6-hourly** (WHO) or infusion **0.5–1 mg/kg/h**. **Omit the loading dose if the child had theophylline/aminophylline or caffeine in the previous 24 h. Apnoea of prematurity: 6 mg/kg IV over 20 min, then 2.5 mg/kg 12-hourly** (WHO) |
| **Maximum** | Loading 300 mg |
| **Side effects** | **Narrow therapeutic index**: vomiting, tachycardia, agitation, hypokalaemia, hyperglycaemia; toxicity → **arrhythmias and seizures** |
| **Contraindications / cautions** | Fast injection can kill; interactions (macrolides and ciprofloxacin raise levels; rifampicin, phenobarbital, phenytoin lower them) |
| **Monitoring** | Pulse (**stop if >180/min**), vomiting, headache, fits; level 10–20 mg/L where available |
| **Toxicity & antidote** | Stop; treat fits with benzodiazepines; multiple-dose charcoal; correct potassium |

### Drug card: Magnesium sulphate (in asthma)

| | |
|---|---|
| **Class** | Electrolyte; calcium antagonist |
| **Mechanism** | Blocks calcium influx in bronchial smooth muscle → bronchodilatation; reduces acetylcholine release |
| **Indications (paeds)** | **Severe or life-threatening asthma not responding to the first hour** of salbutamol, ipratropium and steroids |
| **Dose, route, frequency** | **40 mg/kg IV (range 25–50 mg/kg), max 2 g**, diluted (e.g. in 10 mL/kg saline or to ≤10% concentration), **over 20 min**, single dose |
| **Side effects** | Flushing, hypotension, loss of reflexes and respiratory depression in overdose |
| **Monitoring** | BP, RR, reflexes |
| **Antidote** | **Calcium gluconate 10% 0.5 mL/kg IV** (max 20 mL) |

# Fluids, electrolytes & nutrition

Fluids are drugs. They have a dose (mL/kg), a rate, side effects (overload, hyponatraemia, cerebral oedema) and contraindications (boluses in SAM and severe anaemia). Prescribe them as carefully as an antibiotic.

### Maintenance fluids (UCG 2023)

| Weight | Daily maintenance | Per hour ("4-2-1" rule, Holliday–Segar) |
|---|---|---|
| First 10 kg | **100 mL/kg/day** | 4 mL/kg/h |
| Next 10 kg (11–20 kg) | **+50 mL/kg/day** | +2 mL/kg/h |
| Each kg above 20 | **+25 mL/kg/day** (UCG) (Holliday–Segar: +20) | +1 mL/kg/h |

- **Examples**: 8 kg → 800 mL/day; 15 kg → 1,000 + 250 = **1,250 mL/day**; 30 kg → 1,000 + 500 + 250 = **1,750 mL/day** (UCG) or 1,700 mL (Holliday–Segar).
- Add **10% for each °C of fever** (UCG); give more for ongoing losses (diarrhoea, burns).
- **Give it orally or by NGT whenever possible** (breast milk counts). IV maintenance: **isotonic fluid with 5% dextrose** (e.g. Ringer's lactate or 0.9% saline with 5% dextrose; UCG lists half-normal saline with 5% dextrose). NICE (NG29) and AAP 2018 recommend **isotonic** maintenance fluids to prevent **hospital-acquired hyponatraemia**; sick children (meningitis, pneumonia, post-op) secrete ADH and should often get **two-thirds to three-quarters of maintenance**.
- **Meningitis, cerebral malaria, severe pneumonia**: restrict to **~2/3 to 3/4 of maintenance** unless dehydrated or shocked (NICE approach). In meningitis, WHO and Cochrane (2016) favour **full isotonic maintenance** unless there is hyponatraemia (SIADH, Na <130) or raised ICP: see *Meningitis*.

### Neonatal fluids (UCG 2023)

| Day | Total fluids (oral + IV) | Fluid if IV |
|---|---|---|
| Day 1 | **60 mL/kg/day** | **10% dextrose** |
| Day 2 | 90 mL/kg/day | 10% dextrose |
| Day 3 | 120 mL/kg/day | Half-normal saline + 5% dextrose (or 10% dextrose with electrolytes) |
| Day 4 onwards | **150 mL/kg/day** | When fully fed, up to 180 mL/kg/day |

If only IV fluids are given, **do not exceed 100 mL/kg/day** unless dehydrated, under a radiant heater or on phototherapy (more insensible loss). Reduce if facial oedema develops.

### Shock: fluid boluses (child without SAM or severe anaemia)

1. **Ringer's lactate or 0.9% saline 20 mL/kg as rapidly as possible** (UCG 2023), through an IV or **intraosseous** line.
2. Reassess (pulse, capillary refill, consciousness, urine, **liver size and crackles**). If no improvement, **repeat 10–20 mL/kg** (UCG: up to 3 times, max 60 mL/kg); give **blood 20 mL/kg** if bleeding or severe anaemia.
3. Still shocked → septic shock: further fluids with great caution, **adrenaline or dopamine** infusion where available; ICU.
4. Improving: continue rehydration (for diarrhoea, **70 mL/kg over 5 h in infants, 2.5 h in children**).

UGANDA CONTEXT

The **FEAST trial** (2011, Uganda, Kenya and Tanzania; children with severe febrile illness and impaired perfusion) found that **fluid boluses increased mortality** compared with maintenance fluid alone. WHO's **2016 ETAT update** therefore recommends **10–20 mL/kg of isotonic crystalloid over 30–60 minutes** for children with shock (all of: cold hands, capillary refill >3 s, weak fast pulse), with careful reassessment, and **no boluses for children with impaired perfusion who are not in shock**. UCG 2023 still says "20 mL/kg as rapidly as possible". On the ward: give the bolus to a child who is truly shocked, but **reassess after every 10 mL/kg**, stop at the first sign of overload, and remember that **severe anaemia needs blood**, not saline.

### Shock in a child with severe acute malnutrition

- **15 mL/kg over 1 hour** of **Ringer's lactate with 5% dextrose**, **half-strength Darrow's with 5% dextrose**, or **0.45% saline with 5% dextrose** (UCG 2023; WHO).
- Monitor **pulse and RR every 5–10 minutes**. If improving, **repeat 15 mL/kg over 1 h**, then switch to **oral/NGT ReSoMal 10 mL/kg/h for up to 10 h** and start F-75.
- If no improvement after the first hour: **septic shock**: maintenance IV fluid (4 mL/kg/h), **transfuse 10 mL/kg fresh whole blood slowly over 3 h**, IV antibiotics, start feeding.
- **Stop at once** if the pulse rises by **≥25/min** or RR by **≥5/min** (heart failure).

### Dehydration: Plans A, B and C (UCG 2023)

| Plan | Who | What |
|---|---|---|
| **A** (no dehydration) | Diarrhoea, no signs | Home: **extra fluids** (ORS **50–100 mL after each loose stool under 2 years, 100–200 mL at 2–5 years**), **continue feeding**, **zinc**, **when to return**. Give 2 packets of ORS |
| **B** (some dehydration: 2 of restless/irritable, sunken eyes, drinks eagerly, skin pinch slow) | | **ORS 75 mL/kg over 4 h** in the clinic (<6 kg: 200–400 mL; 6–9.9 kg: 400–700; 10–11.9 kg: 700–900; 12–19 kg: 900–1,400), then reassess |
| **C** (severe: 2 of lethargic/unconscious, sunken eyes, drinks poorly/unable, skin pinch very slow >2 s) | | **Ringer's lactate 100 mL/kg IV**: **<12 months: 30 mL/kg in 1 h, then 70 mL/kg in 5 h; ≥12 months: 30 mL/kg in 30 min, then 70 mL/kg in 2.5 h.** Repeat the first part if radial pulse still weak. Add ORS **5 mL/kg/h** as soon as the child can drink. No IV possible: **ORS 20 mL/kg/h by NGT for 6 h** (120 mL/kg) |

RED FLAG

**Plan C is not for children with SAM.** A malnourished child with diarrhoea gets **ReSoMal slowly** (or the SAM shock protocol), because the standard Plan C volume can tip them into heart failure.

## Drug card: Oral rehydration salts (low-osmolarity ORS)

| | |
|---|---|
| **Composition (per litre)** | **Sodium 75, chloride 65, potassium 20, citrate 10 mmol, glucose 75 mmol; osmolarity 245 mOsm/L** |
| **Mechanism** | **Sodium–glucose co-transport (SGLT1)** in the small bowel still works in secretory diarrhoea: glucose carries sodium, water follows |
| **Indications** | Prevention and treatment of dehydration from diarrhoea (Plans A, B, and C once the child can drink); WHO 2024 reaffirms low-osmolarity ORS for children with acute watery diarrhoea and dehydration |
| **Dose** | As in Plans A–C above; **frequent small sips** from a cup or spoon; if vomiting, wait 10 min and continue slowly |
| **Preparation** | **One sachet in 1 litre of clean (boiled and cooled) water**; discard after 24 h. Too concentrated → hypernatraemia |
| **Side effects** | Vomiting if given too fast; periorbital puffiness = overhydrated (stop, give breast milk/water) |
| **Contraindications** | Ileus, unable to drink (use NGT or IV), SAM (use ReSoMal) |

## Drug card: Zinc

| | |
|---|---|
| **Class** | Micronutrient (zinc sulphate dispersible 20 mg tablets) |
| **Mechanism** | Restores intestinal mucosal repair and immune function; reduces duration and severity of diarrhoea and recurrence for 2–3 months |
| **Indications** | **Every child with diarrhoea** (acute or persistent) |
| **Dose** | **UCG 2023: <6 months 10 mg once daily; ≥6 months 20 mg once daily, for 10 days** (WHO 2013: 10–14 days). **WHO 2024 guideline: suggests a lower dose of 5 mg daily for up to 14 days** for children up to 10 years (equally effective, less vomiting) |
| **Side effects** | Vomiting (especially at 20 mg), metallic taste |
| **Notes** | Dissolve in breast milk, ORS or water |

UGANDA CONTEXT

Uganda's 2023 guideline and drug supply still use **10 mg / 20 mg zinc for 10 days**. The **WHO 2024 pneumonia and diarrhoea guideline** now suggests **5 mg daily** (conditional recommendation, low-certainty evidence), because trials showed similar benefit with about 30% less vomiting. Follow the UCG dose on the ward and know the WHO update for the exam.

## ReSoMal and the SAM feeds

| Product | What it is | Dose |
|---|---|---|
| **ReSoMal** (rehydration solution for malnutrition) | Lower sodium (**45 mmol/L**), higher potassium (**40 mmol/L**), with magnesium and zinc | **5 mL/kg every 30 min for 2 h** (oral/NGT), then **5–10 mL/kg/h in alternate hours with F-75 for up to 10 h**. After SAM shock: 10 mL/kg/h up to 10 h. Stop if overload signs |
| **F-75** (stabilisation phase) | 75 kcal and 0.9 g protein per 100 mL; low protein, low sodium, low lactose | **130 mL/kg/day** (≈100 kcal/kg/day) in **8 feeds 3-hourly** (12 two-hourly feeds at first in very sick children); **100 mL/kg/day if severe (+++) oedema**. Feed day and night (night feeds prevent hypoglycaemia and hypothermia) |

| Product | What it is | Dose |
|---|---|---|
| **F-100** (transition/rehabilitation) | 100 kcal and 2.9 g protein per 100 mL | Transition: replace F-75 with the **same volume of F-100 for 2 days**, then increase by **10 mL per feed** until some is left; rehabilitation **150–220 kcal/kg/day** |
| **RUTF** (ready-to-use therapeutic food, peanut paste) | 92 g sachet ≈ **500 kcal**; contains all minerals, vitamins and iron | **WHO 2023: 150–185 kcal/kg/day** until recovery (or until no longer severely wasted, then 100–130 kcal/kg/day). A 7 kg child needs about **2–2.5 sachets/day**. Give **plenty of clean water**; breastfeed first |
| **F-75 for infants <6 months** | Diluted F-100 or F-75 with supported breastfeeding (supplementary suckling) | Specialist protocol |

### Other SAM medicines (WHO / UCG)

- **Antibiotics for all inpatients** (see Antimicrobials). **Hypoglycaemia <3 mmol/L**: 50 mL of 10% glucose/sucrose orally or by NGT, or **10% dextrose 5 mL/kg IV**, then feed 2-hourly.
- **No iron in the stabilisation phase** (free iron feeds bacteria and causes oxidative damage). Start iron in rehabilitation if on F-100; **RUTF already contains iron**.
- **Vitamin A**: only if **eye signs** of deficiency or **recent measles** (F-75, F-100 and RUTF contain vitamin A): days 1, 2 and 14.
- **Folic acid** and **zinc/copper** come in the mineral mix of F-75/F-100.
- **Measles vaccine** if unvaccinated; **albendazole** in rehabilitation (≥1 year); treat malaria and test for HIV and TB.

## Drug card: Dextrose (glucose)

| | |
|---|---|
| **Preparations** | 5%, 10%, 50% (adult ampoule). **Children: 10%. Neonates: 10%.** 50% is too hypertonic for small veins |
| **Indications** | **Hypoglycaemia** (child <2.5 mmol/L by WHO, <3.0 mmol/L by UCG and in SAM; neonate <2.6 mmol/L), maintenance fluid in neonates, with quinine, with insulin in hyperkalaemia |
| **Dose** | **Child: 10% dextrose 5 mL/kg IV** over 2–5 min (UCG general: 2–5 mL/kg). **Neonate: 10% dextrose 2 mL/kg IV**, then infusion 60 mL/kg/day (day 1). No IV: **sugar water** (4 level teaspoons of sugar in 200 mL water) or 10% dextrose by NGT, or **sugar under the tongue** |
| **Making 10%** | From 50%: **1 mL/kg of 50% + 4 mL/kg of water** = 5 mL/kg of 10%. For a bottle: remove **50 mL** from 500 mL of **5% dextrose** and add **50 mL of 50% dextrose** → 10% |
| **Side effects** | Phlebitis/extravasation injury, hyperglycaemia, rebound hypoglycaemia, hyponatraemia if used alone as maintenance |
| **Monitoring** | **Recheck glucose after 30 min**, then regularly; find the cause (malaria, sepsis, quinine, starvation, SAM) and **feed** |

> **RED FLAG**
> **Never give undiluted 50% dextrose to a child** (UCG 2023 says this in capital letters). It damages veins, causes extravasation necrosis, and sudden hyperosmolality.

## Drug card: Potassium chloride

| | |
|---|---|
| **Preparations** | KCl ampoules (15% = 2 mmol/mL; concentrate), oral KCl solution/tablets; potassium is also in Ringer's lactate (5 mmol/L), half-strength Darrow's (17 mmol/L), ReSoMal and F-75 |

| | |
|---|---|
| **Indications** | Maintenance (after urine is passed), hypokalaemia (diarrhoea, SAM, DKA, salbutamol, diuretics) |
| **Dose** | Maintenance **1–2 mmol/kg/day** (≈ **20 mmol per litre** of maintenance fluid). Hypokalaemia: **oral 1–2 mmol/kg/day** in divided doses (preferred), IV only in a bag. **SAM: 3–4 mmol/kg/day** (via F-75 mineral mix) |
| **Maximum** | Peripheral IV concentration **≤40 mmol/L**; rate **≤0.2 mmol/kg/h** on the ward (up to 0.5 with ECG monitoring in ICU) |
| **Side effects** | Hyperkalaemia → arrhythmia/arrest; phlebitis |
| **Contraindications** | **No urine output** (anuria, AKI) or $K^+$ already high; **never as an IV bolus or push** |
| **Monitoring** | Urine output; potassium daily; ECG if available |
| **Toxicity & antidote** | Hyperkalaemia: stop KCl, **calcium gluconate 10% 0.5 mL/kg**, **salbutamol neb**, **insulin 0.1 unit/kg with 10% dextrose 5 mL/kg**, sodium bicarbonate if acidotic |

### Blood transfusion (the most life-saving "fluid" in Uganda)

- **Indications (WHO/UCG)**: **Hb <4 g/dL**, or **Hb 4–6 g/dL with** respiratory distress, impaired consciousness, shock, heart failure, very high parasitaemia or ongoing loss.
- **Dose**: **whole blood 20 mL/kg** or **packed cells 10 mL/kg over 3–4 h**. **SAM: 10 mL/kg over 3 h**. The **TRACT trial (2019; Uganda and Malawi)** found that **30 mL/kg whole blood** improved survival in afebrile children compared with 20 mL/kg (but not in febrile children), and that children with uncomplicated Hb 4–6 g/dL could be safely observed without immediate transfusion.
- **Furosemide 1 mg/kg IV** only if signs of fluid overload (not routinely).
- Check vital signs at the start, after 15 min and every 30 min; stop if fever, rash, breathlessness or dark urine (transfusion reaction).

## Analgesia & antipyretics

Children feel pain as much as adults, including newborns, and are **under-treated** everywhere. Use the WHO **two-step ladder** for children (WHO 2012 persisting pain guideline; UCG 2023): **step 1 paracetamol ± ibuprofen**, **step 2 morphine**. **Codeine and tramadol are not used in children** (unpredictable metabolism: CYP2D6 ultra-rapid metabolisers have died of respiratory depression).

### Drug card: Paracetamol (acetaminophen)

| | |
|---|---|
| **Class** | Analgesic and antipyretic (central COX inhibition; weak anti-inflammatory) |
| **Indications** | Fever with discomfort, mild–moderate pain, step 1 for all pain; **not** to prevent febrile seizures |
| **Dose, route, frequency** | **15 mg/kg orally 6-hourly** (UCG; 10–15 mg/kg for pain); rectal similar per dose; IV 15 mg/kg (≥10 kg) or 7.5–10 mg/kg (<10 kg). **Neonates: 10 mg/kg 6–8-hourly** (term), longer intervals in preterms |
| **Maximum** | **60 mg/kg/day** (WHO; some UK guidance 75 mg/kg/day short-term) and **4 g/day**; neonates about **40 mg/kg/day** |
| **Formulations** | Syrup **120 mg/5 mL** and **250 mg/5 mL**; 100 mg and 500 mg tablets; suppositories. Write the strength |
| **Side effects** | Very safe at correct doses; hepatotoxicity in overdose (single or repeated) |
| **Contraindications / cautions** | Liver failure; malnutrition, prolonged fasting and dehydration lower glutathione (higher risk) |

| | |
|---|---|
| **Monitoring** | Total daily dose including home doses |
| **Toxicity & antidote** | **Acute ingestion >150 mg/kg** (UCG: toxic dose >150 mg/kg, or **>200 mg/kg in children <6 years**) or repeated supratherapeutic doses → **acetylcysteine** (see Poisoning) |

## Drug card: Ibuprofen

| | |
|---|---|
| **Class** | NSAID (non-selective COX inhibitor) |
| **Indications** | Pain (especially musculoskeletal, post-operative, sickle cell crises with paracetamol), fever, inflammatory conditions; PDA closure in preterms (specialist) |
| **Dose, route, frequency** | **5–10 mg/kg orally 6–8-hourly** with food; **only in children >3 months** (UCG) |
| **Maximum** | **40 mg/kg/day**, 400 mg per dose (adult 2.4 g/day in hospital) |
| **Side effects** | Gastritis/GI bleeding, **AKI in dehydration**, bronchospasm in a minority of asthmatics, platelet dysfunction |
| **Contraindications / cautions** | **Dehydration, shock, AKI, renal disease**; **dengue and viral haemorrhagic fevers** (bleeding); **varicella** (associated with severe skin/soft tissue infection); active GI bleeding; NSAID-sensitive asthma; **<3 months** |
| **Monitoring** | Urine output in sick children |
| **Toxicity & antidote** | Overdose >400 mg/kg: seizures, acidosis, AKI; supportive |

## Drug card: Morphine

| | |
|---|---|
| **Class** | Strong opioid (μ-receptor agonist) |
| **Indications** | Severe pain: sickle cell crisis, burns, fractures, post-operative pain, cancer and palliative care, breathlessness in end-of-life care |
| **Dose, route, frequency** | **Oral (liquid morphine, Uganda's "Hospice" morphine 5 mg/5 mL)**: **1–2 years 0.2–0.4 mg/kg 4-hourly; 2–12 years 0.2–0.5 mg/kg 4-hourly (max 10 mg starting dose)** (UCG 2023); infants <1 year start lower (**about 0.08–0.2 mg/kg 4-hourly**, WHO 2012) with apnoea monitoring. **IV 0.05–0.1 mg/kg** slowly, every 2–4 h (WHO Pocket Book), titrated in small increments. Neonates: 0.025–0.05 mg/kg IV with monitoring |
| **Maximum** | No ceiling dose in palliative care when titrated; increase by **30–50% per 24 h** if pain uncontrolled |
| **Side effects** | **Respiratory depression** (infants, especially <6 months, most sensitive), sedation, **constipation** (always prescribe a laxative except in diarrhoea), nausea, itching, urinary retention |
| **Contraindications / cautions** | Head injury without airway support, severe respiratory depression; renal failure (metabolite accumulates); acute asthma |
| **Monitoring** | **RR, $SpO_2$ and sedation score**; pain score (FLACC <3 years, faces scale 3–7 years, numeric 0–10 older) |
| **Toxicity & antidote** | **Naloxone** (see Emergency drugs): for therapeutic overdose use small doses (1–4 micrograms/kg) to reverse breathing without abolishing analgesia |

UGANDA CONTEXT

Uganda was the first African country to make **oral liquid morphine** freely available through **Hospice Africa Uganda** and authorised nurse prescribers. Liquid morphine is on the essential medicines list and is cheap: use it for children with severe pain from cancer, burns or sickle cell disease. Note that the UCG 2023 children's pain table prints the 1–6 month dose as "0.01 mg/kg", which looks like a misprint; check with your unit's palliative care team for very young infants.

### Aspirin: avoid in children

- **Avoid aspirin under 16 years** because of **Reye's syndrome** (acute encephalopathy + fatty liver after viral illness, especially influenza and varicella).
- Exceptions under specialist care: **Kawasaki disease** (high dose in the acute phase, then **3–5 mg/kg once daily** as an antiplatelet for 6–8 weeks), **rheumatic fever arthritis/carditis** (anti-inflammatory doses, or naproxen), and antiplatelet use after cardiac surgery.
- Many Ugandan homes keep adult aspirin: ask, and counsel against it.

### Other analgesia/sedation drugs

| Drug | Dose | Notes |
|---|---|---|
| **Ketamine** | Procedural sedation **1–2 mg/kg IV** or **4–5 mg/kg IM**; analgesia 0.1–0.3 mg/kg IV | Keeps airway reflexes and BP; emergence reactions; hypersalivation |
| **Lidocaine (lignocaine)** | Local infiltration **max 3 mg/kg** (7 mg/kg with adrenaline) | 1% = 10 mg/mL: a 10 kg child may receive at most **3 mL of 1%** plain |
| **Sucrose 24%** / breastfeeding | 0.1–0.5 mL orally 2 min before heel prick or cannula | Non-drug analgesia for neonates; skin-to-skin also works |
| **Amitriptyline** | 0.2–0.5 mg/kg at night (max 25 mg) | Neuropathic pain (HIV neuropathy, cancer) |

## Emergency & resuscitation drugs

Keep a **weight-based emergency drug chart** at every bedside on the acute ward, and the **paediatric strengths** ready: **1:10,000 adrenaline, 10% dextrose, 10% calcium gluconate, 4.2% bicarbonate for infants**.

### Emergency drug quick table

| Drug | Indication | Dose | Route | Notes |
|---|---|---|---|---|
| **Adrenaline 1:1000 (1 mg/mL)** | **Anaphylaxis** | **0.01 mL/kg (10 mcg/kg), max 0.5 mL: <6 years 0.15 mL; 6–12 years 0.3 mL; >12 years 0.5 mL** | **IM, anterolateral thigh** | Repeat every **5 min** if no improvement |
| **Adrenaline 1:10,000 (0.1 mg/mL)** | **Cardiac arrest** (asystole, PEA; VF after 3rd shock) | **0.1 mL/kg (10 mcg/kg)**, max 10 mL (1 mg) | IV/IO | Every **3–5 min** (every other cycle) |
| **Adrenaline 1:1000** | **Croup** (severe) | **0.4–0.5 mL/kg, max 5 mL** | Nebulised | Observe ≥2–4 h after |

| Drug | Indication | Dose | Route | Notes |
|---|---|---|---|---|
| **Adrenaline infusion** | Fluid-refractory septic/cardiogenic shock | **0.05–0.3 mcg/kg/min** (titrate) | IV (central preferred) | HDU/ICU |
| **Atropine** | Vagal bradycardia (intubation), organophosphate poisoning | **Bradycardia: 20 mcg/kg (0.02 mg/kg)**, max 0.5 mg. **Organophosphate: 0.02–0.05 mg/kg** (UCG 0.05 mg/kg), **double every 5 min** until the chest is clear | IV/IO/IM | Hypoxic bradycardia needs **oxygen and ventilation**, not atropine |
| **Naloxone** | Opioid overdose | **0.01 mg/kg** (UCG), increase to **0.1 mg/kg (max 2 mg)** if no response | IV/IM/IO/intranasal | Short half-life (≈1 h): watch for re-sedation; infusion if long-acting opioid |
| **Calcium gluconate 10%** | Hypocalcaemia (tetany, seizures), **hyperkalaemia**, Mg/CCB toxicity | **0.5 mL/kg (50 mg/kg), max 20 mL** | **Slow IV over 5–10 min** | Watch heart rate (bradycardia); **extravasation causes necrosis**; never with ceftriaxone in neonates; **not with bicarbonate** in the same line |
| **Sodium bicarbonate 8.4% (1 mmol/mL)** | Hyperkalaemia with acidosis, **tricyclic overdose**, prolonged arrest with documented acidosis | **1 mmol/kg = 1 mL/kg of 8.4%** (infants: **2 mL/kg of 4.2%**) | Slow IV | Not routine in arrest; flush before/after calcium or adrenaline (inactivates them) |
| **10% dextrose** | Hypoglycaemia | **5 mL/kg** (neonate 2 mL/kg) | IV/IO | Recheck in 30 min |
| **Amiodarone** | VF/pulseless VT after 3rd and 5th shocks | **5 mg/kg** (max 300 mg) | IV/IO bolus in arrest | Defibrillate at **4 J/kg** |
| **Hydrocortisone** | Anaphylaxis (adjunct), severe asthma, **adrenal crisis** | **4 mg/kg** (asthma, max 100 mg); anaphylaxis UCG: **<1 year 25 mg, 1–5 years 50 mg, 6–12 years 100 mg, >12 years 200 mg**; adrenal crisis 2–4 mg/kg (≈100 mg/m²) | IV/IM | Not a substitute for adrenaline |
| **Chlorphenamine** | Urticaria/itch after anaphylaxis treated | **1–2 years 1 mg 12-hourly; 2–5 years 1 mg 6-hourly; 5–12 years 2 mg 6-hourly** oral (UCG); IV 0.1–0.2 mg/kg | Oral/IV | Adjunct only; sedating |
| **Furosemide** | Fluid overload, heart failure during transfusion | **1 mg/kg** (max 40 mg) | IV | Check potassium |
| **Mannitol 20%** | Raised intracranial pressure (not in cerebral malaria routinely) | 0.25–0.5 g/kg (1.25–2.5 mL/kg) over 20–30 min | IV | Specialist decision |

DRUG DOSING

**Two adrenaline strengths, two routes, two doses.** Anaphylaxis = **1:1000, IM, 0.01 mL/kg**. Cardiac arrest = **1:10,000, IV/IO, 0.1 mL/kg**. Both deliver **10 micrograms/kg**. Giving 1:1000 IV at 0.1 mL/kg is a **tenfold overdose** that can cause fatal arrhythmia. To make 1:10,000: **1 mL of 1:1000 + 9 mL of saline**.

### Paediatric resuscitation in brief (ETAT; ERC/AHA paediatric life support, 2025 updates)

1. Unresponsive and not breathing normally: **5 rescue breaths**, check for signs of life, then **CPR 15 compressions : 2 breaths** (two rescuers), rate **100–120/min**, depth one-third of chest (≈4 cm infant, 5 cm child).
2. **Attach monitor/defibrillator**. Non-shockable (asystole, PEA: most paediatric arrests, usually after hypoxia or shock): **adrenaline 10 mcg/kg as soon as IV/IO access is obtained**, then every 3–5 min. Shockable (VF/pVT, rare): **4 J/kg**, adrenaline and amiodarone after the 3rd shock.
3. **Reversible causes (4 Hs and 4 Ts)**: **hypoxia**, **hypovolaemia**, **hypo/hyperkalaemia, hypoglycaemia, hypocalcaemia**, **hypothermia**; tension pneumothorax, tamponade, toxins, thrombosis.
4. **Intraosseous access** (proximal tibia, 1–2 cm below the tibial tuberosity, medial flat surface) if no IV in 90 seconds: every resuscitation drug and fluid can go through it.

## Neonatal drugs

Newborns get a small set of routine drugs at birth, and a small set of treatment drugs on the neonatal unit. Doses are by **weight and postnatal age**, and many adult drugs are dangerous.

### Routine care at birth (UCG 2023 / Uganda newborn guidelines)

| Drug | Dose | Notes |
|---|---|---|
| **Vitamin $K_1$ (phytomenadione)** | **1 mg IM** (term); **0.5 mg IM** if preterm/LBW | Anterolateral thigh within 1 h of birth; prevents **vitamin K deficiency bleeding** (haemorrhagic disease of the newborn), which is commoner in exclusively breastfed babies |
| **Tetracycline eye ointment 1%** | Both eyes once | Within 1 h; prevents gonococcal and chlamydial conjunctivitis |
| **Chlorhexidine 7.1% digluconate (4% chlorhexidine) gel** | Apply to the cord stump | Uganda policy for cord care (reduces omphalitis); keep the cord dry and clean otherwise |
| **BCG, OPV0, HepB birth dose** | See Vaccines | HepB within 24 h (first 7 days per UNEPI) |
| **NVP (± AZT) prophylaxis** | HIV-exposed infants: see TB & HIV drugs | Start within 6–12 h |

### Neonatal antibiotic doses by postnatal age (WHO Pocket Book; UCG differences flagged above)

| Drug | Week 1 of life | Weeks 2–4 of life | Notes |
|---|---|---|---|
| **Ampicillin** | **50 mg/kg 12-hourly** | **50 mg/kg 8-hourly** | Meningitis: 100 mg/kg per dose. UCG: 6-hourly |
| **Gentamicin** | **5 mg/kg once daily (≥2.5 kg); 3 mg/kg once daily (LBW)** | **7.5 mg/kg once daily** | IV slowly or IM |
| **Benzylpenicillin** | 50,000 IU/kg 12-hourly | 50,000 IU/kg 8-hourly | Double for meningitis |
| **Cloxacillin** | 25–50 mg/kg 12-hourly | 25–50 mg/kg 8-hourly | Skin/umbilical staph |

| Drug | Week 1 of life | Weeks 2–4 of life | Notes |
|---|---|---|---|
| **Cefotaxime** | 50 mg/kg 12-hourly | 50 mg/kg 8-hourly | Preferred in jaundice |
| **Ceftriaxone** | 50 mg/kg once daily (100 mg/kg meningitis) | Same | **Not if jaundiced or with calcium-containing IV fluids** |
| **Metronidazole** | 7.5 mg/kg 12-hourly | 7.5 mg/kg 8–12-hourly | NEC, tetanus |

- **Duration**: clinical sepsis **7–10 days**; culture-negative and well at 48–72 h may stop (where cultures are reliable); **meningitis 14 days (GBS) to 21 days (Gram-negative)**.
- **Simplified outpatient regimens (WHO 2015, when referral is not possible)**: PSBI (any sign except critical illness): **gentamicin IM once daily × 7 days + oral amoxicillin 50 mg/kg 12-hourly × 7 days**; fast breathing alone in a 7–59-day-old: **oral amoxicillin 50 mg/kg 12-hourly × 7 days**.

## Drug card: Caffeine citrate

| | |
|---|---|
| **Class** | Methylxanthine (respiratory stimulant) |
| **Mechanism** | Adenosine receptor antagonist → stimulates the respiratory centre, increases $CO_2$ sensitivity and diaphragm contractility |
| **Indications** | **Apnoea of prematurity** (babies **<34 weeks**); preferred over aminophylline (wider therapeutic index, once daily) |
| **Dose, route, frequency** | **Loading 20 mg/kg caffeine citrate** orally or IV over 30 min; **maintenance 5 mg/kg once daily** starting 24 h later, increased by 5 mg/kg/day to max **20 mg/kg/day** if apnoeas persist (WHO Pocket Book). (Caffeine citrate 20 mg = caffeine base 10 mg: check which salt your unit stocks) |
| **Duration** | Until **4–5 days after the last apnoea** (often to about 34–36 weeks' postmenstrual age) |
| **Side effects** | Tachycardia (>180/min: withhold), feeding intolerance, jitteriness |
| **Monitoring** | Heart rate, apnoea chart, $SpO_2$ (pulse oximeter with alarm) |
| **Availability** | National/regional referral NICUs (Mulago Special Care Unit, Mengo NICU); aminophylline elsewhere |

## Neonatal seizures, hypoglycaemia and other neonatal drugs

| Drug | Dose | Notes |
|---|---|---|
| **Phenobarbital** | **20 mg/kg IV/IM**, further **10 mg/kg** doses up to **40 mg/kg**; maintenance **5 mg/kg daily** | First line for neonatal seizures; check glucose, calcium; consider meningitis. **Avoid diazepam** in neonates |
| **Phenytoin** | 15–20 mg/kg IV in saline over 30 min | If phenobarbital fails; cardiac monitoring |
| **10% dextrose** | **2 mL/kg** bolus, then **60 mL/kg/day** (day 1) | Neonatal hypoglycaemia <2.6 mmol/L |
| **Calcium gluconate 10%** | 0.5–1 mL/kg (diluted) slowly IV over 10 min with heart-rate monitoring (symptomatic hypocalcaemia) | Extravasation → skin necrosis; check the line |

| Drug | Dose | Notes |
|---|---|---|
| **Adrenaline 1:10,000** | 0.1–0.3 mL/kg IV (0.01–0.03 mg/kg); ET 0.5–1 mL/kg | Neonatal resuscitation (see Neonatal resuscitation topic) |
| **Normal saline** | 10 mL/kg over 5–10 min | Hypovolaemia at birth (only if suspected) |
| **Iron (preterm/ LBW)** | **2–4 mg/kg/day elemental iron from 2 weeks (WHO; many units start at 4–6 weeks) to 6 months** | Preterms have low stores |
| **Multivitamins / vitamin D** | Per unit protocol (vitamin D 400 IU daily in preterms) | Preterm metabolic bone disease |
| **Nystatin** | 1 mL (100,000 IU) 6-hourly | Oral thrush |
| **Neonatal tetanus** | **Diazepam (specialist doses by infusion)**, **human tetanus immunoglobulin 500 IU IM** (or equine antitoxin), **metronidazole 7.5 mg/kg 8-hourly** (or benzylpenicillin), NGT feeding, quiet dark room | Prevent with maternal Td vaccination and clean cord care |
| **Surfactant** | 100–200 mg/kg via ETT/LISA | Rarely available in Uganda (national referral, private) |

RED FLAG

**Drugs to avoid in neonates**: **chloramphenicol** in the first week (grey baby), **ceftriaxone** if jaundiced or with calcium, **sulphonamides/cotrimoxazole** in the first weeks (kernicterus), **diazepam** for seizures (use phenobarbital), **codeine**, **aspirin**, **benzyl alcohol-preserved injections** (gasping syndrome), **alcohol and iodine** over large skin areas, **topical steroids** over big areas.

# TB & HIV drugs

Both TB and HIV drugs are given by **weight band** using **child-friendly dispersible fixed-dose combinations (FDCs)**. Reweigh at every visit and move the child up a band as they grow: **under-dosing as children grow** is a classic cause of treatment failure.

## TB: doses and regimens (WHO 2022 consolidated TB guidelines for children and adolescents; UCG 2023)

| Drug | mg/kg (range), once daily | Key side effects |
|---|---|---|
| **Isoniazid (H)** | **10 (7–15)**, max 300 mg | Hepatitis, **peripheral neuropathy** (give **pyridoxine** to HIV-positive, malnourished and breastfed children and adolescents), rash |
| **Rifampicin (R)** | **15 (10–20)**, max 600 mg | **Orange urine, tears and sweat** (warn the family), hepatitis, flu-like illness, thrombocytopenia; **strong enzyme inducer** (DTG, LPV/r, NVP, contraceptives, steroids, phenytoin) |
| **Pyrazinamide (Z)** | **35 (30–40)** | Hepatitis, arthralgia, raised urate |
| **Ethambutol (E)** | **20 (15–25)** | **Optic neuritis** (rare at 20 mg/kg; ask older children about colour vision) |

**Child-friendly FDCs (WHO 2022 weight bands)**: **RHZ 75/50/150 mg** (intensive) and **RH 75/50 mg** (continuation), dispersible; **ethambutol 100 mg** dispersible.

| Weight | RHZ 75/50/150 (intensive) | RH 75/50 (continuation) | Ethambutol 100 mg |
|---|---|---|---|
| **4–7.9 kg** | 1 tablet | 1 tablet | 1 tablet |
| **8–11.9 kg** | 2 tablets | 2 tablets | 2 tablets |
| **12–15.9 kg** | 3 tablets | 3 tablets | 3 tablets |
| **16–24.9 kg** | 4 tablets | 4 tablets | 4 tablets |
| **≥25 kg** | Adult FDCs (RHZE 150/75/400/275) by adult weight bands | | |

| Type of TB (child) | Regimen |
|---|---|
| **Non-severe TB** (peripheral lymph node TB, intrathoracic node TB without airway obstruction, uncomplicated pleural effusion, or paucibacillary non-cavitary disease confined to one lobe), age **3 months–16 years** (UCG: 2 months–16 years) | **4-month regimen: 2HRZ(E)/2HR** (SHINE trial). Ethambutol added where HIV prevalence or isoniazid resistance is high |
| **Severe pulmonary or extrapulmonary TB** (other than meningitis and osteoarticular) | **2HRZE/4HR** (6 months) |
| **TB meningitis, osteoarticular TB** | **2HRZE/10HR** (12 months; UCG). WHO 2022 alternative for TBM: **6-month intensive 6HRZEto** (higher H and R doses + ethionamide) |
| **Adjuncts** | **Prednisolone 2 mg/kg/day (max 60 mg) × 4 weeks then taper over 2 weeks** for TB meningitis and pericarditis; **pyridoxine** (UCG lists 25 mg daily; many paediatric protocols use 5–10 mg in young children and 12.5–25 mg in older ones) |

**TB preventive treatment (TPT)** for **child contacts** (after excluding active TB) and **children living with HIV**: **3HP** (weekly isoniazid + rifapentine × 12 weeks, **≥2 years**), **3RH** (daily rifampicin + isoniazid × 3 months, using RH 75/50 by the bands above), or **6H** (isoniazid **10 mg/kg** daily × 6 months, with pyridoxine). Do not give TPT to contacts of MDR-TB patients without specialist advice.

DRUG DOSING

**TB and ART interactions** (rifampicin is the culprit):

- **Dolutegravir**: give **DTG twice daily** (double the dose) during rifampicin treatment and for 2 weeks after.
- **LPV/r**: "super-boost" with extra ritonavir (or double the dose) per national guidance.
- **Nevirapine**: avoid with rifampicin.
- **EFV**: can be used with rifampicin without dose change (children >3 years).

## HIV: ART for children (Uganda consolidated HIV guidelines 2022; UCG 2023)

| Group | Preferred first line | Alternatives |
|---|---|---|
| **Neonates (<4 weeks)** | **AZT + 3TC + NVP** (or RAL granules) until eligible for DTG | |
| **Children ≥4 weeks and ≥3 kg, <20 kg** | **ABC + 3TC + DTG** (**pDTG 10 mg dispersible** or **pALD** = ABC/3TC/DTG 60/30/5 mg dispersible) | ABC + 3TC + LPV/r granules if DTG unavailable/not tolerated; ABC + 3TC + EFV (>3 years and >10 kg); AZT + 3TC + DTG if ABC contraindicated |
| **Children 20–<30 kg** | **ABC + 3TC + DTG** (DTG 50 mg film-coated tablet from 20 kg) | TAF/FTC/DTG (>6 years and >25 kg); AZT + 3TC + DTG |
| **Adolescents ≥30 kg** | **TDF + 3TC + DTG (TLD)** once daily | TAF/FTC/DTG; TDF/3TC/EFV400 |

### Selected weight-band doses (UCG 2023 dosing tables; check the current national chart)

| Weight | ABC/3TC 120/60 mg dispersible (once daily) | pALD 60/30/5 mg (once daily) | DTG |
|---|---|---|---|
| 3–5.9 kg | 1 tablet | 2 tablets (UCG table) | **pDTG 10 mg**: UCG table lists 1 tablet; WHO/manufacturer dose is **5 mg (½ tablet)** at 3–<6 kg (**verify**) |
| 6–9.9 kg | 1.5 tablets | 3 tablets | pDTG **15 mg (1.5 tablets)** |
| 10–13.9 kg | 2 tablets | 4 tablets | pDTG **20 mg (2 tablets)** |
| 14–19.9 kg | 2.5 tablets | 5 tablets | pDTG **25 mg (2.5 tablets)** |
| 20–24.9 kg | 3 tablets | 6 tablets | **DTG 50 mg film-coated, 1 tablet** |
| ≥25 kg | ABC/3TC 600/300, 1 tablet | | DTG 50 mg |

> **NOTE**
> Dispersible DTG 10 mg and film-coated DTG 50 mg are **not bioequivalent** (the dispersible is better absorbed): **30 mg dispersible ≈ 50 mg film-coated**. Never substitute one for the other milligram for milligram.

**Infant prophylaxis (HIV-exposed infants)**: start **within 6–12 hours of birth**; low risk (mother suppressed): **NVP once daily × 6 weeks**; high risk (mother not suppressed, VL unknown, ART <4 weeks before delivery, diagnosed in labour or while breastfeeding): **AZT + NVP for 6 weeks, then NVP alone for 6 more weeks** (12 weeks; continue NVP if mother still unsuppressed). **NVP 10 mg/mL: birth–6 weeks 2.0–2.49 kg 10 mg (1 mL); ≥2.5 kg 15 mg (1.5 mL); 6–12 weeks 20 mg (2 mL)**; AZT 10 mg/mL: 10 mg (2.0–2.49 kg) or 15 mg (≥2.5 kg) twice daily. Then **cotrimoxazole from 6 weeks**. (Same as the HIV in pregnancy topic.)

## Drug cards in brief: ARVs children take

| Drug | Key side effects | Key point |
|---|---|---|
| **Dolutegravir (DTG)** | Insomnia, headache, weight gain, rare hypersensitivity; hyperglycaemia | High barrier to resistance; **separate from iron, calcium, antacids, zinc** (give DTG 2 h before or 6 h after); **double with rifampicin**; enzyme-inducing anticonvulsants (phenobarbital, phenytoin, carbamazepine) lower levels |
| **Abacavir (ABC)** | **Hypersensitivity reaction** (fever, rash, GI and respiratory symptoms, usually in first 6 weeks; HLA-B*5701, rare in Africans) | **Never rechallenge** after a hypersensitivity reaction (can be fatal) |
| **Lamivudine (3TC)** | Very well tolerated | Also active against hepatitis B |
| **Zidovudine (AZT)** | **Anaemia, neutropenia**, headache, nail pigmentation, lactic acidosis | Check Hb; substitute if Hb <8 g/dL (UCG) |
| **Nevirapine (NVP)** | **Rash/SJS** and **hepatotoxicity** (first 6–8 weeks) | Infant prophylaxis; avoid with rifampicin |
| **Lopinavir/ritonavir (LPV/r)** | Diarrhoea, vomiting, **bitter taste**, dyslipidaemia | Granules/pellets for young children; many interactions |
| **Efavirenz (EFV)** | Vivid dreams, dizziness, psychiatric effects, rash | Only >3 years and >10 kg |

| Drug | Key side effects | Key point |
|---|---|---|
| **Tenofovir (TDF)** | Renal tubular dysfunction, bone density loss | Adolescents ≥30 kg (TLD) |

# Vaccines & supplements

## UNEPI routine schedule (Uganda MoH, 2022 schedule in UCG 2023, plus 2025 malaria vaccine)

| Age | Vaccines | Route / site |
|---|---|---|
| **Birth** (first contact) | **BCG**, **OPV0**, **Hepatitis B birth dose** (within the first 7 days; ideally 24 h) | BCG **0.05 mL intradermal right upper arm** (<12 months; 0.1 mL after 11 months); HepB 0.5 mL IM left thigh; OPV 2 drops |
| **6 weeks** | **OPV1, DPT-HepB-Hib1 (pentavalent), PCV1, Rota1, IPV1** | Penta IM left thigh; PCV IM right thigh; Rota 2 drops oral (inner cheek); IPV IM right thigh 2.5 cm from PCV |
| **10 weeks** | **OPV2, Penta2, PCV2, Rota2** | |
| **14 weeks** | **OPV3, Penta3, PCV3, IPV2** (+ **Rota3** on the 2025/26 MoH poster, brand-dependent; UCG 2023 lists rotavirus at 6 and 10 weeks only) | |
| **6, 7, 8 months and 18 months** | **Malaria vaccine (R21/Matrix-M)**, introduced in 2025 in high-burden districts (WHO 2026: four doses from 5 months, 4th dose 6–18 months after the 3rd; **verify** the current Ugandan districts and ages) | IM |
| **9 months** | **Measles–rubella (MR1)**, **yellow fever** | MR 0.5 mL SC left upper arm; YF 0.5 mL SC right upper arm |
| **18 months** | **MR2** | SC left upper arm |
| **10 years (girls)** | **HPV**: **single dose** (MoH/UNEPI since 2025, per WHO); **2 doses 6 months apart if HIV-positive or immunocompromised**. UCG 2023 still prints 2 doses for all | IM left upper arm |

**Principles**: give each vaccine at the recommended age or at the **first contact after it**; **minor illness is not a contraindication**; admitted severely ill children are vaccinated **at discharge**; **do not give BCG to a child with symptomatic HIV/AIDS**; an exposed unimmunised infant aged 6–9 months can get MR early and then again at 9 months; **live vaccines** (BCG, MR, YF, OPV, rota) are avoided in severe immunosuppression and on high-dose steroids (≥2 mg/kg/day prednisolone for ≥14 days); campaign doses do not replace routine doses.

## Vitamin A

| Use | <6 months | 6–11 months | ≥12 months |
|---|---|---|---|
| **Routine supplementation** (6–59 months, every 6 months) | | **100,000 IU once** | **200,000 IU every 6 months** to 5 years |
| **Measles** (UCG: 3 doses, day 1, day 2 and day 14) | **50,000 IU** | **100,000 IU** | **200,000 IU** |
| **Xerophthalmia / eye signs in SAM** | Same doses on days 1, 2 and 15 | | |
| **Persistent diarrhoea / SAM without eye signs** | Not routinely if on F-75/F-100/RUTF (WHO 2013/2023) | | |

Side effects: transient raised intracranial pressure (bulging fontanelle, vomiting, headache) with large doses; **toxicity** if doses are repeated too close together.

### Iron, folic acid and other supplements

| Supplement | Dose | Notes |
|---|---|---|
| **Iron (treatment of iron deficiency anaemia)** | **Elemental iron 3 mg/kg/day** (UCG: ferrous sulphate 5 mg/kg 8-hourly ≈ 3 mg/kg/day elemental) | Continue **3 months after Hb normalises** (to refill stores); give with deworming and malaria treatment/prevention. Hb rises ~1 g/dL per 2–3 weeks. **Not in SAM stabilisation**; **not with RUTF** |
| **Iron (prophylaxis, preterm/LBW)** | 2–4 mg/kg/day from 2 weeks–6 months | |
| **Folic acid** | Anaemia: **2.5 mg daily** (UCG child); **sickle cell disease: 5 mg daily for life** (≥1 year; 2.5 mg in infants) | High-dose folic acid reduces the effect of SP and other antifolates |
| **Albendazole** | **12–23 months: 200 mg; ≥2 years: 400 mg single dose**, every 6 months (UNEPI/school programmes) | Not under 1 year. Strongyloides: 400 mg 12-hourly × 3 days |
| **Mebendazole** | **500 mg single dose** (250 mg if <2 years; UCG) | Alternative to albendazole |
| **Penicillin V (sickle cell disease)** | **3 months–3 years: 125 mg 12-hourly; 3–5 years: 250 mg 12-hourly** (UCG; at least until 5 years) | Prevents pneumococcal sepsis (functional asplenia) |
| **Hydroxyurea (sickle cell disease)** | Start **20 mg/kg once daily** (UCG), escalate to maximum tolerated dose (≤35 mg/kg) with FBC monitoring | Uganda REACH/NOHARM trials showed it is safe and effective in malaria-endemic Africa; fewer crises, transfusions and deaths |
| **Zinc** | See Fluids | Diarrhoea |
| **Vitamin D** | Rickets: 2,000–6,000 IU daily for ≥12 weeks (or stoss therapy), with calcium | Nutritional rickets is seen in Uganda (covered children, poor diet) |

# Poisoning antidotes

**General management of a poisoned child (UCG 2023 / WHO Pocket Book)**: ABC, glucose, oxygen; **identify the poison** (bring the container); **do not induce vomiting**; **activated charcoal 0.5–1 g/kg (max 50 g)** orally or by NGT **within 1 hour** for adsorbable poisons (not for kerosene, caustics, iron, alcohols, lithium); gastric lavage only for life-threatening ingestions within 1 hour with a protected airway (rarely); remove contaminated clothes and wash skin (organophosphates); observe for at least 6 h; **assess for neglect or deliberate poisoning** in every case.

| Poison | Antidote / specific treatment | Dose | Notes |
|---|---|---|---|
| **Paracetamol** | **Acetylcysteine (NAC)** IV | **150 mg/kg over 15–60 min, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h** (WHO Pocket Book for <20 kg: in **3, 7 and 14 mL/kg** of 5% dextrose) | Toxic dose **>150 mg/kg** (UCG: >200 mg/kg in children <6 years); best **within 8 h**, still useful later. Oral **methionine** if NAC unavailable and <8 h (<6 years: 1 g 4-hourly × 4) |

| Poison | Antidote / specific treatment | Dose | Notes |
|---|---|---|---|
| **Organophosphates / carbamates** | **Atropine** + pralidoxime | **Atropine 0.02–0.05 mg/kg IV/IM**, **double every 5 min** until **atropinised** (chest clear, HR >80, SBP normal, dry axillae), then infusion; **pralidoxime 25–50 mg/kg IV over 30 min**, then 10–20 mg/kg/h (UCG) | Wash the skin; staff wear gloves; treat fits with diazepam; pupils are a poor guide |
| **Opioids** (including codeine, tramadol, morphine) | **Naloxone** | **0.01 mg/kg, then 0.1 mg/kg (max 2 mg)** IV/IM, repeat every 2–3 min | Repeat or infuse (short half-life) |
| **Iron** (≥60 mg/kg elemental is serious; <40 mg/kg unlikely to cause problems) | **Desferrioxamine** | **15 mg/kg/h IV** (max 80 mg/kg/24 h), not >24 h | Charcoal useless; 200 mg ferrous sulphate tablet = **60–65 mg elemental iron**; urine turns "vin rosé" |
| **Benzodiazepines** | Flumazenil (rarely) | 10 micrograms/kg IV | Risk of seizures; support ventilation instead |
| **Warfarin / coumarin rat poison** | **Vitamin $K_1$** | 0.25–0.3 mg/kg IV (max 10 mg) if bleeding or INR raised; FFP for bleeding | Superwarfarin rodenticides need weeks of vitamin K |
| **Methaemoglobinaemia** (dapsone, nitrites, some local anaesthetics) | **Methylene blue** | 1–2 mg/kg IV over 5 min | Avoid in G6PD deficiency |
| **Kerosene (paraffin)** and petrol | **No antidote** | Oxygen, observe ≥6 h, CXR if breathless | **No emesis, no lavage, no charcoal** (aspiration pneumonitis); antibiotics only if secondary infection |
| **Caustics (bleach, battery acid, "jik")** | No antidote | Nil by mouth, analgesia, endoscopy referral | No emesis, no neutralising, no charcoal |
| **Aspirin / salicylates** | Sodium bicarbonate (urinary alkalinisation) | 1–2 mmol/kg IV then infusion | Charcoal; glucose; haemodialysis if severe |
| **Tricyclic antidepressants** | **Sodium bicarbonate** | **1–2 mmol/kg IV** for QRS >100 ms or arrhythmia | Fits: benzodiazepine |
| **Isoniazid** | **Pyridoxine** | Gram for gram of INH ingested (or 70 mg/kg, max 5 g) IV | Refractory fits with acidosis |
| **Carbon monoxide** (charcoal stoves in closed rooms) | **100% oxygen** | Non-rebreather mask | $SpO_2$ falsely normal |
| **Methanol (illicit waragi)** | **Ethanol** (or fomepizole) | UCG: oral 40% alcohol 1.5–2 mL/kg loading | Haemodialysis |
| **Beta-blockers / calcium channel blockers** | Glucagon; calcium; high-dose insulin | Specialist | |

| Poison | Antidote / specific treatment | Dose | Notes |
|---|---|---|---|
| **Snake bite** | **Polyvalent antivenom** (e.g. for East African snakes) | **Same dose as adults** (dose depends on venom, not weight) | Only for systemic envenoming or progressive swelling; have adrenaline ready (reactions common); immobilise, no tourniquet, no cutting |
| **Local herbal medicine** | No antidote | Supportive | Ask specifically; some cause liver failure, AKI or hypoglycaemia |

# Management of drug emergencies & toxicity

These are the drug emergencies you must recognise and start treating on the paediatric ward before the senior arrives.

### 1. Anaphylaxis (ceftriaxone, penicillins, blood, vaccines, peanuts in RUTF, insect stings)

1. **Call for help**; stop the trigger (stop the infusion, keep the line).
2. **Adrenaline 1:1000 IM into the anterolateral thigh**: **0.01 mL/kg** (**<6 years 0.15 mL; 6–12 years 0.3 mL; >12 years 0.5 mL**). **Repeat every 5 min** if no improvement.
3. **Lie the child flat** (legs raised) or sitting if breathless; **high-flow oxygen**; airway support.
4. **IV/IO access: 0.9% saline 20 mL/kg** for hypotension or poor perfusion (repeat).
5. Wheeze: salbutamol nebuliser. Stridor: nebulised adrenaline.
6. Adjuncts after stabilisation: chlorphenamine for itch, hydrocortisone (doses in the emergency table). Observe **at least 6–12 h** (biphasic reactions). Record the allergy and tell the family.

### 2. Respiratory depression after benzodiazepines, phenobarbital or opioids

1. Airway (head tilt, jaw thrust, suction), **oxygen**, **bag-valve-mask ventilation** if RR is low or $SpO_2$ falling.
2. **Opioid**: **naloxone** (0.01 then 0.1 mg/kg). **Benzodiazepine**: support ventilation; flumazenil only in specific cases (not in a child whose seizures may return).
3. Review the chart: **how many doses** of diazepam (max two)? Phenobarbital after two diazepam doses? Did the child also get morphine?
4. Do not give the next sedative dose until the child is reassessed.

### 3. Hypoglycaemia during treatment (quinine, sepsis, SAM, fasting, insulin)

1. Check glucose; **10% dextrose 5 mL/kg IV** (2 mL/kg in neonates) or sugar water by NGT if no IV.
2. Recheck in 30 min; continue a dextrose-containing fluid and **feeds**.
3. On quinine: check glucose **4-hourly**; dilute quinine in **dextrose**.

### 4. Fluid overload (after boluses, transfusion or excessive maintenance)

**Signs**: rising RR and HR, new **crackles**, **enlarging liver**, gallop rhythm, raised JVP (older children), periorbital/facial oedema, falling $SpO_2$. **Children with SAM, severe anaemia and severe malaria are most at risk.**

1. **Stop the fluid** (or slow the blood), sit the child up, oxygen.
2. **Furosemide 1 mg/kg IV** (repeat if needed).
3. Recalculate the fluid plan; weigh daily.

### 5. Gentamicin toxicity

**Rising creatinine, oliguria, or a child who becomes unsteady or deaf.** Stop gentamicin, switch to a non-nephrotoxic antibiotic (ceftriaxone), give appropriate fluids, check potassium; arrange a hearing assessment later. Prevent it: once-daily dosing, correct weight, **no gentamicin in an anuric child**, short courses.

### 6. Extravasation injury (dextrose, calcium, phenytoin, potassium, bicarbonate)

Stop the infusion, aspirate what you can through the cannula, remove it, elevate the limb, mark the area and review; plastic surgery referral for blistering or necrosis. Check cannula sites **hourly** during irritant infusions.

### 7. Stevens–Johnson syndrome / toxic epidermal necrolysis (cotrimoxazole, NVP, phenobarbital, carbamazepine, phenytoin, sulphonamides)

Fever, sore mouth and eyes, then target lesions, blisters and skin peeling with **two or more mucosal surfaces**. **Stop the drug immediately and permanently**, admit, fluids (burns-type care), analgesia, eye care (ophthalmology), mouth care, nutrition, infection surveillance. Record and counsel: the drug (and its class) must never be given again.

### 8. Acute dystonic reaction (metoclopramide, prochlorperazine, haloperidol)

Oculogyric crisis, torticollis, trismus, tongue protrusion. Stop the drug; **diphenhydramine 1 mg/kg IV** or **benztropine/procyclidine** (specialist doses); **diazepam** as an alternative. Avoid metoclopramide as a routine antiemetic in children; use **ondansetron 0.15 mg/kg** (max 4–8 mg) where an antiemetic is genuinely needed.

### 9. Post-artesunate delayed haemolysis (PADH)

New pallor, dark urine, jaundice **7–21 days after IV artesunate** (especially with high parasitaemia). Check Hb; transfuse if severe; it is not a reason to avoid artesunate. Ask children to return for an **Hb check 2–3 weeks** after severe malaria (they also come back for PDMC).

### 10. Paracetamol overdose (accidental or repeated supratherapeutic)

1. Calculate **mg/kg ingested** (all doses over 24 h).
2. Single ingestion ≥150 mg/kg (≥200 mg/kg in <6 years per UCG), or **repeated doses >75–90 mg/kg/day** with symptoms or abnormal LFTs, or unknown amount: **start acetylcysteine** (do not wait for levels you cannot get).
3. Check **glucose, LFTs, INR, creatinine** at baseline and at the end of the infusion.
4. NAC anaphylactoid reactions (flushing, wheeze) are rate-related: pause, give chlorphenamine, restart more slowly.

### 11. Organophosphate poisoning (a Ugandan classic: agricultural sprays, "dudu" killers)

**SLUDGE/DUMBELS**: salivation, lacrimation, urination, diarrhoea, GI cramps, emesis; bradycardia, **bronchorrhoea and bronchospasm**, miosis, then muscle fasciculations, weakness and respiratory failure.

1. Protect staff (gloves), **remove clothes and wash the skin**.
2. ABC, suction, oxygen.
3. **Atropine 0.02–0.05 mg/kg IV**, **double the dose every 5 min until the chest is clear**, then infusion. Expect to need large total doses.
4. **Pralidoxime 25–50 mg/kg IV over 30 min** within 24 h, then infusion.
5. Diazepam for fits; ventilation if weak.

### 12. Kerosene ingestion

**Do not make the child vomit**, no lavage, no charcoal. Oxygen if breathless; observe at least **6 hours**; CXR if symptomatic; antibiotics only for secondary bacterial pneumonia. Teach: store paraffin in labelled containers out of reach, never in soda bottles.

## Emergency drug box for the paediatric acute area

> **UGANDA CONTEXT**
> Many Ugandan paediatric emergency rooms keep a ready box. Check it at the start of every shift.

| Item | Why |
|---|---|
| Weight/age estimation chart or length tape, calculator | Doses |
| 10% dextrose (or 50% + water to dilute), 5% dextrose | Hypoglycaemia |
| Diazepam 10 mg/2 mL, phenobarbital 200 mg/mL, (midazolam) | Convulsions |
| Adrenaline 1:1000 ampoules (+ saline to make 1:10,000) | Anaphylaxis, arrest, croup |
| Salbutamol nebules and MDI with spacer, ipratropium, prednisolone, hydrocortisone | Asthma |
| Ringer's lactate, 0.9% saline, half-strength Darrow's in 5% dextrose, ReSoMal, ORS | Shock, dehydration, SAM |
| Artesunate vials with bicarbonate ampoules | Severe malaria |
| Ampicillin, gentamicin, ceftriaxone | Sepsis, meningitis, pneumonia |
| Calcium gluconate 10%, sodium bicarbonate 8.4%, atropine, naloxone, furosemide | Specific emergencies |
| Paediatric cannulae (22–24 G), **intraosseous needles**, NGTs, giving sets with burettes, syringes, bag-valve-masks (infant and child) with masks, oxygen prongs | Access and airway |

## Ward-round & exam pearls

**Model presentation (drug-focused):** "This is Baby Okello, a **2-year-old boy weighing 10.5 kg** (weight-for-height Z-score –1.5, MUAC 13.2 cm, no oedema), fully immunised for age by card, HIV-negative mother, still breastfeeding plus family foods. He was admitted yesterday with **severe malaria (prostration, two convulsions) and Hb 6.9 g/dL**. He received **rectal diazepam 5 mg** at the HC III and **one dose of IV artesunate 32 mg (3 mg/kg)** here at 22:00; the 12-hour dose is due at 10:00. **Glucose 5.1 mmol/L** this morning. No further fits. He is on **maintenance fluids at 1,025 mL/day** (100 × 10 + 50 × 0.5), mostly breast milk and oral fluids, and **paracetamol 160 mg 6-hourly**. **Plan**: complete artesunate to 24 h then **AL 1 tablet BD × 3 days** once drinking (or DP per UCG at discharge), Hb on day 3 and at 2–3 weeks, PDMC, mosquito net counselling."

**Viva questions with answers**

1. *"How do you estimate the weight of a 4-year-old in resuscitation?"* (2 × 4) + 8 = **16 kg** (APLS), or (4 + 4) × 2 = 16 kg; then weigh as soon as possible.
2. *"Why are neonates at risk of kernicterus with ceftriaxone and cotrimoxazole?"* Both displace unconjugated bilirubin from albumin; free bilirubin crosses the immature blood–brain barrier.
3. *"What is grey baby syndrome?"* Chloramphenicol toxicity in neonates (immature glucuronidation): vomiting, abdominal distension, grey cyanosis, hypothermia, cardiovascular collapse.

4. *"Why does a 2-year-old need a relatively high mg/kg dose of many drugs?"* Higher hepatic clearance per kg (bigger liver relative to body size and higher metabolic rate).
5. *"How do you give IV artesunate to a 12 kg child?"* 3 mg/kg = 36 mg; 60 mg vial + 1 mL bicarbonate + 5 mL saline = 10 mg/mL; give 3.6 mL IV slowly at 0, 12, 24 h then daily, then a full ACT.
6. *"Why not boluses in SAM?"* Their hearts are small and weak (cardiac muscle atrophy), sodium is retained and potassium depleted: standard boluses cause heart failure. Use 15 mL/kg over 1 h of a dextrose-containing fluid and monitor every 5–10 min.
7. *"Why no codeine in children?"* Variable CYP2D6 metabolism to morphine: ultra-rapid metabolisers can stop breathing; poor metabolisers get no analgesia.
8. *"What drug do you give a neonate who is fitting?"* Phenobarbital 20 mg/kg IV/IM (after checking glucose and correcting hypoglycaemia), further 10 mg/kg doses to 40 mg/kg.
9. *"How do you make 1:10,000 adrenaline?"* 1 mL of 1:1000 + 9 mL normal saline.
10. *"What is the first-line treatment of uncomplicated malaria in a 9 kg child?"* AL 1 tablet (20/120 mg) at 0 and 8 h, then twice daily on days 2 and 3 (6 doses), with milk or food.

**OSCE tips**

- In a **prescribing station**: write the **weight**, the **mg/kg**, the **calculated dose in mg**, the **volume in mL** with the **strength**, route, frequency, duration, maximum and your signature. Say your calculation out loud.
- In an **ORS/counselling station**: demonstrate mixing **1 sachet in 1 litre**, frequent small sips, the 4 rules (extra fluid, zinc, continue feeding, when to return) and the danger signs.
- In a **"drug error" station**: identify the error (decimal point, wrong strength, adult ampoule), stop the harm, treat the child, tell the senior and the family honestly, and report it (incident form).

# Mnemonics & memory aids

- **Weight**: "**Double at five, triple at one, quadruple at two**" (months, year, years) for birth weight; resuscitation weight "**(age + 4) × 2**".
- **Gentamicin**: "**3-5-7.5**": 3 mg/kg LBW neonates in week 1, 5 mg/kg term neonates in week 1, 7.5 mg/kg for everyone older, all **once daily**.
- **Ampicillin**: "**50 for the fifty**": 50 mg/kg every 6 h (child), 12 h (week 1 neonate), 8 h (weeks 2–4).
- **Artesunate**: "**3 under 20**": 3 mg/kg if under 20 kg, 2.4 mg/kg if 20 kg or more; "**0-12-24 then daily**".
- **Seizure ladder "D-D-P-P"**: **D**iazepam, **D**iazepam (10 min later), **P**henobarbital, **P**henytoin; glucose at every step.
- **Fluids "100-50-25"**: maintenance per day (UCG).
- **Plan C "30-70"**: 30 mL/kg then 70 mL/kg: "**1 then 5**" hours for infants, "**half then 2½**" for children.
- **Adrenaline "1000 IM, 10,000 IV"**: the higher number (more dilute) is for the vein.
- **Never in children "CCC-AT"**: **C**hloramphenicol (neonates), **C**eftriaxone + calcium (neonates), **C**odeine; **A**spirin; **T**etracyclines (<8 years).
- **Kerosene: "Don't Empty, Don't Wash, Don't Charcoal."**
- **Organophosphate: "DUMBELS"** and treat with "**A-P**": **A**tropine (double until dry) + **P**ralidoxime.

# High-yield summary

## Numbers & doses to know

| Item | Value |
|---|---|
| Weight estimate (1–10 years) | **(age + 4) × 2** kg; 0–12 months (0.5 × months) + 4 |
| Amoxicillin (pneumonia) | **40 mg/kg 12-hourly × 5 days**; 250/500/750 mg BD at 2–12 m / 1–3 y / 3–5 y |
| Ampicillin | **50 mg/kg IV 6-hourly** (neonates 12-hourly week 1, 8-hourly weeks 2–4) |
| Gentamicin | **7.5 mg/kg once daily**; neonates week 1: **5 mg/kg** (≥2.5 kg) / **3 mg/kg** (LBW) |
| Ceftriaxone | **50–80 mg/kg daily**; meningitis **100 mg/kg daily** (max 4 g); not in jaundiced neonates or with calcium |
| Cloxacillin | **25–50 mg/kg 6-hourly** |
| Metronidazole | **7.5 mg/kg 8-hourly** (amoebiasis 10 mg/kg) |
| Benzylpenicillin | **50,000 IU/kg 6-hourly** (meningitis 100,000) |
| Cotrimoxazole prophylaxis | **<5 kg 120 mg; 5–14.9 kg 240 mg; 15–29.9 kg 480 mg; ≥30 kg 960 mg** daily, from 6 weeks |
| IV artesunate | **3 mg/kg (<20 kg) / 2.4 mg/kg (≥20 kg)** at 0, 12, 24 h then daily |
| Rectal artesunate | **10 mg/kg**: 100 mg (<3 y), 200 mg (3–<6 y), pre-referral only |
| AL | 1/2/3/4 tablets for **5–14/15–24/25–34/≥35 kg**, 6 doses over 3 days |
| IV quinine | **10 mg/kg 8-hourly over 4 h in dextrose** (WHO loading 20 mg/kg); glucose 4-hourly |
| Diazepam | **0.5 mg/kg rectal or 0.2–0.3 mg/kg IV**, max 10 mg, max 2 doses |
| Phenobarbital | Child **15–20 mg/kg** (UCG 10–15); neonate **20 mg/kg** (to 40), maintenance 5 mg/kg/day |
| Phenytoin | **15–20 mg/kg IV in saline over ≥20 min**, ≤1 mg/kg/min, ECG |
| Salbutamol | Nebulised **2.5 mg (<5 y) / 5 mg (≥5 y)**; MDI 2–10 puffs via spacer every 20 min × 3 |
| Prednisolone (asthma) | **1–2 mg/kg daily × 3–5 days** (max 20/30/40 mg by age) |
| $MgSO_4$ (asthma) | **40 mg/kg IV over 20 min (max 2 g)** |
| Aminophylline | **5–6 mg/kg IV over ≥20 min** (omit if on theophylline/caffeine), then 5 mg/kg 6-hourly |
| Dexamethasone (croup) | **0.15–0.6 mg/kg** single dose |
| Maintenance fluid (UCG) | **100/50/25 mL/kg/day** |
| Neonatal fluids | **60 → 90 → 120 → 150 mL/kg/day** (days 1–4) |
| Shock bolus | **20 mL/kg** RL/NS (UCG); WHO 2016 10–20 mL/kg over 30–60 min; SAM **15 mL/kg over 1 h** |
| Plan B | **ORS 75 mL/kg over 4 h** |
| Plan C | **RL 100 mL/kg**: 30 then 70 (1 h + 5 h infants; 30 min + 2.5 h children) |
| Zinc | **10 mg (<6 m) / 20 mg (≥6 m) × 10 days** (UCG); WHO 2024 **5 mg daily** |
| F-75 | **130 mL/kg/day** in 8 feeds (100 mL/kg/day if severe oedema) |
| RUTF | **150–185 kcal/kg/day** (WHO 2023); 1 sachet ≈ 500 kcal |
| Hypoglycaemia | **10% dextrose 5 mL/kg** (neonate 2 mL/kg); threshold 2.5 (WHO) / 3.0 (UCG, SAM); neonate 2.6 mmol/L |
| Blood | **20 mL/kg whole blood / 10 mL/kg packed cells** (SAM 10 mL/kg over 3 h) |

| Item | Value |
|---|---|
| Paracetamol | **15 mg/kg 6-hourly**, max 60 mg/kg/day; toxic >150 mg/kg |
| Ibuprofen | **5–10 mg/kg 6–8-hourly**, >3 months, max 40 mg/kg/day |
| Oral morphine | **0.2–0.4 mg/kg 4-hourly** (1–2 y); 0.2–0.5 mg/kg (2–12 y) |
| Adrenaline anaphylaxis | **1:1000 IM 0.01 mL/kg** (0.15 / 0.3 / 0.5 mL) |
| Adrenaline arrest | **1:10,000 IV/IO 0.1 mL/kg** every 3–5 min |
| Atropine | Bradycardia **20 mcg/kg**; organophosphate **0.02–0.05 mg/kg doubling** |
| Naloxone | **0.01 → 0.1 mg/kg** (max 2 mg) |
| Calcium gluconate 10% | **0.5 mL/kg** slow IV (max 20 mL) |
| Sodium bicarbonate 8.4% | **1 mL/kg** (infants 2 mL/kg of 4.2%) |
| NAC | **150 / 50 / 100 mg/kg** over 1 / 4 / 16 h |
| Desferrioxamine | **15 mg/kg/h** (iron >60 mg/kg) |
| Activated charcoal | **0.5–1 g/kg** within 1 h |
| TB (H/R/Z/E) | **10 / 15 / 35 / 20 mg/kg**; RHZ 75/50/150: 1/2/3/4 tabs at 4–8/8–12/12–16/16–25 kg |
| Vitamin A | **50,000 / 100,000 / 200,000 IU** (<6 m / 6–11 m / ≥12 m) |
| Albendazole | **200 mg (12–23 m), 400 mg (≥2 y)** |
| Vitamin K at birth | **1 mg IM** (0.5 mg preterm) |
| Caffeine citrate | **20 mg/kg load, 5 mg/kg daily** |

## Classic exam traps

- **Trap:** "Give 50% dextrose 1 mL/kg IV to a hypoglycaemic child." **Right:** 10% dextrose 5 mL/kg (dilute 50% one part to four parts water).
- **Trap:** "Use 1:1000 adrenaline 0.1 mL/kg IV in cardiac arrest." **Right:** 1:10,000 at 0.1 mL/kg (1:1000 at that volume is a tenfold overdose).
- **Trap:** "Ceftriaxone is the best antibiotic for a jaundiced 3-day-old with sepsis, run with Ringer's lactate." **Right:** ampicillin + gentamicin (or cefotaxime); never ceftriaxone with calcium-containing fluids in neonates.
- **Trap:** "Give 20 mL/kg saline boluses to a child with SAM in shock" or "to a child with Hb 3 g/dL." **Right:** SAM: 15 mL/kg over 1 h of a dextrose-containing fluid; severe anaemia: transfuse.
- **Trap:** "Give diazepam to a fitting neonate." **Right:** phenobarbital 20 mg/kg (and check glucose).
- **Trap:** "A third dose of diazepam because the child is still fitting." **Right:** maximum two benzodiazepine doses, then phenobarbital or phenytoin.
- **Trap:** "Rectal artesunate given, now send the child home with AL." **Right:** rectal artesunate is pre-referral only: refer urgently for injectable artesunate, then a full ACT.
- **Trap:** "Codeine syrup for post-tonsillectomy pain in a 6-year-old." **Right:** paracetamol ± ibuprofen; morphine if severe.

## Questions seniors ask

1. **"What does this child weigh, and what dose is that?"** Always know the weight and do the sum aloud: "13 kg, gentamicin 7.5 mg/kg = 97.5 mg, so about 98 mg IV once daily."
2. **"Why once-daily gentamicin?"** Concentration-dependent killing and post-antibiotic effect; a drug-free trough reduces nephrotoxicity; fewer missed doses.

3. **"Why 3 mg/kg artesunate in small children?"** Under-20 kg children had lower drug exposure at 2.4 mg/kg; WHO raised the dose to match adult exposure.
4. **"The child had two doses of diazepam and is still fitting. What next?"** Phenobarbital 15–20 mg/kg IV over 15–20 min (or phenytoin 15–20 mg/kg in saline over ≥20 min), airway support, glucose, treat the cause.
5. **"Why no aspirin?"** Reye's syndrome (encephalopathy and liver failure after viral infections); use paracetamol or ibuprofen.
6. **"How much maintenance fluid for a 25 kg child?"** UCG: 1,000 + 500 + (5 × 25) = **1,625 mL/day** (Holliday–Segar 1,600 mL/day), about 68 mL/h; give two-thirds if meningitis or severe pneumonia.
7. **"What would you check before giving potassium?"** That the child is passing urine and the serum potassium is not high; never as a bolus; ≤40 mmol/L peripherally.
8. **"What is the antidote to organophosphates and how do you know you've given enough?"** Atropine, doubling every 5 min until the chest is clear of secretions, heart rate >80, BP normal and axillae dry (not by pupil size); plus pralidoxime.

SELF-TEST

# Flashcards

64 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | How do you estimate a child's weight in resuscitation when you cannot weigh? | **APLS**: 0–12 months (0.5 × age in months) + 4; 1–5 years (2 × age) + 8; 6–12 years (3 × age) + 7. Older formula: **(age + 4) × 2**. A length tape is better. Weigh as soon as possible. |
| 2 | Birth weight rules of thumb | Doubles by **5 months**, triples by **1 year**, quadruples by **2 years**. |
| 3 | Why do neonates need longer dosing intervals for gentamicin and ampicillin? | **Low GFR** at birth (matures over the first year) → slower renal clearance. Higher body water also increases the volume of distribution of water-soluble drugs. |
| 4 | Why are ceftriaxone and cotrimoxazole risky in jaundiced neonates? | They **displace bilirubin from albumin** → free bilirubin crosses the immature blood–brain barrier → **kernicterus**. Ceftriaxone also forms fatal precipitates with **IV calcium** (Ringer's lactate) in neonates. |
| 5 | What is grey baby syndrome? | **Chloramphenicol** toxicity in neonates (immature glucuronidation): vomiting, distension, grey cyanosis, hypothermia, shock. Avoid in preterms and the **first week of life**. |
| 6 | Five rules for writing a safe paediatric dose | Weight on the chart; **mg/kg and calculated dose**; **no trailing zero** (1 mg not 1.0); **leading zero** (0.5 mg not .5); write **micrograms/units in full**; give strength and mL for liquids; cap at the **adult maximum**. |
| 7 | Amoxicillin dose for pneumonia (UCG 2023 / WHO 2024) | **40 mg/kg orally 12-hourly × 5 days** (dispersible 250 mg): **2–12 months 250 mg BD; 1–3 years 500 mg BD; 3–5 years 750 mg BD**. WHO 2024: 3 or 5 days for fast breathing only. |
| 8 | Ampicillin dose: child and neonate | Child: **50 mg/kg IV/IM 6-hourly** (max 2 g). Neonate (WHO): **50 mg/kg 12-hourly in week 1, 8-hourly weeks 2–4** (UCG lists 6-hourly). |
| 9 | Gentamicin doses ('3-5-7.5') | **7.5 mg/kg once daily** (children and neonates in weeks 2–4); week 1: **5 mg/kg** (≥2.5 kg) or **3 mg/kg** (low birth weight), once daily. |
| 10 | Why once-daily gentamicin? | **Concentration-dependent killing** + **post-antibiotic effect**; drug-free trough → **less nephrotoxicity**; fewer doses missed. |
| 11 | Ceftriaxone doses | **50–80 mg/kg once daily** (sepsis, pneumonia); **100 mg/kg once daily** for meningitis (max **4 g/day**). Neonates 50 mg/kg (100 meningitis) if not jaundiced; never with calcium-containing fluids. |
| 12 | Cloxacillin dose and indication | **25–50 mg/kg IV/IM 6-hourly** (max 2 g); *Staph aureus*: osteomyelitis, septic arthritis, staph pneumonia, skin infections. |
| 13 | Metronidazole dose | **7.5 mg/kg 8-hourly** (max 500 mg/dose); amoebiasis **10 mg/kg 8-hourly**; giardia 5 mg/kg 8-hourly. |
| 14 | Cotrimoxazole prophylaxis doses (UCG) | Once daily: **<5 kg 120 mg; 5–14.9 kg 240 mg; 15–29.9 kg 480 mg; ≥30 kg 960 mg**. HIV-exposed infants from **6 weeks** until HIV is excluded after breastfeeding stops. |
| 15 | Benzathine penicillin: never by which route, and doses? | **Never IV.** Congenital syphilis (asymptomatic) **50,000 IU/kg IM once**; pharyngitis/rheumatic prophylaxis **600,000 IU (<30 kg) or 1.2 MU (≥30 kg)**, prophylaxis every 4 weeks. |
| 16 | Dysentery treatment (WHO 2024) | **Ciprofloxacin 15 mg/kg 12-hourly × 3 days**; second line ceftriaxone 50–80 mg/kg daily × 3 days. No antibiotics for acute watery diarrhoea (except cholera). |

| # | Question | Answer |
|---|---|---|
| 17 | IV artesunate dose and schedule | **3 mg/kg if <20 kg; 2.4 mg/kg if ≥20 kg**, IV/IM at **0, 12, 24 h then daily** until oral (minimum 24 h), then a **full 3-day ACT**. |
| 18 | How do you prepare IV artesunate? | Vial + **1 mL 5% sodium bicarbonate** (shake until clear) + **5 mL saline/5% dextrose** = **10 mg/mL** IV; for IM add 2 mL saline = 20 mg/mL. Never water for injection; use within 1 h. |
| 19 | Rectal artesunate dose | **10 mg/kg single dose**, children **≤6 years** only: **5–<14 kg 100 mg; 14–19 kg 200 mg**. Re-insert if expelled within 30 min. **Pre-referral only: refer immediately.** |
| 20 | AL weight bands | Artemether–lumefantrine 20/120: **5–14 kg 1 tab; 15–24 kg 2; 25–34 kg 3; ≥35 kg 4**, at 0 and 8 h then BD days 2–3 (6 doses), with milk or fatty food. |
| 21 | IV quinine: dose and main danger | UCG: **10 mg/kg in 5% dextrose over 4 h, 8-hourly** (WHO: loading 20 mg/kg unless quinine in last 12–24 h). Never a bolus. Danger: **hypoglycaemia** (check glucose 4-hourly), cinchonism, arrhythmia. |
| 22 | What is post-artesunate delayed haemolysis? | Haemolytic anaemia **7–21 days** after IV artesunate, especially with high parasitaemia. Check **Hb at 2–3 weeks**; transfuse if severe. |
| 23 | Post-discharge malaria chemoprevention (Uganda) | UCG 2023: severe malaria discharged on **DP**, then **monthly DP courses for 3 months**. WHO: PDMC after admission with severe anaemia in moderate–high transmission areas. |
| 24 | Seizure drug ladder (child >1 month) | **Diazepam 0.5 mg/kg PR** or 0.2–0.3 mg/kg IV (max 10 mg) → repeat once at 10 min → **phenobarbital 15–20 mg/kg** or **phenytoin 15–20 mg/kg in saline** → second agent/ICU. Check glucose. |
| 25 | Neonatal seizure drug and dose | **Phenobarbital 20 mg/kg IV/IM**, further 10 mg/kg to **40 mg/kg**; maintenance **5 mg/kg daily**. Avoid diazepam. Check glucose and calcium. |
| 26 | Phenytoin infusion rules | **Saline only** (precipitates in dextrose), ≤1 mg/kg/min (max 50 mg/min), **ECG/heart-rate monitoring**, good IV line (extravasation necrosis). Never IM. |
| 27 | Buccal midazolam doses (NICE/BNFc bands) | **0.3–0.5 mg/kg** (max 10 mg): 3–11 months 2.5 mg; 1–4 years 5 mg; 5–9 years 7.5 mg; ≥10 years 10 mg. |
| 28 | Nebulised salbutamol dose | **2.5 mg (<5 years), 5 mg (≥5 years)** every 20 min × 3, then 1–4-hourly. MDI + spacer: 2–10 puffs (100 mcg) every 20 min in the first hour. |
| 29 | Prednisolone in acute asthma | **1–2 mg/kg once daily for 3–5 days** (max 20 mg <2 years, 30 mg 2–5 years, 40 mg 6–11 years; UCG 1 mg/kg × 3 days). |
| 30 | IV magnesium sulphate in severe asthma | **40 mg/kg (25–50), max 2 g, IV over 20 min**, after the first hour of failed standard treatment. |
| 31 | Croup treatment doses | **Dexamethasone 0.15–0.6 mg/kg** single dose (oral/IM); severe: **nebulised adrenaline 1:1000 0.4–0.5 mL/kg (max 5 mL)** and observe ≥2–4 h. |
| 32 | Aminophylline loading dose and when to omit it | **5–6 mg/kg IV (max 300 mg) over ≥20 min** then 5 mg/kg 6-hourly; **omit the loading dose** if theophylline/aminophylline or caffeine given in the past 24 h. Stop if pulse >180, vomiting or fits. |
| 33 | Maintenance fluids (UCG 2023) | **100 mL/kg/day** for first 10 kg + **50 mL/kg** for next 10 kg + **25 mL/kg** for each kg above 20 (Holliday–Segar uses 20). Add 10% per °C fever. |
| 34 | Neonatal fluids, days 1 to 4 | **60 → 90 → 120 → 150 mL/kg/day**; day 1–2 **10% dextrose**; IV-only max 100 mL/kg/day unless under phototherapy/radiant heater or dehydrated. |

| # | Question | Answer |
|---|---|---|
| 35 | Plan C rehydration | **Ringer's lactate 100 mL/kg**: infants <12 months 30 mL/kg in 1 h then 70 mL/kg in 5 h; children 30 mL/kg in 30 min then 70 mL/kg in 2.5 h. |
| 36 | Shock in SAM: fluid and rate | **15 mL/kg over 1 h** of Ringer's lactate with 5% dextrose (or half-strength Darrow's/0.45% saline with 5% dextrose); repeat once if improving; then ReSoMal 10 mL/kg/h. Not improving → blood 10 mL/kg over 3 h. |
| 37 | Low-osmolarity ORS composition | Na **75**, Cl 65, K **20**, citrate 10, glucose **75** mmol/L; **245 mOsm/L**. One sachet in 1 litre of clean water. |
| 38 | Zinc in diarrhoea: UCG versus WHO 2024 | UCG 2023: **10 mg (<6 months) / 20 mg (≥6 months) daily × 10 days**. WHO 2024: suggests **5 mg daily for up to 14 days** (less vomiting). |
| 39 | F-75 volume | **130 mL/kg/day** (≈100 kcal/kg/day) in 8 three-hourly feeds (12 two-hourly if very sick); **100 mL/kg/day with severe oedema**. |
| 40 | RUTF dose (WHO 2023) | **150–185 kcal/kg/day** until recovery (may reduce to 100–130 once no longer severely wasted). One 92 g sachet ≈ 500 kcal. |
| 41 | Hypoglycaemia treatment in a child and a neonate | Child: **10% dextrose 5 mL/kg IV** (threshold 2.5 mmol/L WHO, 3.0 UCG and SAM). Neonate: **2 mL/kg 10% dextrose** (<2.6 mmol/L). Never undiluted 50%: dilute 1 mL/kg 50% with 4 mL/kg water. |
| 42 | Potassium chloride rules | Only after **urine output** confirmed; **≤40 mmol/L** peripherally; **≤0.2 mmol/kg/h** on the ward; **never bolus**. Maintenance 1–2 mmol/kg/day. |
| 43 | Paracetamol dose and maximum | **15 mg/kg 6-hourly**, max **60 mg/kg/day** (4 g). Neonate 10 mg/kg 6–8-hourly. Toxic: **>150 mg/kg** (UCG: >200 mg/kg in under-6s). |
| 44 | Ibuprofen dose and contraindications | **5–10 mg/kg 6–8-hourly** (max 40 mg/kg/day), only **>3 months**. Avoid in **dehydration/AKI, dengue/VHF, varicella**, GI bleeding. |
| 45 | Oral morphine doses (UCG 2023) | **1–2 years 0.2–0.4 mg/kg 4-hourly; 2–12 years 0.2–0.5 mg/kg 4-hourly** (max 10 mg start); IV 0.05–0.1 mg/kg. Add a laxative; monitor RR. |
| 46 | Why no codeine or aspirin in children? | Codeine: CYP2D6 ultra-rapid metabolisers → **fatal respiratory depression**. Aspirin: **Reye's syndrome** (<16 years), except Kawasaki/rheumatic fever under specialists. |
| 47 | Adrenaline for anaphylaxis | **1:1000 IM, 0.01 mL/kg** (anterolateral thigh): **<6 years 0.15 mL; 6–12 years 0.3 mL; >12 years 0.5 mL**. Repeat every 5 min. |
| 48 | Adrenaline in cardiac arrest | **1:10,000, 0.1 mL/kg (10 mcg/kg) IV/IO** every 3–5 min (max 1 mg). Make 1:10,000 by 1 mL of 1:1000 + 9 mL saline. |
| 49 | Calcium gluconate 10% dose | **0.5 mL/kg (max 20 mL) slow IV over 5–10 min**: hyperkalaemia, hypocalcaemia, Mg or CCB toxicity. Watch heart rate and the cannula site. |
| 50 | Naloxone dose in children | **0.01 mg/kg** (UCG), increasing to **0.1 mg/kg (max 2 mg)** IV/IM; repeat (short half-life) or infuse. |
| 51 | Acetylcysteine regimen | **150 mg/kg** over 15–60 min, **50 mg/kg** over 4 h, **100 mg/kg** over 16 h (for <20 kg in 3, 7 and 14 mL/kg of 5% dextrose). |
| 52 | Organophosphate poisoning: drugs and end-point | **Atropine 0.02–0.05 mg/kg IV, doubling every 5 min** until **chest clear**, HR >80, BP normal, dry axillae; **pralidoxime 25–50 mg/kg** over 30 min. Wash skin. |
| 53 | Kerosene ingestion: three things NOT to do | **No emesis, no gastric lavage, no activated charcoal** (aspiration pneumonitis). Oxygen, observe ≥6 h. |

| # | Question | Answer |
|---|---|---|
| 54 | Iron poisoning thresholds and antidote | <40 mg/kg elemental unlikely toxic; **>60 mg/kg serious**. **Desferrioxamine 15 mg/kg/h IV** (max 24 h). 200 mg ferrous sulphate = 60–65 mg elemental iron. Charcoal useless. |
| 55 | Activated charcoal dose and exclusions | **0.5–1 g/kg (max 50 g)** within 1 h. Not for kerosene, caustics, iron, alcohols, lithium, or an unprotected airway. |
| 56 | TB drug mg/kg doses in children | **H 10 (7–15), R 15 (10–20), Z 35 (30–40), E 20 (15–25) mg/kg** once daily. |
| 57 | RHZ 75/50/150 weight bands | **4–7.9 kg 1; 8–11.9 kg 2; 12–15.9 kg 3; 16–24.9 kg 4 tablets**; ≥25 kg adult FDCs. Same numbers for RH 75/50 and ethambutol 100 mg. |
| 58 | Which children get the 4-month TB regimen? | **Non-severe TB** (peripheral nodes, uncomplicated intrathoracic nodes or effusion, non-cavitary disease in one lobe) aged 3 months–16 years (UCG: 2 months–16 years): **2HRZ(E)/2HR** (SHINE). |
| 59 | Rifampicin and dolutegravir | Rifampicin induces DTG metabolism: give **DTG twice daily** during rifampicin and for 2 weeks after. |
| 60 | First-line ART for a 12 kg child in Uganda | **ABC + 3TC + DTG** (pDTG 10 mg dispersible 20 mg, or pALD 60/30/5 four tablets). ≥30 kg: **TLD**. |
| 61 | Vitamin A doses | **<6 months 50,000 IU; 6–11 months 100,000 IU; ≥12 months 200,000 IU.** Measles: days 1, 2 and 14 (UCG). Routine: 6–11 months once, then every 6 months to 5 years. |
| 62 | Albendazole and mebendazole doses | Albendazole **200 mg (12–23 months), 400 mg (≥2 years)** single dose; mebendazole **500 mg** (250 mg <2 years). Not under 1 year. |
| 63 | Vitamin K at birth | **1 mg IM** (term), **0.5 mg IM** (preterm/LBW), within 1 h; prevents vitamin K deficiency bleeding. |
| 64 | Caffeine citrate for apnoea of prematurity | **Loading 20 mg/kg**, then **5 mg/kg once daily** (up to 20 mg/kg/day) from 24 h later; stop 4–5 days after the last apnoea. |

SELF-TEST

# Single best answer questions

35 questions. The answers and explanations follow at the end.

**1.** A 2-year-old girl weighing 12 kg is brought to Mengo Hospital with fever for 3 days and two generalised convulsions. She responds only to pain. Temperature 39.4°C, HR 156/min, RR 44/min, capillary refill 2 s, glucose 5.0 mmol/L, mRDT positive. What is the correct first dose of IV artesunate?

- A 29 mg (2.4 mg/kg)
- B 24 mg (2 mg/kg)
- C 60 mg (one full vial)
- D 36 mg (3 mg/kg)
- E 120 mg (10 mg/kg)

**2.** A 4-day-old boy born at 36 weeks weighing 2.2 kg is admitted to the neonatal unit with poor feeding, temperature 38.3°C and RR 72/min. He is visibly jaundiced to the abdomen. An IV line with Ringer's lactate has been started. Which antibiotic plan is MOST appropriate?

- A Ceftriaxone 100 mg/kg IV once daily through the Ringer's lactate line
- B Ampicillin 50 mg/kg IV 12-hourly plus gentamicin 3 mg/kg IV once daily
- C Chloramphenicol 25 mg/kg IV 6-hourly
- D Cotrimoxazole 4 mg/kg trimethoprim IV 12-hourly
- E Amoxicillin 40 mg/kg orally 12-hourly

**3.** A 3-year-old boy (14 kg) is convulsing in the emergency room. He received rectal diazepam 7 mg at arrival and IV diazepam 4 mg ten minutes later. Ten minutes after the second dose he is still convulsing. $SpO_2$ 93% on oxygen, glucose 5.6 mmol/L. What is the MOST appropriate next step?

- A Give a third dose of diazepam 0.5 mg/kg rectally
- B Give paracetamol 15 mg/kg rectally and wait
- C Give phenobarbital 15–20 mg/kg IV over 15–20 minutes
- D Give 10% dextrose 5 mL/kg IV
- E Give phenytoin 15 mg/kg in 5% dextrose as a rapid IV push

**4.** A 10 kg unconscious child with severe malaria has a glucose of 1.9 mmol/L. The emergency trolley has only 50% dextrose ampoules and water for injection. How should you prepare the dose?

- A Give 10 mL of 50% dextrose IV undiluted
- B Give 2 mL/kg of 50% dextrose diluted with an equal volume of water
- C Give 50 mL of 50% dextrose diluted in 500 mL saline over 4 hours
- D Give 5 mL of 50% dextrose IV and recheck after 2 hours
- E Mix 10 mL of 50% dextrose with 40 mL of water and give 50 mL IV over a few minutes

**5.** A 14-month-old girl weighing 6.1 kg (weight-for-length Z-score below −3, MUAC 10.6 cm, no oedema) has had watery diarrhoea for 5 days. She is lethargic with cold hands, capillary refill 4 s, weak fast pulse 168/min. Glucose 4.2 mmol/L. What fluid should you give? Advanced

- A Ringer's lactate 20 mL/kg as rapidly as possible, repeated twice
- B Plan C: Ringer's lactate 100 mL/kg over 3 hours
- C Ringer's lactate with 5% dextrose 15 mL/kg over 1 hour, monitoring pulse and RR every 5–10 minutes
- D ReSoMal 5 mL/kg every 30 minutes orally
- E Packed cells 10 mL/kg over 3 hours as the first fluid

**6.** An 18-month-old boy (10 kg) has cough for 3 days. RR 52/min with lower chest wall indrawing, $SpO_2$ 95% in air, temperature 38.2°C. He is alert, feeding and has no danger signs. His mother lives 2 km from the HC IV. According to WHO 2024 and UCG 2023, what is the MOST appropriate management?

- A Oral amoxicillin 500 mg (two 250 mg dispersible tablets) twice daily for 5 days at home, with review at day 3
- B Admit for IV ampicillin and gentamicin
- C Oral amoxicillin 250 mg three times daily for 10 days
- D IM ceftriaxone 50 mg/kg daily for 5 days as an outpatient
- E Cotrimoxazole 240 mg twice daily for 5 days

**7.** A 4-year-old girl (16 kg) on the ward develops wheeze, facial swelling and a BP of 70/40 mmHg two minutes after starting IV ceftriaxone. What is the MOST appropriate first drug and dose?

- A Adrenaline 1:1000, 0.15 mL IM into the anterolateral thigh
- B Hydrocortisone 50 mg IV
- C Chlorphenamine 1 mg orally
- D Adrenaline 1:1000, 1.6 mL IV
- E Adrenaline 1:10,000, 0.15 mL IM

**8.** A 3-year-old (15 kg) collapses on the ward; there is no pulse and the monitor shows asystole. CPR is in progress and an intraosseous needle is in. Which adrenaline dose should be given?

- A 15 mL of 1:10,000 IO
- B 0.15 mL of 1:1000 IM
- C 1.5 mL of 1:1000 IO
- D 0.5 mL of 1:1000 IO
- E 1.5 mL of 1:10,000 IO, repeated every 3–5 minutes

**9.** A 3-year-old boy (14 kg) swallowed ten 500 mg paracetamol tablets from his grandmother's handbag about 3 hours ago. He is well, alert and has vomited once. Paracetamol levels are not available at this hospital. What is the MOST appropriate management?

- A Reassure and discharge because he is asymptomatic
- B Give activated charcoal and observe for 24 hours without an antidote
- C Start IV acetylcysteine 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h, and check LFTs
- D Induce vomiting with salt water
- E Give oral vitamin K and repeat INR in 24 hours

**10.** A 2-year-old girl (11 kg) drank paraffin (kerosene) from a soda bottle 30 minutes ago. She is coughing, RR 46/min, $SpO_2$ 95% in air. Her mother has already given her milk. What is the MOST appropriate management?

- A Give activated charcoal 1 g/kg by nasogastric tube
- B Induce vomiting with ipecac to remove the kerosene
- C Perform gastric lavage within the first hour
- D Give oxygen as needed, avoid emesis, lavage and charcoal, and observe for at least 6 hours
- E Start IV ceftriaxone immediately for aspiration pneumonia

**11.** A 6-year-old boy (20 kg) was found playing with a bottle of agricultural insecticide. He is drooling, wheezing, has profuse bronchial secretions, pinpoint pupils, HR 52/min and fasciculations. After decontamination you start IV atropine 1 mg. Which finding BEST tells you that atropine dosing is adequate?

A Dilated pupils
B Heart rate above 140/min
C A dry mouth
D Return of normal muscle power
E A clear chest on auscultation with no bronchorrhoea

**12.** A 2-day-old baby born at 35 weeks weighing 2.2 kg is started on ampicillin and gentamicin for suspected sepsis at a regional referral hospital. What is the correct gentamicin dose?

A 6.6 mg IV once daily
B 16.5 mg IV once daily
C 5.5 mg IV 8-hourly
D 11 mg IV 12-hourly
E 22 mg IV once daily

**13.** A 5-year-old (18 kg) with cerebral malaria is on IV quinine because artesunate is out of stock. Six hours into treatment he becomes restless and sweaty and has a brief generalised convulsion. Temperature 38.0°C. What should you do FIRST?

A Increase the quinine infusion rate
B Check the blood glucose and give 10% dextrose 5 mL/kg if low
C Give IV mannitol for raised intracranial pressure
D Start phenytoin 20 mg/kg IV
E Give dexamethasone for cerebral oedema

**14.** An 8-year-old girl (22 kg) with acute asthma has received salbutamol 5 mg nebulised three times, ipratropium 250 micrograms three times and oral prednisolone 40 mg in the last hour. She still has severe indrawing, cannot complete sentences, $SpO_2$ 91% on oxygen, HR 150/min. What is the MOST appropriate next drug? Advanced

A Oral salbutamol syrup
B Repeat prednisolone 40 mg
C IV magnesium sulphate 880 mg (40 mg/kg) over 20 minutes
D IV aminophylline 22 mg/kg as a rapid bolus
E IV ceftriaxone

**15.** A 2-year-old boy (12 kg) has had a barking cough for 2 days and now has stridor at rest, chest indrawing and agitation. $SpO_2$ 95%. He is not drooling. What is the MOST appropriate treatment?

A Examine the throat with a spatula, then give amoxicillin
B Nebulised adrenaline 1:1000 5 mL plus dexamethasone 0.6 mg/kg orally, then observe for at least 2–4 hours
C Nebulised salbutamol 2.5 mg every 20 minutes
D IV ceftriaxone and IV fluids 20 mL/kg
E Discharge with paracetamol and steam inhalation

**16.** A 7-year-old girl (21 kg) with HIV on ABC/3TC/DTG is diagnosed with pulmonary TB and starts rifampicin-containing treatment. What change should be made to her ART? Advanced

A Stop ART until TB treatment is completed
B Switch dolutegravir to nevirapine
C No change is needed
D Give dolutegravir twice daily during rifampicin and for 2 weeks after
E Halve the dolutegravir dose

**17.** A 6-week-old HIV-exposed infant (4.6 kg) attends the young child clinic at Mengo. The mother has been on TLD for 3 years with a suppressed viral load; the infant completed 6 weeks of nevirapine. A DNA PCR sample is taken today. Which drug should be STARTED today?

A Cotrimoxazole 120 mg once daily
B Cotrimoxazole 480 mg once daily
C Zidovudine 15 mg twice daily
D Isoniazid 50 mg daily
E Nothing until the PCR result is back

**18.** A 25 kg child admitted with bacterial meningitis is to receive IV maintenance fluids. His sodium is 128 mmol/L and he is not dehydrated. Using the Uganda Clinical Guidelines 2023 formula, what is the FULL daily maintenance requirement, and what volume would you actually give?

A 1,600 mL/day; give all of it
B 2,500 mL/day; give all of it
C 1,625 mL/day; give about two-thirds to three-quarters (≈1,100–1,200 mL/day)
D 1,250 mL/day; give double because of fever
E 1,000 mL/day; give all of it

**19.** An 8-month-old boy weighing 7 kg has severe dehydration from watery diarrhoea (lethargic, very sunken eyes, unable to drink, skin pinch returns very slowly). He is not malnourished (WFL Z-score −1). How should Ringer's lactate be given? Advanced

A 700 mL over 1 hour
B 210 mL over 30 minutes, then 490 mL over 2.5 hours
C 140 mL over 1 hour only, then oral ORS
D 525 mL (75 mL/kg) of ORS over 4 hours
E 210 mL over 1 hour, then 490 mL over 5 hours

**20.** A 3-year-old girl (14 kg) has bloody diarrhoea for 2 days with fever 38.5°C. She is drinking eagerly with sunken eyes. Which treatment combination is MOST appropriate?

A Metronidazole 7.5 mg/kg 8-hourly for 7 days
B ORS Plan B, zinc, and ciprofloxacin 210 mg (15 mg/kg) twice daily for 3 days
C ORS only; antibiotics are not recommended for diarrhoea in children
D Amoxicillin 40 mg/kg twice daily for 5 days
E Loperamide and ORS

**21.** A 2-day-old term baby (3.2 kg) has repeated episodes of lip-smacking and rhythmic jerking of the right arm. Glucose 3.4 mmol/L, temperature 36.9°C. What is the MOST appropriate first anticonvulsant?

A Phenobarbital 64 mg (20 mg/kg) IV over 15–20 minutes
B Phenytoin 20 mg/kg in 10% dextrose
C Diazepam 0.5 mg/kg rectally
D Carbamazepine 5 mg/kg orally
E Midazolam infusion

**22.** A nurse on the paediatric ward reads a handwritten chart: 'Morphine .5 mg IV 4-hourly' for a 5 kg infant after surgery and draws up 5 mg. Which prescribing rule would have prevented this error?

A Always use a trailing zero, e.g. 5.0 mg
B Always write a leading zero before a decimal point, e.g. 0.5 mg
C Abbreviate micrograms as µg
D Prescribe morphine in mL only
E Write 'u' for units

**23.** A 9-year-old boy (24 kg) with sickle cell disease is admitted with a severe painful crisis in both legs (pain score 8/10). He is well hydrated, $SpO_2$ 97%. Which analgesic is LEAST appropriate?

A Paracetamol 15 mg/kg 6-hourly
B Ibuprofen 5–10 mg/kg 8-hourly with food
C Oral morphine 0.2–0.4 mg/kg 4-hourly with a laxative
D Codeine phosphate syrup 1 mg/kg 6-hourly
E IV morphine 0.05–0.1 mg/kg titrated, with RR and $SpO_2$ monitoring

**24.** A 10-month-old girl (8 kg) has measles with conjunctivitis and a cough. She has not received any vitamin A in the last month. What vitamin A regimen is correct per UCG 2023?

A 50,000 IU once
B 100,000 IU on day 1, day 2 and day 14
C 200,000 IU on day 1 and day 2
D 100,000 IU once only
E 200,000 IU every 6 months

**25.** A 10-year-old girl (28 kg) has rheumatic heart disease with mitral regurgitation. She asks how her monthly injection is dosed. What is her correct secondary prophylaxis?

A Benzathine penicillin 1.2 MU IV every 4 weeks
B Oral amoxicillin 40 mg/kg twice daily
C Benzathine penicillin 2.4 MU IM every 2 weeks
D Benzathine penicillin 50,000 IU/kg IM once only
E Benzathine penicillin 600,000 IU deep IM every 4 weeks

**26.** A 2-year-old boy (11 kg) swallowed about 20 of his mother's ferrous sulphate 200 mg tablets 2 hours ago. He has vomited blood twice, is drowsy and has a capillary refill of 3 s. What is the MOST appropriate specific treatment after resuscitation?

A Activated charcoal 1 g/kg
B Desferrioxamine 15 mg/kg/h IV infusion
C Acetylcysteine 150 mg/kg IV
D Atropine 0.02 mg/kg IV
E Induced vomiting

**27.** A 3-year-old boy (13 kg) on day 4 of ampicillin and gentamicin 7.5 mg/kg daily for pyelonephritis has passed only 0.3 mL/kg/h of urine for 12 hours; creatinine has risen from 30 to 75 µmol/L. He is well perfused. What is the MOST appropriate change?

A Continue gentamicin but give it 12-hourly
B Double the gentamicin dose to overcome the infection
C Add furosemide and continue gentamicin
D Stop gentamicin and switch to ceftriaxone, review fluids and check potassium
E Stop all antibiotics

**28.** A 5-year-old boy weighing 17 kg has uncomplicated malaria (mRDT positive, alert, drinking, no danger signs, Hb 10.5 g/dL). Which regimen is correct?

A Artemether–lumefantrine 1 tablet twice daily for 3 days
B Artemether–lumefantrine 2 tablets at 0 and 8 hours, then 2 tablets twice daily on days 2 and 3, with milk or food
C Artemether–lumefantrine 4 tablets once daily for 3 days
D IV artesunate 3 mg/kg at 0, 12 and 24 hours
E Quinine tablets 300 mg 8-hourly for 7 days

**29.** A 3-year-old girl (13 kg) has a 4 cm matted cervical lymph node mass for 6 weeks, fever and weight loss. Her chest X-ray is normal and Xpert on a node aspirate detects TB, rifampicin-sensitive. She is HIV-negative. What is the MOST appropriate regimen?

A 2HRZE/10HR (12 months)
B RHZ 75/50/150 three tablets daily for 2 months, then RH 75/50 three tablets daily for 2 months
C RHZ 75/50/150 two tablets daily for 6 months
D Isoniazid 10 mg/kg daily for 6 months
E Streptomycin plus HRZE for 2 months then HRE for 6 months

**30.** A 14-year-old girl living with HIV now weighs 42 kg. She is on ABC/3TC/DTG with a suppressed viral load. What is the preferred regimen for her weight according to Uganda guidelines?

A TDF/3TC/DTG (TLD) one tablet once daily
B ABC/3TC + LPV/r
C TDF/3TC/EFV 600
D AZT/3TC/NVP
E Continue paediatric dispersible tablets by weight band

**31.** A 5-year-old boy (18 kg) on cotrimoxazole prophylaxis for HIV presents with fever, red eyes, painful mouth ulcers and blistering target lesions over his trunk 3 weeks after restarting it. What is the MOST appropriate action? Advanced

A Halve the cotrimoxazole dose
B Give chlorphenamine and continue cotrimoxazole
C Stop cotrimoxazole permanently, admit for supportive care, and use dapsone for prophylaxis later
D Switch to sulfadoxine–pyrimethamine for malaria protection
E Give a course of oral prednisolone and restart cotrimoxazole once the rash fades

**32.** A 4-year-old girl (16 kg) with gastroenteritis was given IV metoclopramide 5 mg for vomiting. One hour later her eyes roll upwards and her neck twists to one side; she is conscious and frightened. Glucose 5.4 mmol/L. What is the MOST likely diagnosis and treatment?

A Febrile seizure: give rectal diazepam 0.5 mg/kg
B Hypoglycaemia: give 10% dextrose 5 mL/kg
C Meningitis: give ceftriaxone 100 mg/kg
D Hypocalcaemic tetany: give calcium gluconate
E Acute dystonic reaction: stop metoclopramide and give diphenhydramine (or benztropine)

**33.** A 3-year-old boy (14 kg) drank part of a bottle of his grandmother's oral morphine solution. He is very drowsy with pinpoint pupils, RR 8/min and $SpO_2$ 86%. After opening the airway and starting bag-mask ventilation with oxygen, which drug should you give? Advanced

A Naloxone 0.1 mg/kg (1.4 mg) IV or IM, repeated as needed
B Flumazenil 10 micrograms/kg IV
C Atropine 0.02 mg/kg IV
D Activated charcoal 1 g/kg by mouth
E Diazepam 0.3 mg/kg IV

**34.** At a health centre II in Wakiso, a village health team member brings a 2-year-old (12 kg) with fever, repeated vomiting and inability to sit up. There are no injectable antimalarials. The nearest hospital is 25 km away. What is the MOST appropriate action?

A Give artemether–lumefantrine 1 tablet and review tomorrow
B Give paracetamol and observe for 24 hours
C Give rectal artesunate 200 mg and send home with AL
D Give oral quinine 150 mg and refer if no improvement in 48 hours
E Give rectal artesunate 100 mg (one suppository) and refer immediately to hospital

**35.** A 7-month-old (7.5 kg) is admitted with severe anaemia (Hb 3.6 g/dL), malaria and respiratory distress; HR 170/min, capillary refill 2 s, liver 3 cm. The intern prescribes Ringer's lactate 20 mL/kg as a bolus while waiting for blood. What is the BEST response?

A Agree: give the bolus rapidly to support circulation
B Agree, but give 40 mL/kg instead
C Give 10 mL/kg of 50% dextrose instead
D Cancel the bolus; transfuse whole blood 20 mL/kg (or packed cells 10 mL/kg) urgently, start IV artesunate, and watch for overload
E Give furosemide 1 mg/kg and no blood

## Answers & explanations

> **1. D** · 36 mg (3 mg/kg)
>
> **D** is correct: children **under 20 kg** receive **3 mg/kg** IV artesunate (WHO since 2015, UCG 2023) because small children have lower drug exposure at 2.4 mg/kg: 3 × 12 = 36 mg (3.6 mL of the 10 mg/mL solution). **A** is wrong: 2.4 mg/kg is the dose for children ≥20 kg and adults. **B** is wrong: 2 mg/kg is not a recommended dose. **C** is wrong: giving a whole vial ignores weight-based dosing (5 mg/kg here). **E** is wrong: 10 mg/kg is the **rectal** pre-referral dose, not the IV dose.

**2. B** · Ampicillin 50 mg/kg IV 12-hourly plus gentamicin 3 mg/kg IV once daily

**B** is correct: first-line for neonatal sepsis is **ampicillin + gentamicin**; in the first week a low-birth-weight baby gets **gentamicin 3 mg/kg once daily** and ampicillin **12-hourly** (WHO; UCG uses 6-hourly). **A** is wrong: ceftriaxone displaces bilirubin in a jaundiced neonate and precipitates with the **calcium in Ringer's lactate** (potentially fatal). **C** is wrong: chloramphenicol causes **grey baby syndrome** in the first week of life. **D** is wrong: sulphonamides displace bilirubin (kernicterus) and are not first line for sepsis. **E** is wrong: a febrile, tachypnoeic, poorly feeding neonate needs parenteral antibiotics in hospital.

**3. C** · Give phenobarbital 15–20 mg/kg IV over 15–20 minutes

**C** is correct: after **two benzodiazepine doses** the child is in established status epilepticus; give **phenobarbital 15–20 mg/kg IV (or IM)** over 15–20 min (UCG 10–15 mg/kg) with airway support, or phenytoin. **A** is wrong: more than two benzodiazepine doses greatly increases apnoea risk without much added benefit. **B** is wrong: antipyretics do not stop seizures. **D** is wrong: glucose is 5.6 mmol/L, so dextrose is not indicated. **E** is wrong: phenytoin **precipitates in dextrose** and must be infused slowly (≤1 mg/kg/min) in saline with cardiac monitoring; a rapid push causes arrhythmias.

**4. E** · Mix 10 mL of 50% dextrose with 40 mL of water and give 50 mL IV over a few minutes

**E** is correct: the dose is **10% dextrose 5 mL/kg = 50 mL**; to make it from 50% take **1 mL/kg of 50% (10 mL) + 4 mL/kg of water (40 mL)** (UCG 2023), then recheck glucose in 30 min. **A** is wrong: UCG states never give undiluted 50% dextrose to children (vein damage, extravasation necrosis). **B** is wrong: that makes 25% dextrose (an adult preparation) at double the glucose dose. **C** is wrong: a slow dilute infusion does not correct life-threatening hypoglycaemia quickly. **D** is wrong: 5 mL of 50% is only 2.5 g of glucose given undiluted, and 2 hours is too long to recheck.

**5. C** · Ringer's lactate with 5% dextrose 15 mL/kg over 1 hour, monitoring pulse and RR every 5–10 minutes

**C** is correct: a child with **SAM in shock** gets **15 mL/kg over 1 hour** of a dextrose-containing isotonic or half-strength fluid (Ringer's lactate with 5% dextrose, half-strength Darrow's or 0.45% saline with 5% dextrose), with close monitoring; repeat if improving, then ReSoMal. **A** is wrong: standard rapid boluses cause heart failure in SAM. **B** is wrong: Plan C volumes are for children without SAM. **D** is wrong: ReSoMal is for SAM with dehydration without shock (or after shock is treated); this child is shocked. **E** is wrong: blood is given if there is no improvement after the first hour (or severe anaemia), not as the first fluid.

**6. A** · Oral amoxicillin 500 mg (two 250 mg dispersible tablets) twice daily for 5 days at home, with review at day 3

**A** is correct: chest-indrawing pneumonia **without danger signs** (and $SpO_2$ ≥90%) is treated with **oral amoxicillin 40 mg/kg twice daily for 5 days** as an outpatient (WHO 2024 strong recommendation); UCG band for 1–3 years is **500 mg BD**. **B** is wrong: injectable antibiotics and admission are for severe pneumonia (danger signs or hypoxaemia). **C** is wrong: three-times-daily dosing for 10 days is not the recommended regimen (40 mg/kg twice daily for 5 days); longer courses add side effects, cost and resistance without benefit. **D** is wrong: daily injections are unnecessary and expose the child to injection risks. **E** is wrong: cotrimoxazole is no longer recommended for pneumonia because of resistance.

**7. A** · Adrenaline 1:1000, 0.15 mL IM into the anterolateral thigh

**A** is correct: anaphylaxis is treated with **adrenaline 1:1000 IM 0.01 mL/kg**; children **<6 years get 0.15 mL (150 micrograms)** (UCG 2023), repeated every 5 min if needed, plus oxygen and saline 20 mL/kg for hypotension. **B** is wrong: steroids are slow adjuncts and do not treat the airway or shock. **C** is wrong: oral antihistamine is for itch after stabilisation. **D** is wrong: 1:1000 given IV (and at 0.1 mL/kg) is a dangerous overdose that can cause fatal arrhythmias. **E** is wrong: 1:10,000 at 0.15 mL delivers only 15 micrograms, a tenth of the dose.

**8. E** · 1.5 mL of 1:10,000 IO, repeated every 3–5 minutes

**E** is correct: in cardiac arrest adrenaline is **10 micrograms/kg = 0.1 mL/kg of 1:10,000** IV/IO, repeated every 3–5 min: 0.1 × 15 = 1.5 mL. **A** is wrong: 15 mL of 1:10,000 is 1.5 mg, ten times the dose. **B** is wrong: IM adrenaline is for anaphylaxis, not arrest. **C** is wrong: 1.5 mL of 1:1000 is 1.5 mg, a tenfold overdose. **D** is wrong: 0.5 mL of 1:1000 (0.5 mg) is the adult anaphylaxis dose and too much for a 15 kg child in arrest.

**9. C** · Start IV acetylcysteine 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h, and check LFTs

**C** is correct: he has ingested 5 g = **357 mg/kg**, far above the toxic threshold (**>150 mg/kg**; UCG >200 mg/kg in under-6s). Without levels, treat on the history: **acetylcysteine** is most effective within 8 h; check glucose, LFTs, INR and creatinine. **A** is wrong: patients are usually asymptomatic for the first 24 h while liver injury develops. **B** is wrong: charcoal is most useful within 1 h, and withholding the antidote risks liver failure. **D** is wrong: induced vomiting is not recommended and salt water can cause hypernatraemia. **E** is wrong: vitamin K is the antidote for warfarin, not paracetamol.

**10. D** · Give oxygen as needed, avoid emesis, lavage and charcoal, and observe for at least 6 hours

**D** is correct: the danger of kerosene is **aspiration chemical pneumonitis**; management is supportive (oxygen if $SpO_2$ falls, observation ≥6 h, CXR if breathless) and **no emesis, no lavage, no charcoal** (UCG 2023). **A** is wrong: charcoal does not bind hydrocarbons and provokes vomiting/aspiration. **B** is wrong: vomiting increases aspiration. **C** is wrong: lavage risks aspiration. **E** is wrong: antibiotics are given only if secondary bacterial pneumonia develops.

**11. E** · A clear chest on auscultation with no bronchorrhoea

**E** is correct: the goal of atropine in organophosphate poisoning is **drying of bronchial secretions** (clear chest), with HR >80/min, normal BP and dry axillae; double the dose every 5 min until reached. **A** is wrong: pupil size is a poor guide (miosis may persist). **B** is wrong: tachycardia is not the target and may reflect hypoxia or atropine excess. **C** is wrong: dry mouth occurs early and is not a reliable end-point. **D** is wrong: atropine does not reverse nicotinic muscle weakness; that needs **pralidoxime** and ventilation.

**12. A** · 6.6 mg IV once daily

**A** is correct: in the **first week** a **low-birth-weight** neonate gets **gentamicin 3 mg/kg once daily** (WHO Pocket Book): 3 × 2.2 = 6.6 mg. **B** is wrong: 7.5 mg/kg is the dose for older infants and children (and neonates in weeks 2–4). **C** is wrong: 8-hourly dosing is the old regimen and inappropriate for an immature kidney. **D** is wrong: 5 mg/kg 12-hourly doubles the daily dose. **E** is wrong: 10 mg/kg is an overdose with high oto- and nephrotoxicity risk.

**13. B** · Check the blood glucose and give 10% dextrose 5 mL/kg if low

**B** is correct: **quinine stimulates insulin release**, and malaria itself causes hypoglycaemia; a restless, sweaty, convulsing child on quinine is hypoglycaemic until proven otherwise. Check glucose and give **10% dextrose 5 mL/kg**, then run quinine in dextrose and monitor glucose 4-hourly. **A** is wrong: faster quinine worsens hypoglycaemia and cardiotoxicity. **C** is wrong: mannitol is not routine in cerebral malaria. **D** is wrong: a second-line anticonvulsant is not the first step before excluding hypoglycaemia. **E** is wrong: steroids worsen outcome in cerebral malaria.

**14. C** · IV magnesium sulphate 880 mg (40 mg/kg) over 20 minutes

**C** is correct: severe asthma not responding to the first hour of bronchodilators and steroids → **IV $MgSO_4$ 40 mg/kg (max 2 g) over 20 min**: 40 × 22 = 880 mg. Call for senior help. **A** is wrong: oral salbutamol is slow and ineffective in severe asthma. **B** is wrong: steroids have been given and take hours to work; repeating does not help now. **D** is wrong: the aminophylline loading dose is **5–6 mg/kg over at least 20 min**; 22 mg/kg rapidly could cause fatal arrhythmias and seizures. **E** is wrong: asthma attacks are rarely bacterial; antibiotics do not treat bronchospasm.

**15. B** · Nebulised adrenaline 1:1000 5 mL plus dexamethasone 0.6 mg/kg orally, then observe for at least 2–4 hours

**B** is correct: **severe croup** (stridor at rest, indrawing) needs **nebulised adrenaline 0.4–0.5 mL/kg of 1:1000 (max 5 mL)** for rapid effect plus a **steroid** (dexamethasone 0.15–0.6 mg/kg), with observation for rebound. **A** is wrong: examining the throat can precipitate complete obstruction, and croup is viral. **C** is wrong: salbutamol treats lower-airway bronchospasm, not subglottic oedema. **D** is wrong: antibiotics and boluses are not indicated. **E** is wrong: a child with stridor at rest needs admission.

**16. D** · Give dolutegravir twice daily during rifampicin and for 2 weeks after

**D** is correct: **rifampicin induces** UGT1A1 and CYP3A, lowering DTG levels; give **DTG twice daily** during rifampicin and for 2 weeks after stopping it. **A** is wrong: ART should not be stopped; it is started or continued within 2–8 weeks of TB treatment depending on CD4/TB type. **B** is wrong: nevirapine interacts badly with rifampicin and has a lower barrier to resistance. **C** is wrong: once-daily DTG would be under-dosed. **E** is wrong: halving would make the under-dosing worse.

**17. A** · Cotrimoxazole 120 mg once daily

**A** is correct: HIV-exposed infants start **cotrimoxazole prophylaxis at 6 weeks**; **<5 kg = 120 mg once daily** (UCG). Continue until HIV is excluded after breastfeeding ends. **B** is wrong: 480 mg is the dose for 15–29.9 kg. **C** is wrong: AZT is part of enhanced prophylaxis for high-risk infants only, from birth. **D** is wrong: isoniazid preventive therapy is for TB contacts or children living with HIV after excluding TB. **E** is wrong: prophylaxis should not wait for results; PCP peaks at 2–6 months.

**18. C** · 1,625 mL/day; give about two-thirds to three-quarters (≈1,100–1,200 mL/day)

**C** is correct: UCG: 100 × 10 + 50 × 10 + **25** × 5 = **1,625 mL/day**; in meningitis with hyponatraemia (SIADH) give about **2/3–3/4 of maintenance** unless dehydrated or shocked (without hyponatraemia or raised ICP, WHO and Cochrane 2016 favour full isotonic maintenance; see *Meningitis*). **A** is wrong: 1,600 mL is the Holliday–Segar figure (20 mL/kg above 20 kg), and full maintenance would worsen his hyponatraemia. **B** is wrong: 100 mL/kg for all 25 kg ignores the sliding scale. **D** is wrong: 1,250 mL is the requirement for 15 kg; fever adds about 10% per °C, not double. **E** is wrong: 1,000 mL covers only the first 10 kg.

**19. E** · 210 mL over 1 hour, then 490 mL over 5 hours

**E** is correct: Plan C for an infant **<12 months** is **100 mL/kg Ringer's lactate: 30 mL/kg (210 mL) in 1 hour, then 70 mL/kg (490 mL) in 5 hours**, reassessing frequently and repeating the first part if the radial pulse is still weak. **A** is wrong: the whole volume in 1 h is too fast. **B** is wrong: 30 min + 2.5 h is the schedule for children **≥12 months**. **C** is wrong: 20 mL/kg alone under-treats severe dehydration. **D** is wrong: ORS 75 mL/kg over 4 h is Plan B (some dehydration), and this child cannot drink.

**20. B** · ORS Plan B, zinc, and ciprofloxacin 210 mg (15 mg/kg) twice daily for 3 days

**B** is correct: she has **dysentery with some dehydration**: **Plan B ORS** (75 mL/kg over 4 h), **zinc**, and **ciprofloxacin 15 mg/kg 12-hourly × 3 days** (WHO 2024 first line; ceftriaxone second line). **A** is wrong: metronidazole is for amoebiasis (confirmed trophozoites or failure of shigella treatment), not first-line dysentery. **C** is wrong: WHO 2024 strongly recommends antibiotics for diarrhoea with blood in the stool (it advises against them in acute watery diarrhoea). **D** is wrong: amoxicillin is ineffective against *Shigella*. **E** is wrong: antimotility drugs are contraindicated in children, especially with dysentery.

**21. A** · Phenobarbital 64 mg (20 mg/kg) IV over 15–20 minutes

**A** is correct: **phenobarbital 20 mg/kg** IV (or IM) is first line for neonatal seizures (20 × 3.2 = 64 mg), with further 10 mg/kg doses to 40 mg/kg if needed, after excluding hypoglycaemia (glucose is normal here). **C** is wrong: diazepam is avoided in neonates (apnoea, displacement of bilirubin by the benzoate vehicle, short action). **B** is wrong: phenytoin is second line and must never be given in dextrose. **D** is wrong: oral carbamazepine has no role in acute neonatal seizures. **E** is wrong: midazolam infusions are for refractory seizures in NICU.

**22. B** · Always write a leading zero before a decimal point, e.g. 0.5 mg

**B** is correct: a **leading zero** (0.5 mg) prevents the decimal point being missed; writing **500 micrograms** is even safer. **A** is wrong: trailing zeros (5.0) cause tenfold overdoses when the point is missed (read as 50). **C** is wrong: 'µg' is misread as 'mg' (a thousandfold error); write micrograms in full. **D** is wrong: prescribing in mL alone without strength is a classic error source. **E** is wrong: 'u' is misread as a zero; write 'units'.

**23. D** · Codeine phosphate syrup 1 mg/kg 6-hourly

**D** is correct (least appropriate): **codeine is not used in children** (UCG 2023, WHO): CYP2D6 ultra-rapid metabolisers can develop fatal respiratory depression, and poor metabolisers get no relief. **A** is wrong as an answer: paracetamol is step 1 and appropriate. **B** is wrong as an answer: ibuprofen is appropriate in a hydrated child with normal renal function. **C** is wrong as an answer: oral morphine is appropriate for severe pain. **E** is wrong as an answer: titrated IV morphine with monitoring is standard for severe crisis pain.

**24. B** · 100,000 IU on day 1, day 2 and day 14

**B** is correct: UCG 2023 gives **three doses** of vitamin A in measles (diagnosis, next day and day 14); at **6–12 months the dose is 100,000 IU**. **A** is wrong: 50,000 IU is for infants under 6 months. **C** is wrong: 200,000 IU is for children over 12 months. **D** is wrong: a single dose is routine supplementation, not measles treatment. **E** is wrong: 6-monthly 200,000 IU is routine supplementation from 12 months.

**25. E** · Benzathine penicillin 600,000 IU deep IM every 4 weeks

**E** is correct: for children **<30 kg** secondary prophylaxis is **benzathine penicillin 600,000 IU deep IM every 4 weeks** (1.2 MU from 30 kg), continued for many years (at least 10 years after the last attack/until early adulthood; longer with significant valve disease). **A** is wrong: benzathine penicillin must **never be given IV** (cardiorespiratory arrest). **B** is wrong: amoxicillin is not the prophylactic agent and adherence to twice-daily oral therapy is poor. **C** is wrong: 2.4 MU is an adult syphilis dose. **D** is wrong: 50,000 IU/kg single dose is for asymptomatic congenital syphilis.

**26. B** · Desferrioxamine 15 mg/kg/h IV infusion

**B** is correct: 20 × 60–65 mg = about **1,250 mg elemental iron ≈ 115 mg/kg** (>60 mg/kg is serious) with GI bleeding, drowsiness and poor perfusion: give fluids and **desferrioxamine 15 mg/kg/h IV** (max 80 mg/kg/day, not >24 h). **A** is wrong: charcoal does not bind iron. **C** is wrong: acetylcysteine is for paracetamol. **D** is wrong: atropine is for organophosphates or vagal bradycardia. **E** is wrong: induced vomiting is not recommended and he is already vomiting blood.

**27. D** · Stop gentamicin and switch to ceftriaxone, review fluids and check potassium

**D** is correct: rising creatinine and oliguria on an aminoglycoside suggest **nephrotoxicity**; **stop gentamicin**, switch to a non-nephrotoxic antibiotic (**ceftriaxone** 50–80 mg/kg daily), review fluid balance and check potassium. **A** is wrong: more frequent dosing raises troughs and worsens toxicity. **B** is wrong: higher doses worsen kidney and ear damage. **C** is wrong: furosemide adds ototoxicity and does not fix the cause. **E** is wrong: pyelonephritis still needs treatment.

**28. B** · Artemether–lumefantrine 2 tablets at 0 and 8 hours, then 2 tablets twice daily on days 2 and 3, with milk or food

**B** is correct: AL band **15–24 kg = 2 tablets** per dose, six doses over 3 days (0 h, 8 h, then twice daily), with **fatty food or milk** to improve lumefantrine absorption. **A** is wrong: 1 tablet is the 5–14 kg band. **C** is wrong: AL is given twice daily; once-daily dosing gives inadequate lumefantrine levels. **D** is wrong: IV artesunate is for severe malaria. **E** is wrong: quinine is used only if no ACT is available, and 300 mg is the dose for 24–30 kg.

**29. B** · RHZ 75/50/150 three tablets daily for 2 months, then RH 75/50 three tablets daily for 2 months

**B** is correct: peripheral lymph node TB is **non-severe** TB, so she is eligible for the **4-month regimen 2HRZ/2HR** (WHO 2022; adopted in UCG 2023); at **12–15.9 kg** the dose is **3 tablets** of each dispersible FDC. **A** is wrong: 12 months is for TB meningitis and osteoarticular TB. **C** is wrong: 2 tablets under-doses a 13 kg child, and HRZ for 6 months is not a regimen. **D** is wrong: isoniazid alone is preventive therapy, not treatment of disease. **E** is wrong: streptomycin regimens are obsolete for children.

**30. A** · TDF/3TC/DTG (TLD) one tablet once daily

**A** is correct: adolescents **≥30 kg** move to **TLD** (tenofovir/lamivudine/dolutegravir) once daily, the preferred first line (Uganda consolidated guidelines 2022, UCG 2023); switching a suppressed child on DTG to TLD keeps the DTG anchor and simplifies to one pill. **D** is wrong: NVP-based regimens are no longer preferred (resistance, toxicity). **B** is wrong: LPV/r is an alternative or second-line drug. **C** is wrong: EFV 600 is not preferred; EFV 400 is an alternative only if DTG is contraindicated. **E** is wrong: dispersible paediatric formulations are for <20–25 kg.

**31. C** · Stop cotrimoxazole permanently, admit for supportive care, and use dapsone for prophylaxis later

**C** is correct: this is **Stevens–Johnson syndrome** (mucosal involvement + target lesions/blisters): **stop the drug permanently**, admit (fluids, eye and mouth care, analgesia, infection control) and use **dapsone** as alternative PCP prophylaxis. **A** is wrong: SJS is idiosyncratic; any dose can continue the reaction. **B** is wrong: antihistamines do not treat SJS and continuing the drug can be fatal. **D** is wrong: SP is also a sulphonamide and must be avoided. **E** is wrong: rechallenge after SJS is contraindicated.

**32. E** · Acute dystonic reaction: stop metoclopramide and give diphenhydramine (or benztropine)

**E** is correct: oculogyric crisis and torticollis in a conscious child after metoclopramide is an **acute dystonic reaction** (dopamine blockade); stop the drug and give an anticholinergic such as **diphenhydramine 1 mg/kg IV** (or benztropine/procyclidine); diazepam is an alternative. Avoid metoclopramide in children. **A** is wrong: she is conscious with sustained posturing, not a seizure. **B** is wrong: glucose is normal. **C** is wrong: there are no features of meningitis. **D** is wrong: tetany causes carpopedal spasm, not oculogyric crisis.

**33. A** · Naloxone 0.1 mg/kg (1.4 mg) IV or IM, repeated as needed

**A** is correct: opioid toxidrome (coma, miosis, respiratory depression) is reversed by **naloxone**; UCG starts at 0.01 mg/kg and increases to **0.1 mg/kg (max 2 mg)** if no response; in a severely depressed child give the full dose and repeat or infuse because naloxone wears off in about an hour. **B** is wrong: flumazenil reverses benzodiazepines. **C** is wrong: atropine treats bradycardia/organophosphates. **D** is wrong: oral charcoal in a drowsy child with an unprotected airway risks aspiration. **E** is wrong: diazepam would deepen respiratory depression.

**34. E** · Give rectal artesunate 100 mg (one suppository) and refer immediately to hospital

**E** is correct: a child with danger signs (vomiting everything, prostration) has suspected severe malaria; without injectables, give **rectal artesunate 10 mg/kg** (UCG band 5–<14 kg = **one 100 mg suppository**) and **refer immediately** for IV artesunate and a full ACT. **A** is wrong: a child who vomits everything cannot take oral ACT and needs parenteral therapy. **C** is wrong: 200 mg is the band for 14–19 kg, and rectal artesunate is never a complete treatment (CARAMAL lesson). **D** is wrong: oral treatment is unreliable with vomiting, and delay is dangerous. **B** is wrong: observation without antimalarial treatment risks death.

**35. D** · Cancel the bolus; transfuse whole blood 20 mL/kg (or packed cells 10 mL/kg) urgently, start IV artesunate, and watch for overload

**D** is correct: a child with **severe anaemia** and respiratory distress needs **blood**, not crystalloid: boluses dilute Hb further and precipitate heart failure (FEAST). Transfuse **20 mL/kg whole blood or 10 mL/kg packed cells over 3–4 h** (TRACT supports 30 mL/kg whole blood in afebrile children), treat malaria, monitor for overload. **A** and **B** are wrong: crystalloid boluses worsen oxygen delivery and overload the heart. **C** is wrong: 50% dextrose is never given undiluted and glucose is not the problem. **E** is wrong: furosemide alone does not correct the anaemia; it is used only if overload develops.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Cerebral malaria: artesunate, glucose and the convulsing toddler

A 2-year-old girl is brought to Mengo Hospital from Kasangati HC IV with 3 days of fever and two convulsions this morning. At the HC IV she received rectal diazepam 5 mg and IM artesunate 30 mg 4 hours ago. On arrival she responds only to pain (P on AVPU), temperature 39.2°C, HR 160/min, RR 48/min with deep breathing, capillary refill 3 s, warm hands, mild pallor. Weight 11 kg (WFH Z-score −1), MUAC 13.5 cm, no oedema. Immunisation up to date by card; mother HIV-negative.

**1. What are your immediate priorities in the first 5 minutes?**

> ETAT: airway (recovery position, suction), **oxygen** for deep breathing/low $SpO_2$, IV access, **bedside glucose** and **Hb (HemoCue)**, mRDT. Estimate and confirm weight (11 kg). Check the referral note: she has had **one benzodiazepine dose** and **IM artesunate at hour 0** (next dose due at hour 12, not a new loading dose).

**2. Glucose is 2.0 mmol/L. What do you give and how do you make it?**

> **10% dextrose 5 mL/kg = 55 mL IV** over a few minutes. From 50%: **11 mL of 50% + 44 mL water**. Never undiluted 50%. Recheck glucose in 30 min, then start a dextrose-containing maintenance fluid or NGT feeds; check glucose 4–6-hourly.

**3. She starts convulsing again 20 minutes later and is still fitting at 5 minutes. Glucose now 5.8 mmol/L. What do you give?**

> Second benzodiazepine dose: **diazepam 0.2–0.3 mg/kg IV (about 3 mg)** or **0.5 mg/kg rectally (5.5 mg)**, max 10 mg. This is her **second** dose. If she is still fitting 10 min later: **phenobarbital 15–20 mg/kg (165–220 mg) IV over 15–20 min** (UCG 10–15 mg/kg) with bag-mask ready, or phenytoin 15–20 mg/kg in saline with cardiac monitoring.

**4. Hb is 4.8 g/dL. How do you manage fluids and the anaemia?**

> Hb 4–6 with impaired consciousness and deep breathing = **transfuse**: whole blood **20 mL/kg (220 mL)** or packed cells 10 mL/kg over 3–4 h (TRACT: 30 mL/kg whole blood helped afebrile children). **No crystalloid bolus** (FEAST). Maintenance at about 2/3–3/4 of 1,050 mL/day (100 × 10 + 50 × 1) because of coma; watch for overload (crackles, liver).

**5. How do you continue antimalarial treatment and what follow-up is needed?**

> **IV artesunate 3 mg/kg = 33 mg** at 12 h and 24 h after the first dose, then daily until she can drink (minimum 24 h), then a **full 3-day ACT** (UCG 2023 discharges severe malaria on **DP**). **PDMC**: monthly DP for 3 months. **Hb at 2–3 weeks** for post-artesunate delayed haemolysis; neurological follow-up; mosquito net counselling.

**Take-away:** In cerebral malaria the drug sequence is artesunate (3 mg/kg under 20 kg), glucose (10% dextrose 5 mL/kg), no more than two benzodiazepine doses then phenobarbital, and blood rather than boluses for severe anaemia.

## Case 2: The jaundiced, septic neonate: choosing safe antibiotics

A 5-day-old boy born at 36 weeks (birth weight 2.3 kg, now 2.1 kg) is brought to the Mengo neonatal unit with poor feeding for 1 day, lethargy and a temperature of 38.4°C. RR 70/min with grunting, HR 176/min, capillary refill 3 s, $SpO_2$ 90% in air. He is jaundiced to the knees. The mother had prolonged rupture of

membranes (26 h). She is HIV-negative; the baby received vitamin K, tetracycline eye ointment, BCG and OPV0.

**1. What are the first steps?**

> Warmth (skin-to-skin or radiant warmer), **oxygen** to $SpO_2$ ≥90–95%, IV access, **glucose** (treat <2.6 mmol/L with 2 mL/kg 10% dextrose), blood culture and FBC/CRP if available before antibiotics (do not delay them), serum bilirubin. Weight for dosing: **2.1 kg**.

**2. Which antibiotics and doses?**

> **Ampicillin 50 mg/kg = 105 mg IV 12-hourly** (WHO week 1; UCG 6-hourly) **+ gentamicin 3 mg/kg = 6.3 mg IV once daily** (LBW, week 1). Once he is >7 days old, gentamicin becomes 7.5 mg/kg once daily and ampicillin 8-hourly.

**3. A colleague suggests ceftriaxone through the Ringer's lactate drip because 'it covers meningitis'. Respond.**

> **Not in this baby**: ceftriaxone **displaces bilirubin** (kernicterus risk in a jaundiced preterm) and forms **fatal precipitates with calcium** (Ringer's lactate contains calcium). If meningitis is suspected and a cephalosporin is needed, use **cefotaxime 50 mg/kg 8–12-hourly** or higher-dose ampicillin (100 mg/kg) + gentamicin; and do an **LP** when stable.

**4. What fluids do you prescribe?**

> Day 5: **150 mL/kg/day** total in principle, but if he is on IV fluids only, keep **≤100 mL/kg/day** unless under phototherapy/radiant heater or dehydrated (UCG): about **210 mL/day of 10% dextrose with electrolytes** (or half-normal saline/5% dextrose), plus **expressed breast milk by NGT** as he tolerates; phototherapy increases losses (+10–20%).

**5. He has a 2-minute focal seizure on day 2 of treatment. Glucose 3.8 mmol/L. What next?**

> **Phenobarbital 20 mg/kg = 42 mg IV** over 15–20 min (further 10 mg/kg doses to 40 mg/kg), maintenance 5 mg/kg/day; check calcium and sodium; **LP for meningitis** (if positive: antibiotics for **14–21 days**). Avoid diazepam in neonates. Monitor breathing (apnoea) after phenobarbital.

**Take-away:** Neonatal antibiotics are dosed by weight and postnatal age; ceftriaxone with jaundice or calcium, chloramphenicol in week 1 and diazepam for neonatal fits are classic errors.

## Case 3: Severe acute malnutrition with diarrhoea and shock

A 16-month-old boy is brought to the Mwanamugimu Nutrition Unit at Mulago with diarrhoea for 7 days and vomiting. Weight 6.4 kg, length 74 cm (WFL Z-score < −3), MUAC 10.9 cm, no oedema. He is lethargic, hands cold, capillary refill 4 s, weak pulse 172/min, RR 50/min, temperature 35.6°C. Glucose 2.4 mmol/L. HIV status unknown; mother on ART.

**1. What is your diagnosis and first actions?**

> **Complicated SAM with shock and hypoglycaemia** (and hypothermia). Oxygen, IV access, **10% dextrose 5 mL/kg = 32 mL IV**, warm the child (skin-to-skin, blankets), blood for Hb, malaria test, electrolytes, culture; HIV test.

**2. Which fluid, how much and how fast?**

> **15 mL/kg over 1 hour = 96 mL** of **Ringer's lactate with 5% dextrose** (or half-strength Darrow's with 5% dextrose / 0.45% saline with 5% dextrose). Check **pulse and RR every 5–10 min**: **stop if pulse rises ≥25/min or RR ≥5/min** (overload). If improving, repeat 15 mL/kg over 1 h, then switch to **ReSoMal 10 mL/kg/h orally/NGT for up to 10 h**.

**3. Which antibiotics?**

> All complicated SAM: **ampicillin 50 mg/kg = 320 mg IV/IM 6-hourly × 2 days**, then **oral amoxicillin** to complete 7 days, **plus gentamicin 7.5 mg/kg = 48 mg once daily × 7 days**. Add ceftriaxone if no improvement at 48 h or meningitis suspected. Treat malaria if positive.

**4. How do you start feeding and what micronutrients do you give or withhold?**

> **F-75 130 mL/kg/day = about 830 mL/day**, ~105 mL every 3 h (or 70 mL 2-hourly at first), **day and night**; breastfeed. **No iron** in stabilisation. **Vitamin A only if eye signs or measles**. Zinc/potassium/magnesium come in the mineral mix (no extra KCl bolus). Monitor glucose and temperature.

**5. Day 6: oedema-free, appetite good, eating RUTF. How much RUTF and what else before discharge to OTC?**

> WHO 2023: **RUTF 150–185 kcal/kg/day** → about **960–1,180 kcal ≈ 2–2.5 sachets (92 g, 500 kcal) per day**, with clean water and continued breastfeeding. No extra iron (RUTF has it). **Albendazole** if ≥1 year (200 mg at 12–23 months), measles vaccine if missed, HIV result and ART/CPT as indicated, follow-up weekly at OTC.

**Take-away:** In SAM, fluids are slow (15 mL/kg over 1 h with dextrose), antibiotics are always given, iron is withheld in stabilisation, and F-75 is fed every 2–3 hours day and night.

## Case 4: Organophosphate poisoning in a farming household

A 4-year-old boy (16 kg) is brought to a regional referral hospital 40 minutes after drinking from a soda bottle containing a pesticide used on the family's tomatoes. He is drooling, vomiting, has diarrhoea and is wheezing with copious secretions. Pupils pinpoint, HR 56/min, BP 80/50, RR 36/min, $SpO_2$ 88% in air, fasciculations of the chest wall. Glucose 6.2 mmol/L.

**1. What are your immediate actions?**

> **Protect staff** (gloves), remove contaminated clothes and wash the skin; **airway** (suction), **high-flow oxygen**, IV access. Do **not** induce vomiting. Call for senior help: he may need ventilation. Bring the bottle for identification.

**2. Which drug do you give first, at what dose, and how do you titrate it?**

> **Atropine 0.02–0.05 mg/kg IV** (UCG child 0.05 mg/kg) = about **0.8 mg**, then **double the dose every 5 min** until **atropinised**: chest clear of secretions, HR >80/min, SBP normal, dry axillae. Then an infusion and slow weaning over 24 h+. Pupils are not the end-point.

**3. What else is specific?**

> **Pralidoxime 25–50 mg/kg IV over 30 min** (400–800 mg), then **10–20 mg/kg/h** infusion (UCG), most useful within 24 h. **Diazepam 0.2–0.3 mg/kg IV** for fits. Activated charcoal 1 g/kg only if within 1 h and airway protected.

**4. Two hours later he is agitated, hot (38.9°C), HR 168/min, with dry skin and no bowel sounds; the chest is clear. Interpret.**

> **Atropine toxicity** (anticholinergic: hot, dry, tachycardic, agitated, ileus). **Stop or reduce the atropine infusion**, cool, reassess every 15 min, and restart at a lower rate if secretions return.

**5. What follow-up and prevention?**

> Watch for the **intermediate syndrome** (proximal and respiratory muscle weakness 24–96 h later): check neck flexion and breathing; ICU if needed. Counsel on storing pesticides in labelled original containers, locked away, never in drink bottles; consider whether supervision/neglect needs social work input.

> **Take-away:** Organophosphate poisoning: decontaminate, then atropine doubling every 5 minutes until the chest is dry, plus pralidoxime; watch for atropine toxicity and the intermediate syndrome.

## Case 5: Life-threatening asthma in a school child

An 8-year-old girl (24 kg) is brought to Mengo emergency by her teacher with breathlessness for 6 hours. She uses a salbutamol inhaler 'sometimes' without a spacer and has no steroid inhaler. She can say only single words, RR 46/min, severe subcostal indrawing, HR 152/min, $SpO_2$ 88% in air, quiet chest with scattered wheeze.

**1. Classify the attack and give initial treatment.**

> **Severe, bordering on life-threatening** (single words, $SpO_2$ <92%, quiet chest). Sit her up, **oxygen** to $SpO_2$ 94–98%, **salbutamol 5 mg nebulised (oxygen-driven) + ipratropium 250 micrograms**, repeated every 20 min × 3 (or MDI 10 puffs via spacer); **prednisolone 1–2 mg/kg (max 40 mg) orally** or **hydrocortisone 4 mg/kg = 96 mg IV** if she cannot swallow. Call a senior.

**2. After 1 hour she is no better: $SpO_2$ 91% on oxygen, HR 160. What next?**

> **IV magnesium sulphate 40 mg/kg = 960 mg** (max 2 g) diluted, **over 20 min**; continue frequent/continuous salbutamol. Check potassium (salbutamol lowers it) and glucose; blood gas if available; prepare HDU care.

**3. Still deteriorating. The intern proposes aminophylline 120 mg IV push. Respond.**

> Aminophylline is second line, with a **narrow therapeutic index**: **loading 5–6 mg/kg (max 300 mg) = 120–144 mg over at least 20 min (preferably 1 h)**, never a push, and **omit the load** if she took theophylline in the past 24 h; then 5 mg/kg 6-hourly. Stop if pulse >180, vomiting or fits. Consider IV salbutamol in HDU; anaesthetist for possible ventilation.

**4. She improves. What do you check before reducing treatment, and what are the warning signs of drug toxicity?**

> Talking in sentences, RR falling, $SpO_2$ ≥94% in air, minimal indrawing. Salbutamol toxicity: **tremor, tachycardia, hypokalaemia, lactic acidosis** (rising RR despite less wheeze). Aminophylline toxicity: vomiting, arrhythmia, fits. Magnesium: hypotension, weakness.

**5. Plan her discharge medicines.**

> Complete **prednisolone 3–5 days**; salbutamol MDI **with a spacer** (taught and checked); start **controller therapy** (GINA 2025 from 6 years: low-dose ICS daily or ICS-formoterol per track; in Uganda often beclometasone/budesonide where available); written action plan; review in 1 week; avoid triggers (smoke).

> **Take-away:** Acute asthma: oxygen, salbutamol + ipratropium, early steroids, then IV $MgSO_4$ 40 mg/kg; aminophylline only slowly with a load of 5–6 mg/kg and never as a push.

# References


1. Uganda Clinical Guidelines 2023: National Guidelines for Management of Common Health Conditions. *Ministry of Health Uganda*, 2023. https://library.health.go.ug/sites/default/files/resources/Uganda%20Clinical%20Guidelines%202023.pdf
2. Pocket book of hospital care for children: guidelines for the management of common childhood illnesses, 2nd edition. *World Health Organization*, 2013. https://iris.who.int/handle/10665/81170
3. Guideline on management of pneumonia and diarrhoea in children up to 10 years of age. *World Health Organization*, 2024. https://iris.who.int/handle/10665/380059
4. WHO guidelines for malaria, 10 September 2026. *World Health Organization*, 2026. https://iris.who.int/handle/10665/387504
5. WHO guideline on the prevention and management of wasting and nutritional oedema (acute malnutrition) in infants and children under 5 years. *World Health Organization*, 2023. https://iris.who.int/handle/10665/376075
6. WHO consolidated guidelines on tuberculosis. Module 5: management of tuberculosis in children and adolescents. *World Health Organization*, 2022. https://iris.who.int/handle/10665/352522
7. Updated guideline: paediatric emergency triage, assessment and treatment (ETAT): care of critically ill children. *World Health Organization*, 2016. https://iris.who.int/handle/10665/204463
8. Consolidated guidelines for the prevention and treatment of HIV and AIDS in Uganda. *Ministry of Health Uganda*, 2022.
9. Gentamicin: dosing, indications, interactions, adverse effects (pediatric dosing). *Medscape Drugs & Diseases*, 2025. https://reference.medscape.com/drug/gentamicin-342517
10. Ceftriaxone: dosing, indications, interactions, adverse effects (pediatric dosing). *Medscape Drugs & Diseases*, 2025. https://reference.medscape.com/drug/ceftriaxone-342510
11. Global Strategy for Asthma Management and Prevention (2025 update). *Global Initiative for Asthma (GINA)*, 2025. https://ginasthma.org/reports/
12. Mortality after fluid bolus in African children with severe infection (FEAST trial). *New England Journal of Medicine*, 2011. https://doi.org/10.1056/NEJMoa1101549


A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.